

POLITECNICO DI MILANO

Facoltà di Ingegneria Industriale

Corso di Laurea Magistrale in
Ingegneria Aeronautica



Analytical and Numerical Periodic Stability Analysis of Wind Turbines

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Anno Accademico 2012 - 2013

Acknowledgements

The first acknowledgements goes to my parents, for all the love and the support that they donate me every day.

I want to thank Professor Carlo Luigi Bottasso, for having trusted in my capabilities, and having given me the opportunity to work at an intriguing research project. His ample competence of the subject and ability to direct the research had greatly improved this thesis.

My sincere gratitude goes to Ph.D. Stefano Cacciola, I want to thank him for all the things he taught me, the interesting hours spent together at solving this or that problem, the amusing time, and every smile with which he welcomed me. This work would have not been possible without him. A big thanks goes also to his colleagues of the Poliwind Research Group: Carlo Riboldi, Federico Gualdoni and Filippo Campagnolo, for their suggestions and for letting me feel part of their group.

I wish to thank all the friends that accompanied me during this wonderful adventure, and that have shared with me unforgettable moments. A particular thanks goes to Laura, who in these months has become really important in my life; to Galeazzo, for his constant presence and encouragement; to Marcello for his beautiful character; and to Selene and Martina, that made even the heaviest days light-hearted.

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Abstract

The development of current, and of the near future, wind turbines require to accurately evaluate the aero-servo-elastic stability characteristics of the system. During this theses we treated the problem of performing the stability analysis by considering wind turbines periodic systems. This task has been tackled under two points of view, each intended to solve one of the most important needs appeared in this field.

In the first part we compared the periodic analysis performed with the Coleman approximation, with that performed with the more rigorous Floquet theory. The comparison has been done by means of a simplified wind turbine model, build so as to accentuate the differences between the two techniques. In particular, the model is able to represent the principal normal modes of vibration of the turbine and various aerodynamic phenomena. We also applied to the model a periodic controller, capable independently regulate the pitch of the blades. After a large number of analysis we concluded that the two techniques provide perfectly similar results, and that the only limit of the Coleman approximation lies in its inability to see more than three harmonics.

The second part has been dedicated to development of a procedure that would allow to perform the periodic stability analysis of real wind turbines, such procedure extends a recent article of Bottasso and Cacciola to the case of wind turbines operating in turbulent wind conditions. The procedure is based on the identification of a reduced order model that is able to represent one single normal mode of vibration, by using only input-output time histories. The procedure is model-independent, and is applicable to wind turbines of any class, and whether to numerical models or real wind turbines. The procedure has been developed by studying the first blade edgewise mode, that is known for being characterized by a low damping level, and we were thus able to draw the periodic Campbell diagram for that mode. We have also ascertained that the turbulence level doesn't influence the frequencies nor the dampings of that mode.

Keywords: Floquet, Coleman, stability, identification, PARMAX, turbulence.

Abstract

Sommario

Lo sviluppo delle turbine eoliche odierne, e del prossimo futuro, richiede di valutare accuratamente le caratteristiche di stabilità aero-servo-elastiche del sistema. In questa sede si è trattato il problema di effettuare l'analisi di stabilità considerando le turbine eoliche dei sistemi periodici. Questo compito è stato affrontato sotto due punti di vista, ciascuno dei quali volto a risolvere una delle più importanti necessità apparse in questo campo.

Nella prima parte sono state messe a confronto l'analisi periodica eseguita con l'approssimazione di Coleman, e quella eseguita con la più rigorosa teoria di Floquet. Il confronto è stato eseguito usando un modello semplificato di turbina eolica, costruito in modo da accentuare le differenze fra le due tecniche. In particolare, il modello è in grado di rappresentare i principali modi di vibrare della turbina e diversi fenomeni aerodinamici. Al modello è stato inoltre applicato un controllore periodico capace di regolare individualmente il passo delle pale. Al termine di un ampio numero di analisi si è constatato che le due tecniche forniscono risultati del tutto simili, e che l'unico limite dell'approssimazione di Coleman risiede nella sua incapacità di prevedere più di tre armoniche.

La seconda parte è stata invece dedicata allo sviluppo di una procedura che permettesse di eseguire l'analisi di stabilità periodica su turbine reali, tale procedura estende un recente articolo di Bottasso e Cacciola al caso di turbine operanti in condizioni di vento turbolento. La procedura si basa sull'identificazione di un modello ridotto in grado di rappresentare un singolo modo di vibrare, utilizzando unicamente storie temporali di ingresso e uscita. La procedura risulta indipendente dal modello adottato, ed è applicabile a turbine eoliche di qualunque classe, e sia a modelli numerici che a turbine reali. La procedura è stata sviluppata studiando il primo modo nel piano della pala, che è noto per essere caratterizzato da un basso livello di smorzamento, e siamo stati così in grado di disegnare il diagramma di Campbell periodico per tale modo. Abbiamo inoltre appurato che il livello di turbolenza non influenza né le frequenze né gli smorzamenti di tale modo.

Parole chiave: Floquet, Coleman, stabilità, identificazione, PARMAX, turbolenza.

Chapter 1

Introduction

In these years the need of energy coming from renewable resources has become more and more important. A variety of technologies have been developed for this purpose, and wind turbines are attracting lots of investments.

During the design and the certification of wind turbines it is required to perform the modal analysis, and so compute the natural frequencies, modal dampings and mode shapes [25]. The modal analyses serves for a number of purposes: to assess the stability of low damped modes, to understand the causes of vibration phenomena, to compute the flutter boundaries, The modal analysis can be done either numerically or experimentally, for an explanation of various experimental techniques see [34].

Because of the rotation of the rotor, wind turbines linear models are characterized by periodic rather than constant coefficients [23, 35]. For this reason the modal analysis can't be done in the standard way.

Before proceed with the discussion it is useful to review the sources of periodicity. Two sources of periodicity can be found in the structure itself. Consider a rotary wing system, when the blade is directed towards the earth its own weight acts as a preload that increase the blade natural frequency. When the blade is in the upper position the opposite happens, and the phenomenon has a cyclic nature. Also the flexibility of the support is able to cause periodic loads. The atmospheric boundary layer is responsible of the variation with the altitude of the mean wind speed. This variation is named wind shear, and can either be modeled with Prandtl's log law or the power law. In any case the mean wind speed raises with height, and hence each blade, and more important each blade section feels a different wind. Gusts, wakes, cross-flows, and the interaction with the tower manifest as azimuth dependent components of the wind velocity acting on each airfoil. Periodic loads can also be generated by the control system, where an important example is given by the Individual Pitch Controller.



Figure 1.1: Photo of an offshore wind farm.

1.1 State of the art

The state of the art in the modal analysis of wind turbine can be divided in three strategies, the choice of which depends from the result that the engineer wants to achieve.

1.1.1 Linear time invariant analysis

The first way is to perform an eigenvalue analysis of the wind turbine in the undeformed configuration, with the blades preloaded by the inertia forces that would result from the motion of the rotor. With this kind of analysis the wind

turbine is considered to operate in vacuo, without the controller, and possibly without the gravity. `Cp-Lambda`, one of the software that will be used through this thesis, performs this task by writing the structural matrices, and using the method of Arnoldi to compute the eigenvalues. `GH Bladed` uses an alternative approach, in fact it is based on the *component mode synthesis* [12], in this way the modal properties of the rotating and non-rotating parts are computed independently.

The linear time invariant analysis is fast to execute, widely implemented in industry employed software, and provides results that are easy to understand, however it is completely incapable of calculating the periodic content of the modes. Moreover it doesn't see the whirling modes.

1.1.2 Coleman based analysis

The Coleman based analysis lies between the LTI analysis and the full periodic analysis; for a recent review see [7].

In [40] it is stated that the modal analysis of a numerical model is composed by the following steps:

1. location of a steady state operating condition;
2. linearization of the equations of motion around that steady state;
3. modal decomposition of the linearized system providing modal frequencies, modal dampings and mode shapes.

The Coleman based approach consists in applying steps 1 and 2, but before applying step 3 a coordinate transformation with the Coleman matrix is done [21]. The Coleman transformation is able to change the harmonic content of the structural matrices, by leaving only a small 3 per rev periodicity (in the case of a three-bladed rotor). The remaining periodicity is then removed by an average of the structural matrices over the period. Once a constant dynamic matrix is obtained the eigenvalues and eigenvectors can be computed in the standard way.

With the Coleman based analysis it is possible to assess the stability of medium-large multi-body systems, but its implementation in high-fidelity multi-body software is cumbersome, and moreover it is difficult to bound a priori the error committed by neglecting the remaining periodicity of the system. Another aspect that should not be underestimated is that the Coleman based analysis require to linearize the system, a task which can introduce non-negligible errors.

1.1.3 Full periodic analysis

The full periodic analysis employs the rigorous Floquet theory to perform the modal analysis. Because of the high computational cost of evaluating the state

transition matrix, Floquet theory can be applied directly only to systems having a limited number of degrees of freedom. For this reason its application to systems having a large number of degrees of freedom, such as the multi-body models, required the development of some ad hoc techniques.

Bauchau and Nikishkov created a technique named *implicit Floquet analysis* [6], that makes use of the Arnoldi algorithm to extract the dominant eigenvalues of the monodromy matrix without explicitly computing it. This techniques has been applied to study the stability of wind turbines in [41].

The frequency domain approach has been pursued in [3], where the authors studied a wind turbine by extending the Operational Modal Analysis (OMA) to time periodic systems, by using the concept of harmonic transfer function. In [3] the simple peak picking method has been used, and subsequently the work has been improved in [2] by extending to the periodic case more advanced OMA techniques, like the curve fitting approaches.

In [5] it has been presented a technique that is aimed at performing periodic stability analyses over large multi-body systems, and also experimental data. The technique is based on two classes of algorithms based on the partial Floquet and the autoregressive approaches, and focuses on the monodromy matrix of the system.

In [16] the periodic stability analysis of wind turbine has been accomplished by means of system identification techniques. In this article input-output time histories has been used to identify, in a model-independent way, reduced order models capable of representing one single mode of an aero-servo-elastic system. The technique exposed however is subject to the strong hypothesis of non-turbulent wind.

1.2 Motivations for this work

Nowadays there are two important needs in the field of periodic stability analysis of wind turbines. The first concerns the assessment of the error committed by using the Coleman approximation instead of the Floquet theory, while the second is related to execute periodic stability analysis on real wind turbines.

As explained before the Coleman approximation represents a valuable tool to do the modal analysis of the system, however the error committed in using it can't be quantified a priori. In the past there has been some attempts to compare the two theories, see for example [26], [42] and [16], but in all cases the studies focused on highly simplified models, neglecting the aerodynamics and the control system, and in some cases also the gravity. Here we wanted to do the periodic stability analyses in more complex scenarios, that could accentuate the errors committed by the Coleman approximation. For this reason we required that our model must be governed by a periodic control system, and for its performance we choose the

Individual Pitch Controller [13].

With [16] Bottasso and Cacciola demonstrated that it is possible to do periodic stability analysis of real wind turbines, without having to apply the Coleman approximation or using expensive hardware. In the course of this thesis we want to carry on their work, and perform periodic stability analysis of wind turbines in turbulent wind conditions.

1.3 Main obtained results and innovative content of this thesis

During this thesis many efforts were made in performing periodic stability analysis of wind turbines. We decided to examine the problem from two different points of view: the analytical and the numerical. The first served primarily to compare the results obtained by employing the Floquet theory with those obtained with the Coleman approximation, while in the second we developed a procedure to perform periodic stability analyses in realistic wind conditions.

On the analytical side we created a new simplified wind turbine model, that conjugates: the modeling of the main modes of wind turbines, the aerodynamics and the IPC. We employed this model in a variety of scenario, and we discovered that in all cases the Coleman approximation provided results that almost match the one of the Floquet theory.

On the numerical side we invented a new way of identifying Periodic AutoRegressive Moving Average with eXogenous input models. Our approach exploit the Prediction Error Method, and minimize the cost function in a constrained way, so as to guarantee the stability of the predictor.

The task of performing periodic stability analyses of wind turbines operating in realistic, i.e. turbulent, wind conditions has been accomplished by establishing a procedure that takes into account the various sources of errors, and the statistical nature of the problem. This procedure allowed us to do the periodic stability analysis of the first blade edgewise mode, and compare the result with the ideal case. What we have discovered is that the turbulence level doesn't influence the frequencies nor the dampings of this mode.

1.4 Organization of this thesis

We will now give an overview of how the material of this thesis is organized.

CHAPTER 3 The chapter starts by exposing the Lyapunov-Floquet theory of linear time periodic systems, applied to the continuous time and the discrete time

cases. Next the Coleman approximation is used to render the system time invariant. The last section introduces a tool which will be widely used in this thesis: the periodic Campbell diagram.

CHAPTER 4 Here the Floquet and Coleman based modal analyses are compared. For this purpose an eight degrees of freedom wind turbine analytical model has been built. The model has then be linearized, and an individual pitch controller has been applied to both the linear and nonlinear models. Lastly the stability analysis with both techniques has been performed in a variety of scenarios.

CHAPTER 5 This chapter explains the system identification techniques used in this thesis. To identify PARX models the well known equation-error and the output-error techniques are used, while to identify PARMAX models a constrained minimization approach is adopted. The last section covers the equivalence between the state space and time series representations.

CHAPTER 6 The procedure developed to perform the stability analysis in turbulent wind conditions is explained in detail here. The procedure is then applied to draw the periodic Campbell diagram of the first blade edgewise mode, in turbulent and non-turbulent winds.

Capitolo 2

Introduzione (introduction in italian)

In questi anni la necessità di ottenere energia proveniente da fonti rinnovabili è considerevolmente aumentata. Per soddisfare questa richiesta è stato sviluppato un gran numero di tecnologie, ed i generatori eolici stanno attirando molti investimenti.

Durante il progetto e la certificazione delle turbine eoliche le normative richiedono l'esecuzione dell'analisi modale, e quindi il calcolo delle frequenze e degli smorzamenti propri e delle forme modali [25]. L'analisi modale serve per diversi scopi: per valutare la stabilità dei modi poco smorzati, per capire le cause di fenomeni vibratorii, per calcolare i confini delle regioni di instabilità aeroelastica, L'analisi modale può essere eseguita sia tramite il calcolo numerico che tramite degli esperimenti, per una presentazione di varie tecniche sperimentali si veda [34].

A causa della rotazione del rotore, i modelli lineari di turbine eoliche sono caratterizzati da coefficienti periodici invece che costanti [23, 35]. Per questo motivo l'analisi modale non può essere eseguita nel modo consueto.

Prima di procedere con la discussione è utile richiamare brevemente le fonti di periodicità. Due fonti di periodicità si trovano nella struttura stessa. Si consideri un sistema ad ala rotante, quando la pala è diretta verso la terra il suo peso agisce come un precarico che ne aumenta la frequenza naturale di vibrare. Quando la pala è nella posizione superiore accade l'opposto, e il fenomeno ha una natura ciclica. Anche la flessibilità del supporto è in grado di causare carichi periodici. Lo strato limite atmosferico è responsabile della variazione della velocità del vento medio all'aumentare della quota. Questa variazione prende il nome di "wind shear", e può essere modellata sia con la legge logaritmica di Prandtl che con la legge di potenza. In ogni caso la velocità del vento medio aumenta con la quota, e quindi ciascuna pala, e a maggior ragione ciascuna sezione della pala è soggetta ad un vento diverso. Raffiche, scie, "cross-flows", e l'interazione con la torre si manifestano come componenti della velocità del vento agente su ciascun

profilo dipendenti dall'azimuth. Carichi periodici possono anche essere generati dal sistema di controllo; in tal senso un importante esempio è dato dall'*Individual Pitch Controller*.

2.1 Stato dell'arte

Lo stato dell'arte nell'analisi modale delle turbine eoliche può essere suddiviso in tre strategie, la scelta della quale dipende dal risultato che l'ingegnere desidera ottenere.

2.1.1 Analisi lineare tempo invariante

La prima strategia consiste nell'eseguire l'analisi agli autovalori della turbina nella configurazione indeformata, con le pale precaricate dalle forze d'inerzia che risulterebbero dal moto del rotore. Con questo tipo d'analisi si suppone che la turbina stia operando nel vuoto, senza il controllore ed eventualmente senza la gravità. Cp-Lambda, uno dei software che verrà impiegato in questa tesi, svolge questo compito scrivendo le matrici strutturali, ed usando il metodo di Arnoldi per calcolare gli autovalori. GH Bladed utilizza un approccio alternativo, la *component mode synthesis* [12], in questo modo le proprietà modali delle parti rotanti e non rotanti vengono calcolate indipendentemente.

L'analisi lineare tempo invariante è veloce da eseguire, ampiamente implementata nei software usati nell'industria, e fornisce risultati facilmente interpretabili, tuttavia è del tutto incapace di calcolare il contenuto periodico dei modi. In più non vede i modi di whirl.

2.1.2 Analisi basata sulla trasformata di Coleman

L'analisi basata sulla trasformata di Coleman si colloca fra quella tempo invariante e quella periodica; per una rassegna recente si veda [7].

In [40] viene affermato che l'analisi modale di un modello numerico è composta dai seguenti passi:

1. individuazione di una condizione di funzionamento di equilibrio;
2. linearizzazione delle equazioni di moto nell'intorno di tale condizione;
3. decomposizione modale del sistema linearizzato, e ottenimento delle frequenze e smorzamenti modali e delle forme modali.

L'approccio basato sulla trasformata di Coleman consiste nell'applicare i passi 1 e 2, ma prima di applicare il passo 3 viene fatta una trasformazione di coordinate con la matrice di Coleman [21]. La trasformazione di Coleman è in grado di modificare il contenuto armonico delle matrici strutturali, lasciando solo una piccola periodicità alla 3 per giro (nel caso di un rotore tripala). La periodicità rimanente viene poi rimossa da una media delle matrici strutturali sul periodo. Una volta che è stata una matrice della dinamica costante gli autovalori e gli autovettori possono essere calcolati nel modo consueto.

Attraverso l'analisi basata sulla trasformata di Coleman è possibile valutare la stabilità di sistemi multi corpo aventi dimensioni medio-grandi, ma la sua implementazione in software multi corpo ad alta fedeltà risulta macchinosa, e per di più è difficile valutare a priori l'errore commesso dall'aver trascurato parte della periodicità del sistema. Un altro aspetto che non va sottovalutato è che l'analisi di stabilità basata sulla trasformata di Coleman richiede di linearizzare il sistema, un processo che può introdurre errori non trascurabili.

2.1.3 Analisi periodica

L'analisi periodica utilizza la più rigorosa teoria di Floquet per eseguire l'analisi modale. A causa dell'alto costo computazionale necessario per calcolare la matrice di transizione dello stato, la teoria di Floquet può venire applicata solo a sistemi con un numero limitato di gradi di libertà. Per questo motivo la sua applicazione a sistemi aventi un grande numero di gradi di libertà, come ad esempio i modelli multi corpo, ha richiesto lo sviluppo di tecniche ad hoc.

Bauchau e Nikishkov hanno sviluppato una tecnica che prende il nome di *analisi di Floquet implicita* [6], che fa uso dell'algoritmo di Arnoldi per estrarre gli autovalori dominanti della matrice di monodromia, senza aver bisogno di calcolarla esplicitamente. Questa tecnica è stata applicata alle turbine eoliche in [41].

L'approccio nel dominio delle frequenze è stato intrapreso in [3], ove gli autori hanno studiato una turbina eolica estendendo l'*Operational Modal Analysis* (OMA) ai sistemi periodici, facendo uso del concetto di funzione di trasferimento armonica. In [3] è stato usato il semplice metodo *peak picking*, e successivamente il lavoro è stato migliorato in [2] estendendo al caso periodico tecniche OMA più avanzate, come gli approcci all'interpolazione tramite curve.

In [5] è stata presentata una tecnica volta ad eseguire analisi di stabilità periodica su sistemi multi-corpo, ed anche dati sperimentali. La tecnica si basa su due classi di algoritmi basati sugli approcci Floquet parziale e autoregressivo, ed è focalizzata sulla matrice di monodromia del sistema.

In [16] l'analisi di stabilità periodica delle turbine eoliche è stata compiuta tramite tecniche di identificazione dei sistemi. In questo articolo storie temporali di ingressi e uscite sono state usate per identificare, in un modo indipendente dal

modello, modelli ridotti di turbine eoliche, capaci di rappresentare un singolo modo di sistemi aero-servo-elastici. La tecnica esposta soffre tuttavia dell'ipotesi particolarmente restrittiva di vento non turbolento.

2.2 Motivazioni per questo lavoro

Oggigiorno vi sono due importanti necessità nel campo dell'analisi periodica di turbine eoliche. La prima concerne la misurazione dell'errore commesso usando l'approssimazione di Coleman invece della teoria di Floquet, mentre la seconda è collegata all'eseguire analisi di stabilità periodica su turbine eoliche reali.

Come si è spiegato prima l'approssimazione di Coleman rappresenta un valido strumento per fare l'analisi modale del sistema, tuttavia l'errore commesso nell'usarla non è quantificabile a priori. Nel passato sono stati fatti vari tentativi di confrontare le due teorie, si vedano ad esempio [26], [42] e [16], ma in tutti i casi gli studi si sono focalizzati su modelli molto semplificati, che trascuravano l'aerodinamica e il sistema di controllo, e in alcuni casi anche la gravità. In questa sede abbiamo voluto fare le analisi periodiche in scenari più complessi, che potessero accentuare l'errore commesso dall'approssimazione di Coleman. Per questa ragione abbiamo imposto che il nostro modello fosse governato da un sistema di controllo periodico, e per le sue prestazioni abbiamo scelto il controllore a passo individuale [13].

Tramite [16] Bottasso e Cacciola hanno dimostrato come sia possibile eseguire analisi di stabilità periodica su turbine reali, senza dover applicare l'approssimazione di Coleman o usare hardware costoso. Nel corso di questa tesi abbiamo voluto proseguire il loro lavoro, ed eseguire analisi di stabilità periodica in condizioni di vento turbolento.

2.3 Principali risultati ottenuti e contenuto innovativo di questa tesi

Durante questa tesi sono stati compiuti molti sforzi per compiere analisi di stabilità periodica su turbine eoliche. Abbiamo deciso di esaminare il problema sotto due differenti punti di vista: quello analitico e quello numerico. Il primo è servito principalmente per confrontare i risultati ottenuti usando la teoria di Floquet con quelli ottenuti usando l'approssimazione di Coleman, mentre nel secondo abbiamo sviluppato una procedura per eseguire le analisi di stabilità periodica in condizioni di vento realistiche.

Sul fronte analitico abbiamo creato un nuovo modello semplificato di generatore eolico, che coniuga: la modellazione dei principali modi di vibrare della turbina,

l'aerodinamica e l'IPC. Abbiamo impiegato questo modello in una varietà di scenari, e abbiamo scoperto che in tutti i casi l'approssimazione di Coleman ha fornito risultati quasi identici a quelli della teoria di Floquet.

Sul fronte degli studi numerici abbiamo inventato un nuovo modo di identificare modelli periodici autoregressivi a media mobile con ingresso esogeno (PARMAX). Il nostro approccio sfrutta il metodo dell'errore di predizione (PEM), e minimizza la funzione di costo in un modo vincolato, così da garantire la stabilità del predittore.

Il compito di eseguire analisi di stabilità periodica su turbine eoliche operanti in condizioni di vento realistiche, ossia turbolente, è stato portato a termine stabilendo una procedura che tenesse in conto delle varie fonti di errore, e la natura statistica del problema. Questa procedura ci ha consentito di fare l'analisi di stabilità periodica del primo modo laterale della pala, e confrontare il risultato con il caso ideale. Così facendo abbiamo scoperto che il livello di turbolenza non influenza né le frequenze né gli smorzamenti di tale modo.

2.4 Organizzazione di questa tesi

Daremo ora un quadro generale di come è stato organizzato il materiale che compone questa tesi.

CAPITOLO 3 Il capitolo inizia con una esposizione della teoria di Lyapunov-Floquet dei sistemi lineari tempo periodici, applicata ai casi di tempo continuo e discreto. Successivamente l'approssimazione di Coleman è usata per rendere il sistema tempo invariante. L'ultima sezione introduce uno strumento che verrà ampiamente usato in questa tesi: il diagramma di Campbell periodico.

CAPITOLO 4 Qui vengono confrontate le analisi modali basate sulla teoria di Floquet e la trasformata di Coleman. Per questo scopo è stato costruito un modello analitico di turbina eolica ad otto gradi di libertà. Il modello è stato poi linearizzato, e il sistema di controllo a passo individuale è stato applicato sia al modello nonlineare che a quello linearizzato. Infine l'analisi di stabilità con entrambe le tecniche è stata eseguita in diversi casi.

CAPITOLO 5 Questo capitolo spiega le tecniche di identificazione dei sistemi usate in questa tesi. Per identificare modelli PARX sono state usate le ben note tecniche *equation-error* e *output-error*, mentre per identificare modelli i PARMAX è stato adottato un approccio a minimizzazione vincolata. L'ultima sezione tratta l'equivalenza fra le rappresentazioni in storia temporale e nello spazio degli stati.

CAPITOLO 6 La procedura sviluppata per eseguire le analisi di stabilità periodica in condizioni di vento turbolento viene spiegata in dettaglio qui. La procedura viene poi applicata per disegnare il diagramma di Campbell periodico del primo modo nel piano della pala, sia per venti turbolenti che non.

Chapter 3

Stability analysis of linear time periodic systems

During this thesis we will make a large use of linear time periodic systems, and this chapter will serve to recall their basic properties. The exposition is based on [20], [9], and also [16]. From the latter we will show some recently found results.

The chapter will start by covering the basic features of LTP systems in continuous and discrete time, and then will go on by introducing the Coleman transformation. Lastly we will introduce a recently presented tool named *periodic Campbell diagram*, that has been widely used in [16], and has revealed to be a fundamental tool during the development of this thesis.

3.1 Floquet–Lyapunov theory

The theory of Linear Time Periodic (LTP) systems has been developed by many mathematicians from 19th century until the present day. Gaston Floquet and Aleksander Mikhailovich Lyapunov played an important role in this field; this section will review a part of their contributions.

3.1.1 Continuous time

The state-space form of a continuous time linear periodic system is given by

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t) + \mathbf{B}(t)\mathbf{u}(t), \quad (3.1a)$$

$$\mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t) + \mathbf{D}(t)\mathbf{u}(t). \quad (3.1b)$$

In (3.1) t is the time, $\mathbf{x}$ is the state, $\mathbf{u}$ is the input and $\mathbf{y}$ is the output. In order for system (3.1) to be defined as periodic the matrices must obey the followings

$$\begin{aligned} \mathbf{A}(t+T) &= \mathbf{A}(t), & \mathbf{B}(t+T) &= \mathbf{B}(t), \\ \mathbf{C}(t+T) &= \mathbf{C}(t), & \mathbf{D}(t+T) &= \mathbf{D}(t). \end{aligned} \quad (3.2)$$

The system's period T is defined as the smallest time interval for which the relations (3.2) are all satisfied.

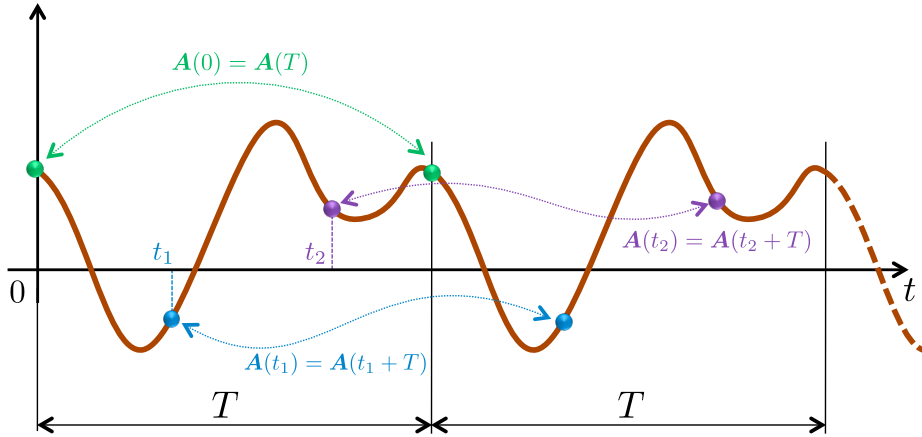


Figure 3.1: Evolution with time of a component of the $\mathbf{A}(t)$ matrix. Reproduced from [15]

As for the time invariant case the stability of system (3.1) is studied through the $\mathbf{A}(t)$ matrix. To accomplish this task we consider the autonomous version of (3.1), that is to say with the input $\mathbf{u}$ set to zero, and we associate to it an initial condition

$$\dot{\mathbf{x}}(t) = \mathbf{A}(t)\mathbf{x}(t), \quad \mathbf{x}(0) = \mathbf{x}_0. \quad (3.3)$$

Given the state at a time τ , the state at a subsequent time t is obtained by using the *state transition matrix* $\Phi(t, \tau)$

$$\mathbf{x}(t) = \Phi(t, \tau)\mathbf{x}(\tau). \quad (3.4)$$

The evolution of the state transition matrix is governed by the following ordinary differential equation

$$\dot{\Phi}(t, \tau) = \mathbf{A}(t)\Phi(t, \tau), \quad \Phi(\tau, \tau) = \mathbf{I}, \quad (3.5)$$

in which $\mathbf{I}$ is the identity matrix. It can be proved that the state transition matrix is periodic in both its arguments: $\Phi(t+T, \tau+T) = \Phi(t, \tau)$, and that swapping the time instants t and τ causes its inversion $\Phi(t, \tau)^{-1} = \Phi(\tau, t)$. The reversibility

of system (3.1) is demonstrated by proving that Φ is non-singular, and this can be easily seen through the Liouville-Jacoby formula

$$\det(\Phi(t, \tau)) = \exp \left(\int_{\tau}^t \text{tr}(\mathbf{A}(\sigma)) d\sigma \right). \quad (3.6)$$

The state transition matrix over one period is particularly important, and is named *monodromy matrix*

$$\Psi(\tau) \equiv \Phi(\tau + T, \tau). \quad (3.7)$$

By using the previous relations it can be proved that in continuous time $\Psi(\tau)$ is periodic, and non-singular $\forall t$.

Consider now the following state transformation

$$\mathbf{y}(t) = \mathbf{Q}(t)\mathbf{x}(t). \quad (3.8)$$

If $\mathbf{Q}(t)$ is T -periodic and invertible we can derive (3.8) obtaining

$$\dot{\mathbf{y}}(t) = \mathbf{Q}(t)\dot{\mathbf{x}}(t) + \dot{\mathbf{Q}}(t)\mathbf{x}(t), \quad (3.9)$$

upon substituting (3.3), and using (3.8) we obtain the transformed dynamical system

$$\dot{\mathbf{y}}(t) = \mathbf{R}(t)\mathbf{y}(t), \quad (3.10)$$

where the matrix $\mathbf{R}$, termed *Floquet factor*, is defined as

$$\mathbf{R}(t) \equiv \mathbf{Q}(t)\mathbf{A}(t)\mathbf{Q}^{-1}(t) + \dot{\mathbf{Q}}(t)\mathbf{Q}^{-1}(t). \quad (3.11)$$

The Floquet problem is that of finding a constant transformation matrix $\mathbf{R}(t) = \mathbf{R}$, so that system (3.10) becomes time invariant. If the Floquet factor is constant from its definition follows that the transformation matrix is governed by the following matrix ode

$$\dot{\mathbf{Q}}(t) = \mathbf{R}\mathbf{Q}(t) - \mathbf{Q}(t)\mathbf{A}(t) \quad (3.12)$$

whose solution is

$$\mathbf{Q}(t) = e^{\mathbf{R}(t-\tau)}\mathbf{Q}(\tau)\Phi(\tau, t). \quad (3.13)$$

By imposing on $\mathbf{Q}(t)$ the periodicity constraint we get

$$\mathbf{Q}(\tau) = e^{\mathbf{R}T}\mathbf{Q}(\tau)\Phi(\tau, \tau + T), \quad (3.14)$$

which allows to write the monodromy matrix as

$$\Psi(\tau) = \mathbf{Q}(\tau)^{-1}e^{\mathbf{R}T}\mathbf{Q}(\tau). \quad (3.15)$$

The previous equation can be solved for $\mathbf{R}$ giving

$$\mathbf{R} = \frac{1}{T} \ln [\mathbf{Q}(\tau) \Psi(\tau) \mathbf{Q}(\tau)^{-1}]. \quad (3.16)$$

We can now state that there are infinite matrices $\mathbf{R}$ satisfying (3.21), because $\mathbf{Q}(t)$ can be any invertible T -periodic matrix, and because in the complex plane the matrix logarithm has infinite possible solutions.

By introducing the *periodic eigenvector* $\mathbf{P}(t) \equiv \mathbf{Q}^{-1}$, and using (3.13) we can express the state transition matrix as

$$\Phi(t, \tau) = \mathbf{P}(t) e^{\mathbf{R}(t-\tau)} \mathbf{P}(\tau)^{-1}. \quad (3.17)$$

Choosing $\tau = 0$ and $\mathbf{P}(0) = \mathbf{I}$ leads to

$$\Phi(t, 0) = \mathbf{P}(t) e^{\mathbf{R}t}. \quad (3.18)$$

The eigenvalues of $\mathbf{R}$ are termed *characteristic exponents* and will be indicated in the following by η_j , with $j = 1 \dots N_s$, N_s being the order of the system. Equation (3.18) highlights that any periodic system is stable if all the characteristic exponents have negative real part. In light of this result we can see that the Floquet factor $\mathbf{R}$ captures the contractivity of the solution.

Instead of adopting the Floquet factor the system's stability can be assessed by using the monodromy matrix. Starting from an initial time τ the monodromy matrix can be used to obtain the system's response sampled at times $\tau + kT$, $k \in \mathbb{N}$, see figure 3.2.

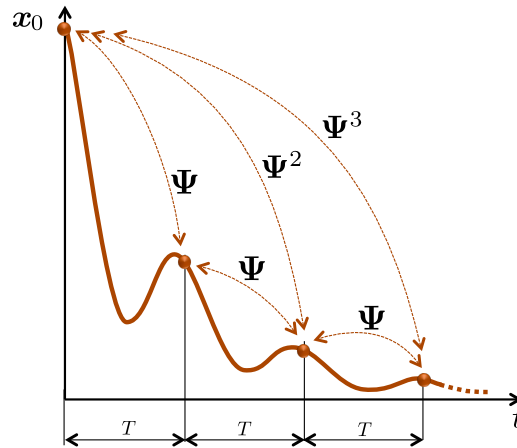


Figure 3.2: State of an LTP system and relation with the monodromy matrix. Reproduced from [15]

By defining $\hat{\mathbf{x}}(k) \equiv \mathbf{x}(\tau + kT)$ we have

$$\hat{\mathbf{x}}(k) = \Psi(\tau)^k \hat{\mathbf{x}}_0, \quad (3.19)$$

that is a discrete time system. This system is asymptotically stable if and only if the eigenvalues θ_j of Ψ , called *characteristic multipliers*, have modulus less than 1. It is possible to prove that the characteristic multipliers and their multiplicity are time invariant, even if the monodromy matrix isn't constant [9].

The factorization of Ψ is

$$\Psi(\tau) = \mathbf{S} \text{diag}(\theta_j) \mathbf{S}^{-1}, \quad (3.20)$$

and it is closely related to the one of $\mathbf{R}$

$$\mathbf{R} = \mathbf{Q}(\tau) \mathbf{S} \text{diag}(\eta_j) \mathbf{S}^{-1} \mathbf{Q}(\tau)^{-1}. \quad (3.21)$$

In turn the characteristic exponents are related to the characteristic multipliers by

$$\theta_j = e^{\eta_j T}. \quad (3.22)$$

By solving (3.22) for η_j we get

$$\eta_j = \frac{1}{T} \ln(\theta_j) = \frac{1}{T} (\ln|\theta_j| + i(\angle(\theta_j) + 2\ell\pi)), \quad \ell \in \mathbb{Z}. \quad (3.23)$$

The indeterminacy in η_j is only apparent since the monodromy matrix is uniquely defined, and so are the characteristic multipliers.

We now set for each mode j the integer ℓ equal to 0, and we indicate the corresponding characteristic exponent by $\hat{\eta}_j$. All others characteristic exponents of mode j can be recovered through

$$\eta_j = \hat{\eta}_j + i \frac{2m\pi}{T}, \quad m \in \mathbb{Z}. \quad (3.24)$$

By inserting the definition of the Floquet factor (3.16) into the one of the state transition matrix (3.17) we obtain

$$\Phi(t) = \sum_{j=1}^{N_s} \mathbf{Z}_j(t) e^{\hat{\eta}_j t}, \quad (3.25)$$

where $\mathbf{Z}_j(t)$ is defined as

$$\mathbf{Z}_j(t) \equiv \mathbf{P}(t) \mathbf{Q}(\tau) \mathbf{S} \mathbf{I}_{jj} \mathbf{S}^{-1}, \quad (3.26)$$

and $\mathbf{I}_{jj}$ is a matrix of appropriate size filled with zeros and the sole element (j, j) equal to 1. $\mathbf{Z}_j(t)$ is a periodic matrix, and we expand it in Fourier series

$$\mathbf{Z}_j(t) = \sum_{n=-\infty}^{+\infty} \mathbf{Z}_{jn} e^{in \frac{2\pi}{T} t}. \quad (3.27)$$

$\mathbf{Z}_{jn}$ is the matrix which contains the complex amplitudes of the n^{th} harmonic of $\mathbf{Z}_j(t)$. The previous expression allows us to write the state transition matrix in a new and meaningful way

$$\Phi(t) = \sum_{j=1}^{N_s} \sum_{n=-\infty}^{+\infty} \mathbf{Z}_{jn} e^{(\hat{\eta}_j + in\frac{2\pi}{T})t}. \quad (3.28)$$

This expression of the state transition matrix lead us to some consideration. At each mode j is associated an infinite number of periodic eigenvalues η_j , thus it is not characterized by a single frequency and damping, but instead by an infinite number of frequencies and dampings. The periodic eigenvalues belonging to mode j have the same real part, and are spaced along the imaginary axis by $2\pi/T$, as figure 3.3 shows. For this reason a mode is stable or not, in agreement with the previous considerations based on the characteristic multipliers.

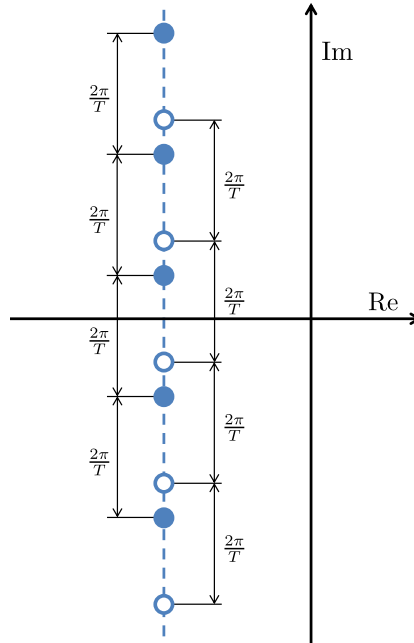


Figure 3.3: Representation of the periodic eigenvalues characterizing the response of a periodic continuous time system. A couple of complex conjugate poles is shown. Reproduced from [20]

From the knowledge of the periodic eigenvalue η_{jn} we can recover the associ-

ated frequency ω_{j_n} , and damping factor ξ_{j_n}

$$\omega_{j_n} \equiv \sqrt{(\text{Re}(\eta_{j_n}))^2 + (\text{Im}(\eta_{j_n}))^2}, \quad (3.29a)$$

$$\xi_{j_n} \equiv -\frac{\text{Re}(\eta_{j_n})}{\omega_{j_n}}. \quad (3.29b)$$

The harmonics $\mathbf{Z}_{j_n}$ are used to obtain the strength of each harmonic with respect to the others. This quantity is called *participation factor*, and will be indicated in the following by ϕ_{j_n}

$$\phi_{j_n} \equiv \frac{\|\mathbf{Z}_{j_n}\|}{\sum_n \|\mathbf{Z}_{j_n}\|}. \quad (3.30)$$

We have used $\|\cdot\|$ to denote the matrix norm, chosen here as the Frobenius one.¹

The harmonics $\mathbf{Z}_{j_n}$ can be pre-multiplied by the *constant* matrix $\mathbf{C}$ to get the participation of a particular output y , this quantity is termed *output-specific participation factor*, and is defined as

$$\phi_{j_n} \equiv \frac{\|\mathbf{C}\mathbf{Z}_{j_n}\|}{\sum_n \|\mathbf{C}\mathbf{Z}_{j_n}\|}. \quad (3.31)$$

The triads $\{\omega_{j_n}, \xi_{j_n}, \phi_{j_n}\}$ completely describe the dynamics of the periodic system. Each of the periodic eigenvalues η_{j_n} is present in the response and participate with a different strength. When looking at the spectra of the response many peaks are present at the frequencies ω_{j_n} , and their height is related to ϕ_{j_n} , it is thus of paramount importance to compute the harmonics with the highest participation factors. In [11, 38] have been observed that the choice of the integer ℓ translates in a frequency shift in $\mathbf{Z}_j$, so that the triads $\{\omega_{j_n}, \xi_{j_n}, \phi_{j_n}\}$ don't change.

Consider now the case in which for a certain mode j the participation factor associated to an harmonic is almost 1, and so all the others are close to 0. In this case the behaviour of that mode will be characterized by a single frequency and damping, and hence it will behave as time invariant. This example suggest that the participation factor could be interpreted as a periodicity indicator.

Another result reported in [16] is related to the case of time periodic output matrix $\mathbf{C}$. Let us consider a strictly proper system (i.e. with $\mathbf{D}(t) = \mathbf{0}$), the output will be given by

$$\mathbf{y}(t) = \mathbf{C}(t)\mathbf{x}(t). \quad (3.32)$$

¹The Frobenius norm of an $m \times n$ matrix A is defined as the square root of the sum of the absolute squares of its elements

$$\|A\| \equiv \sqrt{\sum_{i=1}^m \sum_{j=1}^n |a_{ij}|^2}.$$

$C(t)$ is a periodic matrix and can represent for example the passage from a fixed reference frame to a rotating one. The transition matrix between the initial state and the output is obtained by pre-multiplying the state transition matrix by $C(t)$

$$\mathbf{y}(t) = C(t)\Phi(t)\mathbf{x}(0) = \Phi_y(t)\mathbf{x}(0). \quad (3.33)$$

The output specific transition matrix $\Phi_y(t)$ can be expanded in terms of characteristic exponents and Fourier's harmonics as done in equation (3.28) for the matrix Φ , giving rise to

$$\Phi_y(t) = C(t) \sum_{j=1}^{N_s} \sum_{n=-\infty}^{+\infty} \mathbf{Z}_{j_n} e^{(\eta_j + in\frac{2\pi}{T})t}. \quad (3.34)$$

Also $C(t)$ can be expanded in Fourier series

$$C(t) = \sum_{m=-\infty}^{+\infty} C_m e^{im\frac{2\pi}{T}t}, \quad (3.35)$$

and upon substituting into (3.34) we get

$$\Phi_y(t) = C(t) \sum_{j=1}^{N_s} \sum_{m=-\infty}^{+\infty} \sum_{n=-\infty}^{+\infty} C_m \mathbf{Z}_{j_n} e^{(\eta_j + i(n+m)\frac{2\pi}{T})t}. \quad (3.36)$$

The expression can be simplified by defining a new index $r \equiv n + m$, so that the previous equation becomes

$$\Phi_y(t) = C(t) \sum_{j=1}^{N_s} \sum_{r=-\infty}^{+\infty} \mathbf{G}_{j_r} e^{(\eta_j + ir\frac{2\pi}{T})t}. \quad (3.37)$$

where

$$\mathbf{G}_{j_r} \equiv \sum_{m=-\infty}^{+\infty} C_m \mathbf{Z}_{j_{r-m}}. \quad (3.38)$$

By making use of $\mathbf{G}_{j_r}$ we are now able to define the output specific participation factor

$$\phi_{j_r}^y \equiv \frac{\|\mathbf{G}_{j_r}\|}{\sum_r \|\mathbf{G}_{j_r}\|}. \quad (3.39)$$

Looking back at demonstration we can see that characteristic exponents do not depend on the periodic output matrix $C(t)$, and so also the frequencies and dampings. On the contrary the participations depends on the particular reference frame in which the output is measured, and therefore a mode can appear more or less periodic when viewed in different reference frames. This should not be surprising, since if we express the output in multi-blade coordinates by using the Coleman matrix the periodicity is strongly reduced.

3.1.2 Discrete time

If we indicate with k the *index* of a generic time instant, we can write a linear discrete time periodic system in state space form as

$$\mathbf{x}(k+1) = \mathbf{A}(k)\mathbf{x}(k) + \mathbf{B}(k)\mathbf{u}(k), \quad (3.40a)$$

$$\mathbf{y}(k) = \mathbf{C}(k)\mathbf{x}(k) + \mathbf{D}(k)\mathbf{u}(k). \quad (3.40b)$$

The matrices in (3.40) are subject to constraints similar to the ones of the continuous time case

$$\begin{aligned} \mathbf{A}(k+K) &= \mathbf{A}(k), & \mathbf{B}(k+K) &= \mathbf{B}(k), \\ \mathbf{C}(k+K) &= \mathbf{C}(k), & \mathbf{D}(k+K) &= \mathbf{D}(k), \end{aligned} \quad (3.41)$$

where K is the number of time instants in a period.

The stability of (3.40) is studied by considering its autonomous version

$$\mathbf{x}(k+1) = \mathbf{A}(k)\mathbf{x}(k), \quad \mathbf{x}(0) = \mathbf{x}_0. \quad (3.42)$$

The associated state transition matrix obeys an equation analogous to (3.5)

$$\Phi(k+1, \kappa) = \mathbf{A}(k)\Phi(k, \kappa), \quad \Phi(\kappa, \kappa) = \mathbf{I}, \quad (3.43)$$

but differently from the continuous time case the reversibility is not guaranteed. In particular the state transition matrix is singular whenever the dynamic matrix $\mathbf{A}(k)$ is singular for some k . In this thesis we will consider only reversible systems, that is to say those for which the determinant of the state transition matrix is nonzero for all time instants.

If the discrete time periodic system is reversible the state transition matrix can be decomposed according to

$$\Phi(k, \kappa) = \mathbf{P}(k)\mathbf{R}^{k-\kappa}\mathbf{P}(\kappa)^{-1}, \quad (3.44)$$

where $\mathbf{P}(k)$ is a periodic matrix representing the periodicity of the solution and $\mathbf{R}$ is a constant matrix representing the contractivity of the solution.

Like before the monodromy matrix is defined as the state transition matrix over one period, that is to say $\Psi \equiv \Phi(\kappa+K, \kappa)$, and possess the following spectral decomposition

$$\Psi(\kappa) = \mathbf{S} \text{diag}(\theta_j) \mathbf{S}^{-1}, \quad (3.45)$$

where again θ_j are the characteristic multipliers. Similarly to the continuous time case the characteristic multipliers and their multiplicity are independent from κ .

The characteristic exponents are related to the characteristic multipliers through

$$\theta_j = \eta_j^K, \quad \forall j, \quad (3.46)$$

which stems from the definition of the monodromy matrix.

In the continuous time case the indeterminacy of the characteristic exponents appeared in equation (3.23) when we calculated the logarithm of the complex number θ_j ; in this case equation (3.46) lead us to compute the root of the complex number θ_j , and instead of an indeterminacy in the imaginary part we face an indeterminacy in the phase

$$\eta_j = \sqrt[K]{|\theta_j|} \left(\cos \left(\frac{\angle(\theta_j) + 2\ell\pi}{K} \right) + i \sin \left(\frac{\angle(\theta_j) + 2\ell\pi}{K} \right) \right), \quad (3.47)$$

where $\ell = 0, \dots, K-1$ is an arbitrary integer. An illustration of this phenomenon is given in figure 3.4.

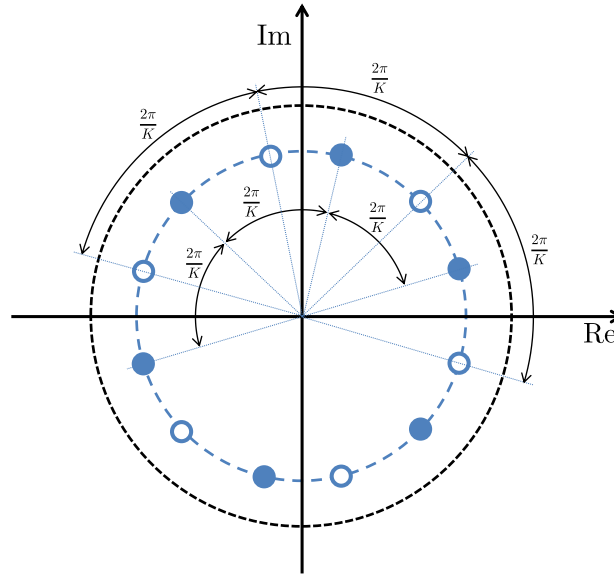


Figure 3.4: Representation of the periodic eigenvalues characterizing the response of a periodic discrete time system. A couple of complex conjugate poles is shown. Reproduced from [20]

The consideration made about the multiplicity of the characteristic exponents in the continuous time case still hold, hence also in the discrete time case there is no inconsistency on the frequencies, dampings and participation factors.

By repeating the same passages of the continuous time case the state transition matrix is expressed as

$$\Phi(k) = \sum_{j=1}^{N_s} \sum_{n=0}^{K-1} \mathbf{z}_{j_n} \left(|\eta_j| e^{i\angle(\eta_j) + n\frac{2\pi}{K}} \right)^k. \quad (3.48)$$

K characteristic exponents are present in the response of the j^{th} mode, and have equal module but different phase. Again we can observe that the characteristic exponents are all together stable or not.

Each of the discrete time characteristic exponents can be converted in its continuous time version by employing the following

$$\eta_{j\text{continuous}} = \frac{1}{\Delta t} \ln(\eta_{j\text{discrete}}). \quad (3.49)$$

The continuous time characteristic exponents, and the discrete time $\mathbf{Z}_{j_n}$ are then used to compute the triads $\{\omega_{j_n}, \xi_{j_n}, \phi_{j_n}\}$, for which the previous formulae still hold.

3.2 Coleman-based analysis

The stability of bladed rotors have been an active research field for a long time. Many researchers studied various combinations of anisotropic rotors and anisotropic supports, for a review see [50]. Among the others Coleman and Feingold succeeded in finding a coordinate transformation which strongly reduce the periodicity of the system [21, 24, 22].

The linearized equation of motion of a rotor rotating at a constant angular speed and not subject to any input can be written as

$$\mathbf{M}(t)\ddot{\mathbf{q}} + \mathbf{L}(t)\dot{\mathbf{q}} + \mathbf{K}(t)\mathbf{q} = \mathbf{0}, \quad (3.50)$$

where $\mathbf{M}(t)$, $\mathbf{L}(t)$ and $\mathbf{K}(t)$ are respectively the mass, damping and stiffness matrices. The vector $\mathbf{q}$ contains the physical degrees of freedom that belong to the blades and the support. In the following we will assume that all the blades are characterized by the same set of degrees of freedom, and that these degrees of freedom are expressed in a rotating reference frame. The motion of the support is instead given in terms of another set of variables, expressed in a fixed reference frame. From these assumption we can decompose the vector $\mathbf{q}$ in the following way

$$\mathbf{q} = (\mathbf{q}_1^T, \mathbf{q}_2^T, \dots, \mathbf{q}_B^T, \mathbf{q}_f^T)^T, \quad (3.51)$$

in which $\mathbf{q}_b$ contains the degrees of freedom of the b^{th} blade, so that $\mathbf{q}_b = (q_{1,b}, q_{2,b}, \dots, q_{N_f,b})^T$, and $\mathbf{q}_f$ contains the N_f degrees of freedom of the support.

The Coleman transformation matrix is used to express the vector of the degrees of freedom $\mathbf{q}$ into the multi-blade coordinates. This task is accomplished through the following change of variables

$$\mathbf{q} = \mathcal{C}(\psi)\mathbf{q}_c. \quad (3.52)$$

The azimuth angle of the blades ψ , is given by

$$\psi_1 = \psi_0 + \Omega t, \quad (3.53a)$$

$$\psi_b = \psi_1 + (b - 1) \frac{2\pi}{B}, \quad b = 2, \dots, B. \quad (3.53b)$$

The Coleman matrix has different expressions in accordance to the number of blades. Here we will show its expression only for a three-bladed rotor, for the general case see [42].

$$\mathcal{C}(\psi) = \begin{bmatrix} \mathbf{I}_{N_r} & \mathbf{I}_{N_r} \cos \psi_1 & \mathbf{I}_{N_r} \sin \psi_1 & \mathbf{0} \\ \mathbf{I}_{N_r} & \mathbf{I}_{N_r} \cos \psi_2 & \mathbf{I}_{N_r} \sin \psi_2 & \mathbf{0} \\ \mathbf{I}_{N_r} & \mathbf{I}_{N_r} \cos \psi_3 & \mathbf{I}_{N_r} \sin \psi_3 & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_{N_f} \end{bmatrix} \quad (3.54)$$

System (3.50) can be written as

$$\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x}, \quad (3.55)$$

with $\mathbf{x} = (\mathbf{q}^T, \dot{\mathbf{q}}^T)^T$ and

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{I} \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{L} \end{bmatrix}. \quad (3.56)$$

Following [42] we demonstrate that, given a generic time varying system $\dot{\mathbf{x}} = \mathbf{A}(t)\mathbf{x}$, and a transformation rule

$$\mathbf{x} = \mathbf{T}(t)\mathbf{y}, \quad (3.57)$$

the derivative of the state is obtained by the chain rule

$$\dot{\mathbf{x}} = \dot{\mathbf{T}}\mathbf{y} + \mathbf{T}\dot{\mathbf{y}}, \quad (3.58)$$

and by substitution the transformed system is

$$\dot{\mathbf{y}} = \mathbf{T}^{-1}(\mathbf{AT} - \dot{\mathbf{T}})\mathbf{y}. \quad (3.59)$$

By substituting $\mathbf{T}(t)$ with $\mathcal{C}(\psi(t))$ the transformation of system (3.55) becomes

$$\dot{\mathbf{y}} = \mathbf{A}_c(t)\mathbf{y}, \quad (3.60)$$

with $\mathbf{A}_c(t) = \mathcal{C}^{-1}(\mathbf{AC} - \dot{\mathcal{C}})$. Only in some special cases the Coleman transformation renders the system time invariant, thus in general an average over the period must be made

$$\bar{\mathbf{A}}_c = \frac{1}{T} \int_0^T \mathbf{A}_c(t) dt. \quad (3.61)$$

We will now attempt to study the stability of the following approximated system

$$\dot{\mathbf{y}} = \bar{\mathbf{A}}_{\mathcal{C}} \mathbf{y}, \quad (3.62)$$

and enlighten the relations with Floquet theory.

The dynamic matrix $\bar{\mathbf{A}}_{\mathcal{C}}$ has the following spectral decomposition

$$\bar{\mathbf{A}}_{\mathcal{C}} = \bar{\mathbf{V}} \text{diag}(\bar{\boldsymbol{\lambda}}) \bar{\mathbf{V}}^{-1}, \quad (3.63)$$

where $\bar{\mathbf{V}}$ is a matrix whose columns are the eigenvectors of $\bar{\mathbf{A}}_{\mathcal{C}}$, and $\bar{\boldsymbol{\lambda}}$ is the vector of the eigenvalues. The state transition matrix of (3.62) is

$$\bar{\Phi}_{\mathcal{C}}(t) = e^{\bar{\mathbf{A}}_{\mathcal{C}} t} = \bar{\mathbf{V}} e^{\bar{\boldsymbol{\lambda}} t} \bar{\mathbf{V}}^{-1}, \quad (3.64)$$

and allows to compute the response of the system

$$\mathbf{y}(t) = \bar{\mathbf{V}} e^{\bar{\boldsymbol{\lambda}} t} \bar{\mathbf{V}}^{-1} \mathbf{y}(0). \quad (3.65)$$

The state vector in multi-blade coordinates $\mathbf{y}$ can be transformed into the physical one by using again the Coleman matrix

$$\mathbf{x}(t) = \mathcal{C}(\boldsymbol{\psi}(t)) \bar{\mathbf{V}} e^{\bar{\boldsymbol{\lambda}} t} \bar{\mathbf{V}}^{-1} \mathcal{C}^{-1}(\boldsymbol{\psi}(0)) \mathbf{x}(0), \quad (3.66)$$

and in turn the approximated state transition matrix becomes

$$\bar{\Phi}(t) = \mathcal{C}(\boldsymbol{\psi}(t)) \bar{\mathbf{V}} e^{\bar{\boldsymbol{\lambda}} t} \bar{\mathbf{V}}^{-1} \mathcal{C}^{-1}(\boldsymbol{\psi}(0)). \quad (3.67)$$

The matrix $\bar{\Phi}(t)$ can be rewritten as

$$\bar{\Phi}(t) = \sum_{j=1}^{N_s} \mathcal{C}(\boldsymbol{\psi}(t)) \bar{\mathbf{V}} \mathbf{I}_{jj} \bar{\mathbf{V}}^{-1} \mathcal{C}^{-1}(\boldsymbol{\psi}(0)) \cdot e^{\bar{\lambda}_j t}, \quad (3.68)$$

which allows to define the periodic matrix $\bar{\mathbf{Z}}_j(t)$ exactly in the same way of equation (3.26)

$$\bar{\mathbf{Z}}_j(t) = \mathcal{C}(\boldsymbol{\psi}(t)) \bar{\mathbf{V}} \mathbf{I}_{jj} \bar{\mathbf{V}}^{-1} \mathcal{C}^{-1}(\boldsymbol{\psi}(0)). \quad (3.69)$$

This matrix has a limited frequency content, in fact because of the nature of $\mathcal{C}$ it has only one harmonic besides the average value. By expanding $\bar{\mathbf{Z}}_j(t)$ in Fourier series as done in (3.27) and substituting we finally get

$$\bar{\Phi}(t) = \sum_{j=1}^{N_s} \sum_{n=-1}^{+1} \bar{\mathbf{Z}}_{jn} e^{(\bar{\lambda}_j + in\Omega)t}. \quad (3.70)$$

We can now see that when the rotor is studied with Floquet theory each mode j is characterized by an infinite number of harmonics, while with the Coleman approximation only 3 are visible.

The three characteristic exponents associated to each mode are given by

$$\bar{\eta}_{j_n} = \bar{\lambda}_j + in\frac{2\pi}{T}, \quad n = -1, 0, 1. \quad (3.71)$$

To obtain the frequencies and dampings formulae (3.29) can be used. The definition (3.30) can be used to compute the participation factors, but in this work we will compare the results obtained with Floquet theory with those obtained with the Coleman approximation, hence we will use the following

$$\bar{\phi}_{j_n} = \frac{\|\bar{\mathbf{z}}_{j_n}\|}{\sum_n \|\mathbf{z}_{j_n}\|}. \quad (3.72)$$

3.3 The periodic Campbell diagram

The Campbell diagram, also known as fan plot, is a standard tool for evaluating at first sight the dynamics of rotary wing systems such that wind turbines. In the Campbell diagram the natural frequencies of the wind turbine's modes are plotted against the rotor's angular speed (or the wind velocity), together with rays from the origin representing integer multiples of the rotor's speed, see §5.8.5 of [18] and §7.5.1 of [35].

The usual way of drawing the Campbell diagram is to employ a multi-body software to obtain the variation of the natural frequencies with the rotor's speed, when the wind turbine is operating in vacuo. This method rely on the linear time invariant approximation of the system, and hence furnish only one line of frequencies per mode (see figure 3.5a). With the Coleman approximation three frequencies are visible, as in figure 3.5b, and with the periodic one an infinite number of frequencies is associated to each mode cf. figure 3.5c.

To ease the bridge between what has proved to be a valuable tool for the design of wind turbine, and this new tool, we decided to call the LTP frequency line that is closer to the MBC-LTI the *principal harmonic*. The other lines of the fan are drew on the sides of the principal harmonic.

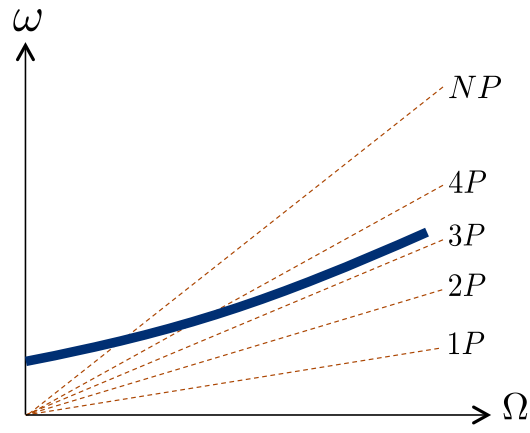
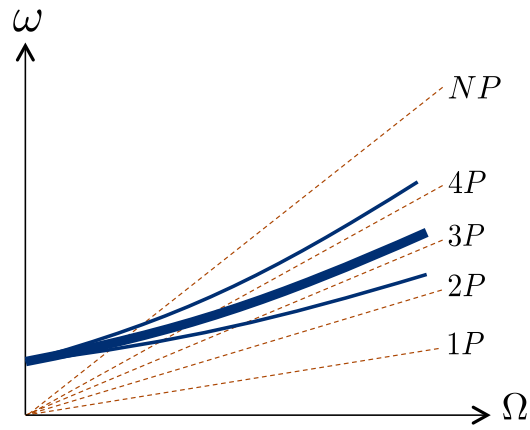
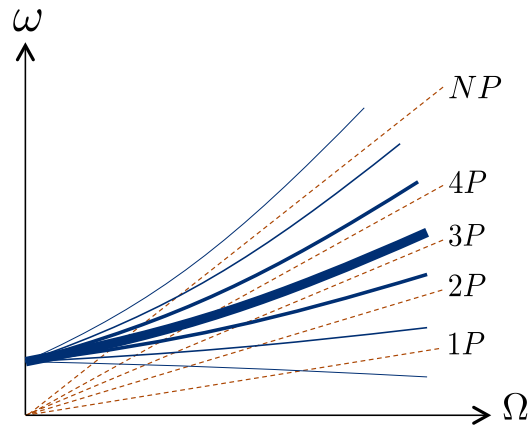

 (a) *MBC-LTI.*

 (b) *Coleman approximation.*

 (c) *LTP.*

Figure 3.5: Qualitative Campbell diagram obtained with the standard method (top), the Coleman approximation (middle), and the new periodic method (bottom). For clarity only one mode has been shown. This figure has been partly reproduced from [16].

Chapter 4

Stability analysis of a wind turbine analytical model with individual pitch controller

In this chapter we will compare the stability analyses performed with Floquet's theory with those obtained by using the Coleman's approximation, with the aim of quantifying the error committed by using the latter technique. To achieve this result we will first construct an analytical model of wind turbine, then we will increase its anisotropy by applying to it a control system, finally we will perform the stability analyses in different scenarios.

Because of the complexity of the calculus developed in the next pages we chose to rely on Wolfram Mathematica[®] for the symbolic computations, and on MathWorks[®] MATLAB[™] for the numerical evaluations.

4.1 Description of simplified wind turbine analytical models

To date only a few simplified wind turbine analytical models exist, however since wind turbines have an elevated similitude with helicopters and tilt-rotors aircrafts, the engineers involved in the design of wind turbines can chose from a variety of models.

One of the first models to study the stability of helicopter rotors has been presented by Coleman and Feingold in [21, 24, 22], and later retaken in [26]. This is a linear model that allow the in-plane motion of the hub, and consider each blade constituted by two rigid beams joined by a lag hinge. It also consider the structural damping of the hub and the blades, but neglects the aerodynamics.

In [48] the author managed to write the nonlinear equations of motion of a helicopter, by considering the couplings between the rotor and the fuselage. He considered the horizontal translation of the hub and its pitch and roll rotations. Each blade have been modeled by two rigid beams joined by a flap-lag hinge. Also this models don't consider the aerodynamics.

A tilt-rotor model has been created by Howard and presented in [31, 32]. It is a semispan model, which simulate the wing deflection with finite element elastic beams, clamped at the fuselage. The pylon is supposed to be rigid and attached to the wing tip by two springs. The blades are rigid, but can rotate about the flap and lag hinges. This linearized models includes the aerodynamics, which is inserted through the blade element momentum method.

A comprehensive wind turbine model has been published in [23]. The authors concentrated their efforts on the simulation of the blade loads, they supposed each blade formed by two rigid beams joined by a flap-lag hinge. To include the aerodynamics they applied the beam element momentum method, and they inserted in the model several characteristics of the wind.

In [37] the authors exposed a wind turbine model thought for control law testing. Differently from the models exposed so far their assume that the rotor is a flat disc swept by the blades, which are rigid, in the sense that they are not allowed to flap, lag or twist. Nevertheless the rotor has six degrees of freedom, and include the aerodynamics.

In [42] Skjoldan and Hansen proposed a wind turbine model in which the blades are rigid beams joined at the root by flap hinges. The nacelle is rigid and can tilt and yaw on a tower supposed rigid. This model doesn't include the aerodynamics.

Hansen proposed another wind turbine model in [28]. Here the model has seven degrees of freedom. The nacelle is free to move in the horizontal plane, and can tilt and yaw about the tip of the tower. The shaft bending is modeled by two springs so that the entire rotor can rotate out of its plane; also the shaft and drive train torsion is included. Differently from the other models here the blades are assumed flexible, but the aerodynamics has not been included, and the model is linear.

4.2 Model description

In this section we present an analytical model suited for the study of B -bladed wind turbines, that aims to be a bridge between those of Eggleston and Stoddard [23] and Hammond [26].

Each blade is modeled through two rigid beams joined by a hinge, that allow the in-plane and the out of plane rotations (see figure 4.1). The hub is modeled with a point mass, that is constrained to move in the horizontal plane, so as to model the

horizontal in-plane and the fore-aft tower modes. On each of the three hinges two spring-damper systems constrain the flap and lag rotations, and the same has been done for the hub displacements. The rotor is forced to rotate at a constant angular speed, and is subject to inertial, gravitational and aerodynamic loads.

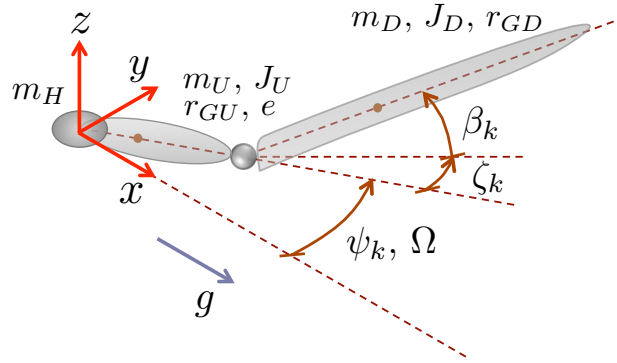


Figure 4.1: Blade's degrees of freedom (adapted from [48]).

The degree of freedom selected for this model will allow to see the following modes (sorted in order of ascending non-rotating frequency):

1. longitudinal tower,
2. lateral tower,
3. flapwise backward whirling,
4. flapwise forward whirling,
5. symmetric flapwise,
6. edgewise backward whirling,
7. symmetric edgewise,
8. edgewise forward whirling.

In order to write the equations of motion we have used only one reference frame, shown in figure 4.2 and defined as follows:

- the origin is located on the hub;
- the x axis points downward;
- the z axis coincide with the shaft and is directed from the tower to the rotor;
- the y axis forms a right hand triad with the other two, and is thus directed to the right.

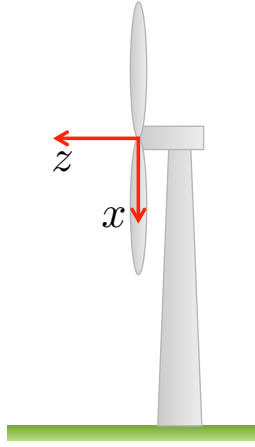


Figure 4.2: Rotor's reference frame. The y axis is directed toward the reader.

4.3 Derivation of the equations of motion

Between the various principle belonging to the field of classical mechanics we chose to use the Lagrange's equations. In the following we will write each of the terms needed by Lagrange's equations. During the demonstration we will drop the subscript k , which denote the blade number, wherever possible.

4.3.1 Blade kinematic

The part of the blade between the hub and the hinge is an homogeneous beam that rotates around the hub. Let r be the dimensional abscissa along this part of the blade, then we can write the position of a generic point belonging to it as

$$\mathbf{r}_U = \begin{pmatrix} r \cos \psi \\ y_H + r \sin \psi \\ z_H \end{pmatrix}. \quad (4.1)$$

The time derivative of $\mathbf{r}_U$ gives the velocity of the first part of the blade.

$$\mathbf{v}_U = \begin{pmatrix} -r\Omega \sin \psi \\ \dot{y}_H + r\Omega \cos \psi \\ \dot{z}_H \end{pmatrix}. \quad (4.2)$$

By indicating again with r the position of a point lying on the second part of the blade we can write the second position vector as

$$\mathbf{r}_D = \begin{pmatrix} e \cos \psi + r \cos \beta \cos(\psi + \zeta) \\ y_H + e \sin \psi + r \cos \beta \sin(\psi + \zeta) \\ z_H + r \sin \beta \end{pmatrix}. \quad (4.3)$$

The expression of $\mathbf{r}_D$ can be proved through the use of three-dimensional rotations, but also by looking at figure 4.1 and observing that the abscissa r projected in the rotor's plane has a length of $r \cos \beta$, while in the z direction its length is $r \sin \beta$. We can now derive $\mathbf{r}_D$ with respect to time to get the velocity of the second part of the blade

$$\mathbf{v}_D = \begin{pmatrix} -e\Omega \sin \psi - r(\Omega + \dot{\zeta}) \cos \beta \sin(\psi + \zeta) - r\dot{\beta} \sin \beta \cos(\psi + \zeta) \\ \dot{y}_H + e\Omega \cos \psi + r(\Omega + \dot{\zeta}) \cos \beta \cos(\psi + \zeta) - r\dot{\beta} \sin \beta \sin(\psi + \zeta) \\ \dot{z}_H + r\dot{\beta} \cos \beta \end{pmatrix}. \quad (4.4)$$

4.3.2 Kinetic energy

The kinetic energy of the whole rotor is obtained by summing the kinetic energy of the hub and of the three blades

$$T = T_H + \sum_{k=1}^B (T_{U_k} + T_{D_k}). \quad (4.5)$$

The hub's kinetic energy can be written directly

$$T_H = \frac{1}{2} m_H (\dot{y}_H^2 + \dot{z}_H^2), \quad (4.6)$$

while the blade's kinetic energy will require some more effort.

The kinetic energy of the first part of the blade is written by computing the following line integral, in which $\rho(r)$ is the linear density,

$$T_U = \frac{1}{2} \int_0^e \rho(r) \mathbf{v}_U(r) \cdot \mathbf{v}_U(r) dr, \quad (4.7)$$

which gives

$$T_U = \frac{1}{2} (J_U \Omega^2 + 2m_U r_{GU} \Omega \dot{y}_H \cos \psi + m_U (\dot{y}_H^2 + \dot{z}_H^2)). \quad (4.8)$$

The same procedure is applied to get the kinetic energy of the second part of the blade, but in this case the integration goes from the hinge till the tip:

$$T_D = \frac{1}{2} \int_e^R \rho(r) \mathbf{v}_D(r) \cdot \mathbf{v}_D(r) dr. \quad (4.9)$$

By carrying out the computations we arrive at

$$\begin{aligned}
T_D = \frac{1}{2} & \left(m_D (e^2 \Omega^2 + 2e\Omega \dot{y}_H \cos \psi + \dot{y}_H^2 + \dot{z}_H^2) \right. \\
& + J_D (\Omega^2 \cos^2 \beta + \dot{\beta}^2 + 2\Omega \dot{\zeta} \cos^2 \beta + \dot{\zeta}^2 \cos^2 \beta) \\
& + 2m_D r_{GD} (e\Omega^2 \cos \beta \cos \zeta + \dot{z}_H \dot{\beta} \cos \beta \\
& + \Omega (\dot{y}_H \cos \beta \cos(\psi + \zeta) - e\dot{\beta} \sin \beta \sin \zeta + e\dot{\zeta} \cos \beta \cos \zeta) \\
& \left. + \dot{y}_H (\dot{\zeta} \cos \beta \cos(\psi + \zeta) - \dot{\beta} \sin \beta \sin(\psi + \zeta)) \right). \tag{4.10}
\end{aligned}$$

The rotor's kinetic energy is obtained by inserting (4.6), (4.8) and (4.10) into (4.5). The result can be simplified by remembering that for $B > 1$

$$\sum_{k=1}^B \sin \psi_k = 0, \quad \sum_{k=1}^B \cos \psi_k = 0, \tag{4.11}$$

as done in [26], so that the second addendum in (4.8) simplifies. The reader should also observe that in the case of constant rotor speed the inertia of the first part of the blade J_U doesn't contribute to the dynamics.

4.3.3 Potential energy

The contributions to the potential energy of the system comes from the springs and the weight, so that

$$V = V_{y_H} + V_{z_H} + \sum_{k=1}^B (V_{\beta_k} + V_{\zeta_k} + V_{U_k} + V_{D_k}). \tag{4.12}$$

The cone angle has not been included in the model, thus none of the springs have been preloaded, and their potential energies are given by

$$V_{y_H} = \frac{1}{2} K_y y_H^2, \tag{4.13}$$

$$V_{z_H} = \frac{1}{2} K_z z_H^2, \tag{4.14}$$

$$V_{\beta} = \frac{1}{2} K_{\beta} \beta^2, \tag{4.15}$$

$$V_{\zeta} = \frac{1}{2} K_{\zeta} \zeta^2. \tag{4.16}$$

The potential energy of the weight can be easily written as

$$V_U = -m_U g x_{CG_U} = -m_U g r_{GU} \cos \psi, \quad (4.17)$$

$$V_D = -m_D g x_{CG_D} = -m_D g r_{GD} \cos \beta \cos(\psi + \zeta). \quad (4.18)$$

The rotor's potential energy is obtained by inserting the previous into (4.12).

4.3.4 Damping function

The derivation of the damping function is rather similar to that of the potential energy, in fact

$$D = D_{y_H} + D_{z_H} + \sum_{k=1}^B (D_{\beta_k} + D_{\zeta_k}), \quad (4.19)$$

and

$$D_{y_H} = \frac{1}{2} C_y \dot{y}_H^2, \quad (4.20)$$

$$D_{z_H} = \frac{1}{2} C_z \dot{z}_H^2, \quad (4.21)$$

$$D_{\beta} = \frac{1}{2} C_{\beta} \dot{\beta}^2, \quad (4.22)$$

$$D_{\zeta} = \frac{1}{2} C_{\zeta} \dot{\zeta}^2. \quad (4.23)$$

4.3.5 Generalized forces

The aerodynamic load enters the model through the generalized forces. The aerodynamic model is taken from [23] with the little modification of adding the hub velocity ($\dot{y}_H$, $\dot{z}_H$) in the inflow and crossflow terms, and setting the yaw rate to zero.

The aerodynamic model presented in [23] use the beam element momentum method to get the root loads caused by the air. The authors considered the following contributions:

- Wind's relative velocity caused by the rotation of the blade and the flap velocity.
- Uniform inflow over the rotor disc.
- Horizontal crosswind and yaw rate.
- Vertical wind shear.

In section 8.5 of [23] the aerodynamic distributed load is integrated along the blade to get the hub shears and the hinge moments, for convenience we will report them here, with the aforementioned changes.

Table 4.1: Definitions of the symbols.

Symbol	Meaning	Expression
λ	nondimensional inflow	$(V_0 - v_i - \dot{z}_H)/(\Omega R)$
$\bar{U}_0$	nondimensional crossflow	$(U_0 - \dot{y}_H)/(\Omega R)$
$\bar{V}_0$	nondimensional free stream	$V_0/(\Omega R)$
γ	Lock number	$\rho C_{l_\alpha} c R^4 / J_D$

The hub shear in the fore-aft direction is

$$S_\beta^{\text{aero}} = \frac{1}{2} \gamma J_D \frac{\Omega^2}{R} \left\{ \frac{\lambda}{2} + \frac{\theta_p}{3} - \frac{\dot{\beta}/\Omega}{3} - \sin \psi \left[\frac{\bar{U}_0 \beta}{2} \right] - \cos \psi \left[\bar{U}_0 \left(\lambda - \frac{\dot{\beta}/\Omega}{2} - \theta_p \right) + \frac{K_1 \bar{V}_0}{3} \right] \right\}. \quad (4.24)$$

The hinge out of plane moment is ¹

$$M_\beta^{\text{aero}} = \frac{1}{2} \gamma J_D \Omega^2 \left\{ \frac{\lambda}{3} + \frac{\theta_p}{4} - \frac{\dot{\beta}/\Omega}{4} - \sin \psi \left[\frac{\bar{U}_0 \beta}{3} \right] - \cos \psi \left[\bar{U}_0 \left(\frac{\lambda}{2} - \frac{\dot{\beta}/\Omega}{3} + \frac{2\theta_p}{3} \right) + \frac{K_1 \bar{V}_0}{4} \right] \right\}. \quad (4.25)$$

The hub shear in the direction parallel to the blade's chord, and pointing toward the leading edge, is

$$S_\zeta^{\text{aero}} = \frac{1}{2} \gamma J_D \frac{\Omega^2}{R} \left\{ \lambda \left(\lambda + \frac{\theta_p}{2} \right) - \frac{\dot{\beta}}{\Omega} \left(\lambda + \frac{\theta_p}{3} \right) - \sin \psi \left[\beta \bar{U}_0 \left(2\lambda + \frac{\theta_p}{2} - \frac{\dot{\beta}}{\Omega} \right) \right] - \cos \psi \left[K_1 \bar{V}_0 \left(\lambda + \frac{\theta_p}{3} \right) + \bar{U}_0 \theta_p \lambda - \frac{\dot{\beta}}{\Omega} \left(\frac{2}{3} K_1 \bar{V}_0 + \frac{\bar{U}_0 \theta_p}{2} \right) \right] \right\}. \quad (4.26)$$

¹In [23] there was a sign error, later corrected in [35] and [15].

The hinge moment in the edgewise direction is

$$\begin{aligned}
 M_{\zeta}^{\text{aero}} = & \frac{1}{2} \gamma J_D \Omega^2 \left\{ \lambda \left(\frac{\lambda}{2} + \frac{\theta_p}{3} \right) - \frac{\dot{\beta}}{\Omega} \left(\frac{2}{3} \lambda + \frac{\theta_p}{4} \right) \right. \\
 & - \sin \psi \left[\beta \bar{U}_0 \left(\lambda + \frac{\theta_p}{3} - \frac{2}{3} \frac{\dot{\beta}}{\Omega} \right) \right] \\
 & \left. - \cos \psi \left[K_1 \bar{V}_0 \left(\frac{2}{3} \lambda + \frac{\theta_p}{4} \right) + \frac{\bar{U}_0 \theta_p \lambda}{2} - \frac{\dot{\beta}}{\Omega} \left(\frac{K_1 \bar{V}_0}{2} + \frac{\bar{U}_0 \theta_p}{3} \right) \right] \right\}.
 \end{aligned} \tag{4.27}$$

The forces and moments multiplied by the virtual variations of the degrees of freedom forms the aerodynamic virtual work

$$\delta W^{\text{aero}} = \sum_{k=1}^B \left(S_{\zeta_k}^{\text{aero}} \cos(\psi_k + \zeta_k) \delta y_H + S_{\beta_k}^{\text{aero}} \delta z_H + M_{\beta_k}^{\text{aero}} \delta \beta_k + M_{\zeta_k}^{\text{aero}} \delta \zeta_k \right). \tag{4.28}$$

The generalized forces are obtained by deriving the aerodynamic virtual work by the virtual variations of the degrees of freedom

$$Q_{y_H} = \sum_{k=1}^B S_{\zeta_k}^{\text{aero}} \cos(\psi_k + \zeta_k), \tag{4.29}$$

$$Q_{z_H} = \sum_{k=1}^B S_{\beta_k}^{\text{aero}}, \tag{4.30}$$

$$Q_{\beta} = M_{\beta}^{\text{aero}}, \tag{4.31}$$

$$Q_{\zeta} = M_{\zeta}^{\text{aero}}. \tag{4.32}$$

$$\tag{4.33}$$

4.3.6 Nonlinear equations of motion

We now have all the ingredients to write the equations of motion. The Lagrange equation written for the generic free coordinate q_k is

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{q}_k} \right) - \frac{\partial T}{\partial q_k} + \frac{\partial V}{\partial q_k} + \frac{\partial D}{\partial \dot{q}_k} = Q_{q_k}. \tag{4.34}$$

By substituting q_k with β we get the flap equation of motion

$$\begin{aligned}
 J_D \ddot{\beta} + C_{\beta} \dot{\beta} + K_{\beta} \beta = & M_{\beta}^{\text{aero}} \\
 & - J_D (\Omega + \dot{\zeta})^2 \cos \beta \sin \beta \\
 & - m_D r_{GD} (g \cos(\psi + \zeta) \sin \beta + e \Omega^2 \cos \zeta \sin \beta \\
 & - \ddot{y}_H \sin(\psi + \zeta) \sin \beta + \ddot{z}_H \cos \beta).
 \end{aligned} \tag{4.35}$$

The lag equation is obtained by substituting q_k with ζ

$$\begin{aligned}
J_D \cos^2 \beta \ddot{\zeta} + C_\zeta \dot{\zeta} + K_\zeta \zeta &= M_\zeta^{\text{aero}} \\
&+ 2J_D(\Omega + \dot{\zeta})\dot{\beta} \cos \beta \sin \beta \\
&- m_D r_{GD} \cos \beta (g \sin(\psi + \zeta) + e\Omega^2 \sin \zeta \\
&+ \ddot{y}_H \cos(\psi + \zeta)).
\end{aligned} \tag{4.36}$$

Similarly the hub fore-aft equation is

$$\begin{aligned}
(m_H + B(m_U + m_D))\ddot{z}_H + C_z \dot{z}_H + K_z z_H \\
= \sum_{k=1}^B S_{\beta_k}^{\text{aero}} - m_D r_{GD} \sum_{k=1}^B (\ddot{\beta}_k \cos \beta_k - \dot{\beta}_k^2 \sin \beta_k),
\end{aligned} \tag{4.37}$$

and the hub side-side equation is

$$\begin{aligned}
(m_H + 3(m_U + m_D))\ddot{y}_H + C_y \dot{y}_H + K_y y_H &= \sum_{k=1}^B S_{\zeta_k}^{\text{aero}} \cos(\psi_k + \zeta_k) \\
&+ m_D r_{GD} \sum_{k=1}^B (\Omega^2 \cos \beta_k \sin(\psi_k + \zeta_k) + \dot{\beta}_k^2 \cos \beta_k \sin(\psi_k + \zeta_k) \\
&+ 2\dot{\beta}_k \dot{\zeta}_k \sin \beta_k \cos(\psi_k + \zeta_k) + \dot{\zeta}_k^2 \cos \beta_k \sin(\psi_k + \zeta_k) \\
&+ 2\Omega(\dot{\beta}_k \sin \beta_k \cos(\psi_k + \zeta_k) + \dot{\zeta}_k \cos \beta_k \sin(\psi_k + \zeta_k)) \\
&+ \ddot{\beta}_k \sin \beta_k \sin(\psi_k + \zeta_k) - \ddot{\zeta}_k \cos \beta_k \cos(\psi_k + \zeta_k)).
\end{aligned} \tag{4.38}$$

It is possible to prove that the equations (4.35), (4.36) and (4.38) are equivalent to the ones presented in [23] and [26], under the hypotheses of small angles and displacements, and the removal of the appropriate degrees of freedom.

The parameters that appear in the equation of motion have been set accordingly with table 4.2, which extends the one presented in [16]. The system composed by equations 4.35–4.38 has been integrated with MATLAB's function `ode15i`, under the following conditions:

- null initial conditions for the degrees of freedom and their first derivatives,
- collective pitch set to -0.54 deg,

- frontal wind of 9 m/s;

and the result can be seen in figures 4.3 and 4.4.

Table 4.2: Analytical wind turbine model: main parameters and their numerical values

Parameter	Symbol	Value	
Number of blades	B	3	
Rotor radius	R	75	[m]
Rated rotor speed	Ω_r	11.5	[rpm]
Hinge offset	e	25.651	[%R]
Mass of hub	m_H	7.5000E+4	[kg]
Blade mass (movable part)	m_D	1.4482E+4	[kg]
Blade mass (fixed part)	m_U	1.0873E+4	[kg]
Blade CG after hinge	r_{GD}	18.724	[m]
Blade moment of inertia	J_D	7.4881E+6	[kg m ²]
Edgewise spring stiffness	K_ζ	2.1192E+8	[N m]
Edgewise spring damper	C_ζ	1.7555E+6	[N m s]
Flapwise spring stiffness	K_β	5.2147e7	[N m]
Flapwise spring damper	C_β	1.7555E+6	[N m s]
Tower SS spring stiffness	K_y	7.3116E+5	[N m ⁻¹]
Tower SS spring damper	C_y	1.3294E+4	[N s m ⁻¹]
Tower FA spring stiffness	K_z	6.58046E+5	[N m ⁻¹]
Tower FA spring damper	C_z	1.3294E+4	[N s m ⁻¹]
Lock number	γ	20	
Wind shear gradient	K_1	0.018	[s ⁻¹]

The time histories of the degrees of freedom allow some considerations.

- After about ten periods the solution becomes periodic. This suggest that the best way of linearizing the equations of motion is in the neighborhood of the periodic steady state.
- The flap mode gets quickly damped by the aerodynamics, while the same doesn't happen for the lag mode, for which the aerodynamic damping is comparable in magnitude with the structural one.
- The flap angle, and accordingly the hub fore-aft displacement, are fed by the aerodynamic, and when the wind turbine is operating in vacuo they go to zero. The same doesn't happen to the lag angle and the hub side-side displacement.

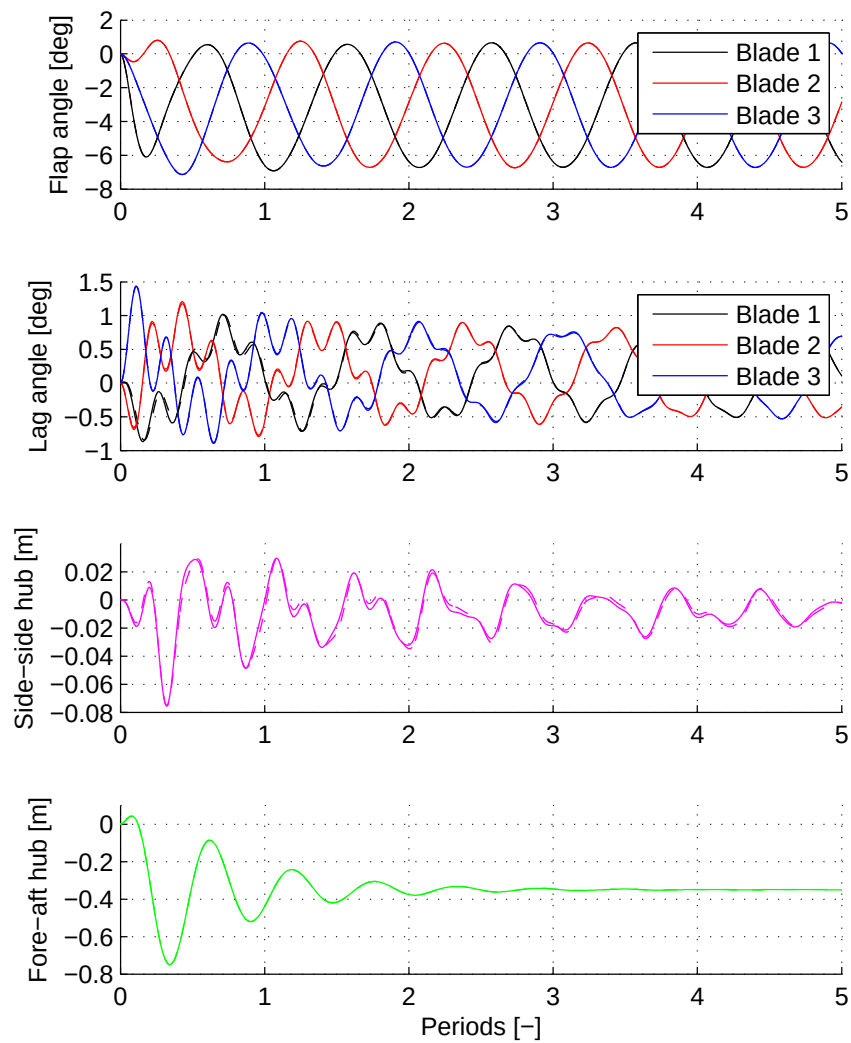


Figure 4.3: Comparison between the nonlinear (in continuous line) and the linearized (in dashed line) solutions for the uncontrolled wind turbine model. 30 periods were simulated.

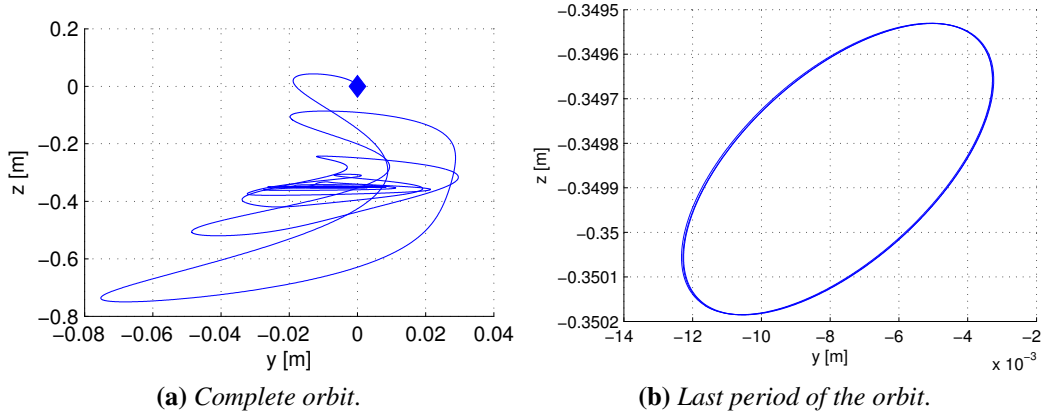


Figure 4.4: Top view of the hub motion for the nonlinear simulation of figure 4.3, with magnification of the last period (right).

- The time histories of the hub in-plane and fore-aft displacement become sinusoidal with the same phase and 3P frequency, and that's the reason for which figure 4.4b shows an ellipse. This is in agreement with figure 3 of [29], where the authors examined a multibody model.

4.4 Periodic linearization

The equations of motion of our analytical model form a system of: implicit, second order, ordinary, nonlinear differential equations. The system can be written concisely as

$$F(\xi, \dot{\xi}, \ddot{\xi}, \nu, t) = 0, \quad (4.39)$$

where ξ is the vector of the degrees of freedom, namely

$$\xi = (\beta_1, \beta_2, \beta_3, \zeta_1, \zeta_2, \zeta_3, y_h, z_H)^T, \quad (4.40)$$

and ν is the vector of the inputs

$$\nu = (\theta_{p1}, \theta_{p2}, \theta_{p3})^T. \quad (4.41)$$

Although in principle system (4.39) could be written in the following form

$$\ddot{\xi} + \bar{f}(\xi, \dot{\xi}, \nu, t) = 0, \quad (4.42)$$

we haven't pursued this path. Instead we have combined the technique of the periodic linearization exposed in [9] with the linearization of implicit ODE used in [44].

Previously we have observed that by integrating system (4.39) with a periodic input for a sufficiently long time, the solution becomes periodic. Thus the equations of motion can be linearized for small displacement from it.

We indicate the periodic trajectory with $(*)$, so that the periodic state vector is $\tilde{\xi}(t)$, and the periodic input is $\tilde{\nu}(t)$. The linearized state vector and input are defined as

$$\hat{\xi}(t) \equiv \xi(t) - \tilde{\xi}(t), \quad \hat{\nu}(t) \equiv \nu(t) - \tilde{\nu}(t). \quad (4.43)$$

The mass, damping and stiffness matrices are obtained by deriving the vectorial functional F in (4.39) with respect to the state vector, its time derivatives, and the input, and evaluating the result over the periodic trajectory.

$$\begin{aligned} M(t) &\equiv \left. \frac{\partial F}{\partial \ddot{\xi}} \right|_{\tilde{\xi}, \dot{\tilde{\xi}}, \ddot{\tilde{\xi}}, \tilde{\nu}}, & D(t) &\equiv \left. \frac{\partial F}{\partial \dot{\xi}} \right|_{\tilde{\xi}, \dot{\tilde{\xi}}, \ddot{\tilde{\xi}}, \tilde{\nu}}, \\ K(t) &\equiv \left. \frac{\partial F}{\partial \xi} \right|_{\tilde{\xi}, \dot{\tilde{\xi}}, \ddot{\tilde{\xi}}, \tilde{\nu}}, & U(t) &\equiv \left. \frac{\partial F}{\partial \nu} \right|_{\tilde{\xi}, \dot{\tilde{\xi}}, \ddot{\tilde{\xi}}, \tilde{\nu}}. \end{aligned} \quad (4.44)$$

The periodically linearized equation of motion can now be written as

$$M(t)\ddot{\hat{\xi}} + D(t)\dot{\hat{\xi}} + K(t)\hat{\xi} + U(t)\hat{\nu} = 0. \quad (4.45)$$

From a rigorous mathematical point of view, we have performed the periodic linearization of an implicit system of ordinary, second order, differential equations.

The linearized system has been integrated, and the solution superimposed to the nonlinear one in figure 4.3. The reader can notice the goodness of the result, that confirms the correctness of the linearization procedure and the validity of the stability analysis that will be performed in the following.

4.5 Individual pitch control

We decided to control the analytical wind turbine model with a kind of control algorithm called individual pitch control (IPC). For an extended explanation of this type of control see [13, 14, 18]. During the mathematical passages that will follow we will often omit the dependence from the time to don't weigh down the notation.

The individual pitch control algorithm can be summarized as follows.

1. Get the three blade root moments in the out of plane direction, expressed in the rotating reference frame.
2. Transform them in the non-rotating frame through the inverse Coleman's transformation matrix

$$\begin{pmatrix} L_d \\ L_q \end{pmatrix} = \frac{2}{3} \begin{bmatrix} \cos \psi_1 & \cos \psi_2 & \cos \psi_3 \\ \sin \psi_1 & \sin \psi_2 & \sin \psi_3 \end{bmatrix} \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix}. \quad (4.46)$$

3. Apply to these loads two standard decoupled proportional–integral (PI) controller, with the reference set to the null value, and obtain two pitches in the non-rotating frame

$$\begin{pmatrix} \theta_d \\ \theta_q \end{pmatrix} = \begin{bmatrix} K_P & 0 & K_I & 0 \\ 0 & K_P & 0 & K_I \end{bmatrix} \begin{pmatrix} L_d \\ L_q \\ \int L_d dt \\ \int L_q dt \end{pmatrix}. \quad (4.47)$$

4. Transform the two pitches in the rotating reference frame through the Coleman's transformation matrix

$$\begin{pmatrix} \theta_1^{\text{IPC}} \\ \theta_2^{\text{IPC}} \\ \theta_3^{\text{IPC}} \end{pmatrix} = \begin{bmatrix} \cos \psi_1 & \sin \psi_1 \\ \cos \psi_2 & \sin \psi_2 \\ \cos \psi_3 & \sin \psi_3 \end{bmatrix} \begin{pmatrix} \theta_d \\ \theta_q \end{pmatrix}. \quad (4.48)$$

5. Add them to the collective pitch

$$\theta_{p_k} = \theta_p^C + \theta_k^{\text{IPC}}, \quad \text{for } k = 1, 2, 3. \quad (4.49)$$

4.5.1 Application to the nonlinear model

In our analytical model the three flap moments are obtained by multiplying the three flap angles by the stiffness of the flap spring

$$\begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} = K_\beta \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}, \quad (4.50)$$

therefore the transformation operated in (4.46) is straightforward. The application of the PI controller pose a problem because in (4.47) appear the integral moments $\int L_d dt$ and $\int L_q dt$. This difficulty has been overcome by adding two other equations to the system, namely

$$\frac{d}{dt} \begin{pmatrix} \int \beta_d dt \\ \int \beta_q dt \end{pmatrix} = \mathcal{C}^{-1} \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}, \quad (4.51)$$

where $\mathcal{C}^{-1}$ is the inverse of the Coleman's matrix, already appeared in (4.46),

$$\mathcal{C}^{-1} \equiv \begin{bmatrix} \cos \psi_1 & \cos \psi_2 & \cos \psi_3 \\ \sin \psi_1 & \sin \psi_2 & \sin \psi_3 \end{bmatrix}. \quad (4.52)$$

The linearized system has been integrated under the same conditions of before, with the IPC's gains set to

$$K_P = -2 \times 10^{-8}, \quad K_I = -5 \times 10^{-9}.$$

The result of the numerical integration can be seen in figure 4.5, which makes it clear that the IPC caused a 100% reduction of the 1P out of plane moment, while leaving the other loads almost unaffected.

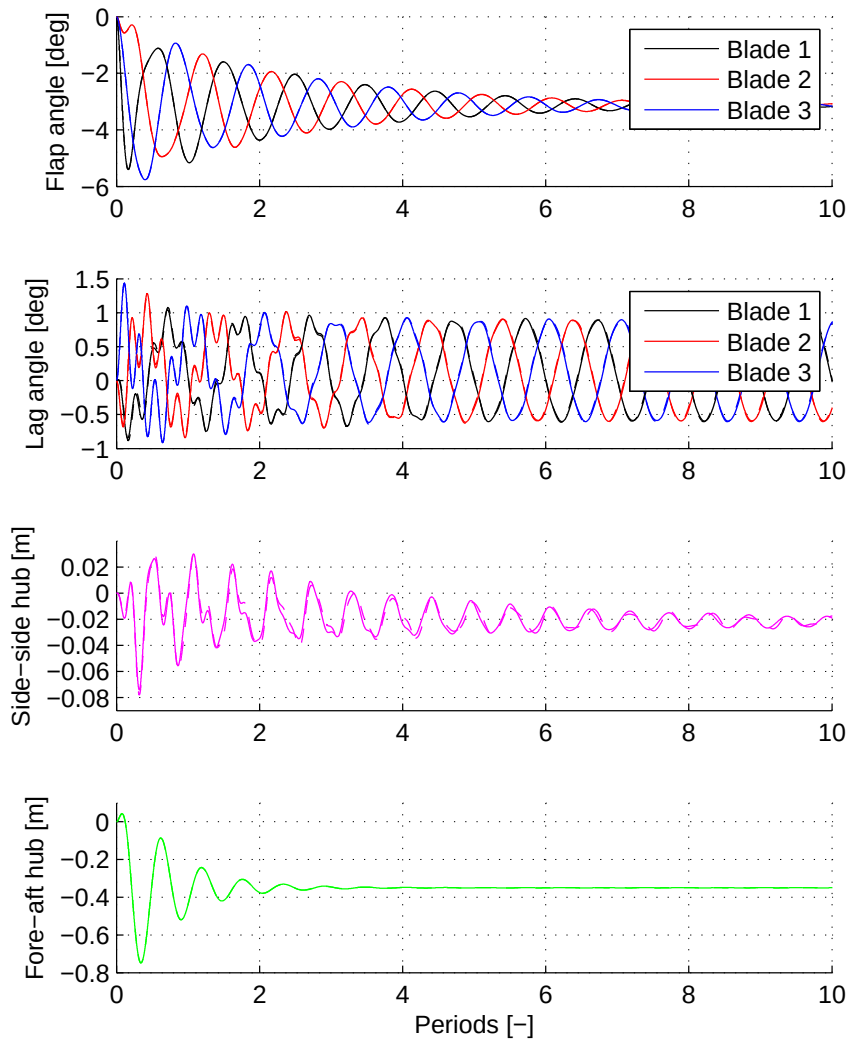


Figure 4.5: Comparison between the nonlinear (in continuous line) and the linearized (in dashed line) solutions for the controlled wind turbine model. 50 periods were simulated.

4.5.2 Application to the linearized model

There are two ways to derive the linearized equations of motion of the controlled system:

1. directly linearize the controlled nonlinear system;
2. linearize the uncontrolled nonlinear system, and then build the IPC over it.

The two ways are not equivalent because of the different periodic trajectory around which the linearizations are performed. In particular the first way present a good steady state response, while the second present the best performance in the transitory regime. These reasons convinced us to employ the second way for the stability evaluation.

In order to apply the IPC to the linearized system (4.45) we have to write it in state-space form. By using the standard formulas we write

$$\dot{\mathbf{x}}_{\text{dyn}} = \mathbf{A}_{\text{dyn}}(t)\mathbf{x}_{\text{dyn}} + \mathbf{B}_{\text{dyn}}(t)\mathbf{u}(t), \quad (4.53)$$

in which the state and the input² vectors are

$$\mathbf{x}_{\text{dyn}} \equiv \begin{pmatrix} \hat{\xi} \\ \dot{\xi} \end{pmatrix}, \quad \mathbf{u}(t) = \hat{\mathbf{v}} = \begin{pmatrix} \theta_1^{\text{IPC}} \\ \theta_2^{\text{IPC}} \\ \theta_3^{\text{IPC}} \end{pmatrix}, \quad (4.54)$$

and the matrices are

$$\mathbf{A}_{\text{dyn}}(t) \equiv \begin{bmatrix} \mathbf{0}_{8 \times 8} & \mathbf{I}_8 \\ -\mathbf{M}^{-1}\mathbf{K} & -\mathbf{M}^{-1}\mathbf{D} \end{bmatrix}, \quad \mathbf{B}_{\text{dyn}}(t) \equiv \begin{bmatrix} \mathbf{0}_{8 \times 3} \\ -\mathbf{M}^{-1}\mathbf{U} \end{bmatrix}. \quad (4.55)$$

Similarly to what has been done for controlling the nonlinear system we add the two integral states. This operations is performed by hemming $\mathbf{A}_{\text{dyn}}$ with the inverse of the Coleman's matrix, and hemming $\mathbf{B}_{\text{dyn}}$ accordingly

$$\mathbf{A}(t) = \begin{bmatrix} \mathbf{A}_{\text{dyn}} & \mathbf{0}_{16 \times 2} \\ \mathbf{C}^{-1} & \mathbf{0}_{2 \times 13} & \mathbf{0}_{2 \times 2} \end{bmatrix}, \quad \mathbf{B}(t) = \begin{bmatrix} \mathbf{B}_{\text{dyn}} \\ \mathbf{0}_{2 \times 3} \end{bmatrix}. \quad (4.56)$$

The state vector is now

$$\mathbf{x} = (\beta_1, \beta_2, \beta_3, \zeta_1, \zeta_2, \zeta_3, y_H, z_H, \dot{\beta}_1, \dot{\beta}_2, \dot{\beta}_3, \dot{\zeta}_1, \dot{\zeta}_2, \dot{\zeta}_3, \dot{y}_H, \dot{z}_H, \int \beta_d dt, \int \beta_q dt)^T. \quad (4.57)$$

²It would have been an error to include the collective pitch in the input of the linearized system, since it is already considered in the periodic response.

The pitches in the direct and quadrature axis are obtained through the PI controller

$$\begin{pmatrix} \theta_d \\ \theta_q \end{pmatrix} = \begin{bmatrix} K_P & 0 & K_I & 0 \\ 0 & K_P & 0 & K_I \end{bmatrix} \begin{pmatrix} L_d \\ L_q \\ \int L_d dt \\ \int L_q dt \end{pmatrix} = \mathbf{G}K_\beta \begin{pmatrix} \beta_d \\ \beta_q \\ \int \beta_d dt \\ \int \beta_q dt \end{pmatrix}, \quad (4.58)$$

where $\mathbf{G}$ is the gain matrix. Then we substitute the three flap angles, so that

$$\begin{pmatrix} \theta_d \\ \theta_q \end{pmatrix} = \mathbf{G}K_\beta \text{blkdiag}(\mathbf{C}^{-1}, \mathbf{I}_2) \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \\ \int \beta_d dt \\ \int \beta_q dt \end{pmatrix}. \quad (4.59)$$

The three pitches are obtained by premultiplying the previous by the Coleman matrix, and after inserting the state vector we obtain

$$\begin{pmatrix} \theta_1^{\text{IPC}} \\ \theta_2^{\text{IPC}} \\ \theta_3^{\text{IPC}} \end{pmatrix} = \mathbf{C}\mathbf{G}K_\beta \text{blkdiag}(\mathbf{C}^{-1}, \mathbf{I}_2)\mathbf{E}(\mathbf{x} + \tilde{\mathbf{x}}), \quad (4.60)$$

in which $\mathbf{E}$ is the “state extraction” matrix

$$\mathbf{E} \equiv \begin{bmatrix} \mathbf{I}_3 & \mathbf{0}_{3 \times 13} & \mathbf{0}_{3 \times 2} \\ \mathbf{0}_{2 \times 3} & \mathbf{0}_{2 \times 13} & \mathbf{I}_2 \end{bmatrix}. \quad (4.61)$$

A note must be made about equation (4.60). The load that enters the IPC must be the complete one, and not that of the linearized solution, in another words the load that enters the IPC must the approximation of the vector $\boldsymbol{\xi}(t)$. For this reason we added to $\mathbf{x}(t)$ the periodic state vector $\tilde{\mathbf{x}}(t)$, however since the inclusion of the integral states $\tilde{\mathbf{x}}(t)$ is no more periodic in all of its components. To better understand this point the reader should look at figure 4.6, which shows the integral states for $\mathbf{x}(t)$, $\tilde{\mathbf{x}}(t)$ and $\mathbf{x}(t) + \tilde{\mathbf{x}}(t)$, the latter being close to the nonlinear integral states $\boldsymbol{\xi}(t)$. We can see that the linearized and the “periodic” solutions diverge, but their sum has the shape of a sinusoid having a 3P frequency.

The matrix product in (4.60) can be thought as a time dependent (1P periodic) gain matrix

$$\mathbf{K}(t) \equiv \mathbf{C}\mathbf{G}K_\beta \text{blkdiag}(\mathbf{C}^{-1}, \mathbf{I}_2)\mathbf{E}. \quad (4.62)$$

We can now write the linearized equations of motion of the controlled system

$$\begin{aligned} \dot{\mathbf{x}} &= \mathbf{A}\mathbf{x} + \mathbf{B}u \\ &= \mathbf{A}\mathbf{x} + \mathbf{B}\mathbf{K}(\mathbf{x} + \tilde{\mathbf{x}}) \\ &= (\mathbf{A} + \mathbf{B}\mathbf{K})\mathbf{x} + \mathbf{B}\mathbf{K}\tilde{\mathbf{x}}. \end{aligned} \quad (4.63)$$

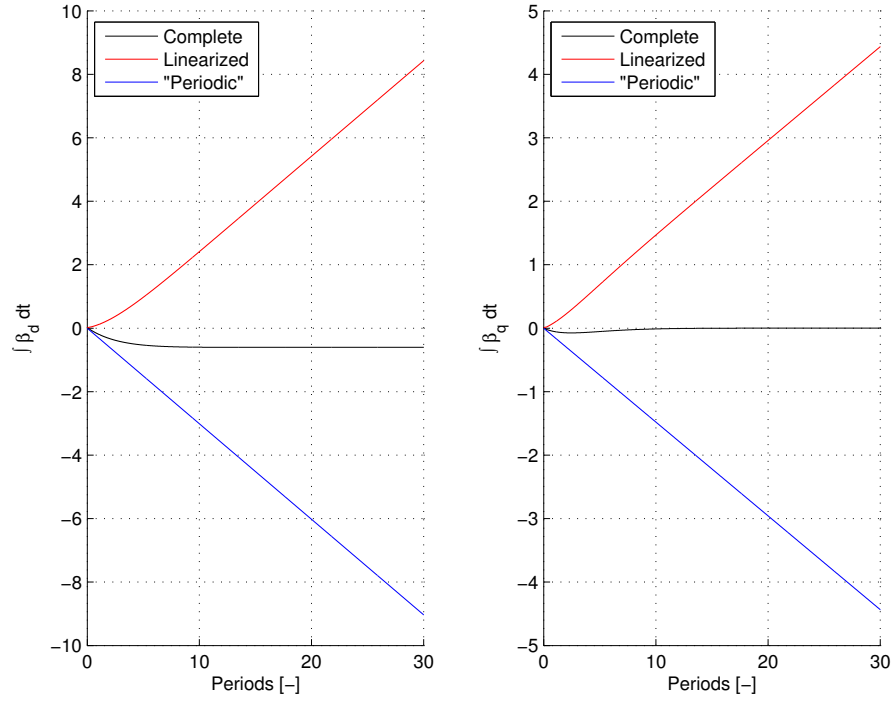


Figure 4.6: Time histories of $\int \beta_d dt$ and $\int \beta_q dt$.

The linearized controlled system is written as

$$\dot{\mathbf{x}} = \mathbf{A}_C \mathbf{x} + \mathbf{B}_C \mathbf{u}, \quad (4.64)$$

where the input $\mathbf{u}$ is the periodic state $\tilde{\mathbf{x}}$, and the initial conditions are

$$\mathbf{x}(0) = \begin{pmatrix} \boldsymbol{\xi}(0) - \tilde{\boldsymbol{\xi}}(0) \\ \dot{\boldsymbol{\xi}}(0) - \dot{\tilde{\boldsymbol{\xi}}}(0) \\ 0 \\ 0 \end{pmatrix}. \quad (4.65)$$

The result of the numerical integration is shown in figure 4.5, where the reader can notice again the agreement with the nonlinear solution.

4.6 Coleman transformation

To assess the validity of the Coleman's transformation we have converted the degrees of freedom in the multiblade coordinates, that is to say in the average, direct and quadrature axis.

In our case the state vector (4.57) contains both the degrees of freedom and their derivatives, and we chose to rewrite it in an abbreviate form

$$\mathbf{x} = \begin{pmatrix} \beta \\ \zeta \\ \mathbf{s}_H \\ \dot{\beta} \\ \dot{\zeta} \\ \dot{\mathbf{s}}_H \\ \dot{\mathbf{i}}_{dq} \end{pmatrix}. \quad (4.66)$$

To write the transformation matrix it is useful to define the Coleman's matrix ³

$$\mathcal{C}(t) \equiv \begin{bmatrix} 1 & \cos \psi_1 & \sin \psi_1 \\ 1 & \cos \psi_2 & \sin \psi_2 \\ 1 & \cos \psi_3 & \sin \psi_3 \end{bmatrix}, \quad (4.67)$$

whose derivative is given by

$$\dot{\mathcal{C}}(t) = \Omega \begin{bmatrix} 0 & -\sin \psi_1 & \cos \psi_1 \\ 0 & -\sin \psi_2 & \cos \psi_2 \\ 0 & -\sin \psi_3 & \cos \psi_3 \end{bmatrix}. \quad (4.68)$$

In this way the transformation matrix employed in the transformation (3.57) becomes

$$\mathbf{T} = \begin{bmatrix} \mathcal{C} & & & & & \\ & \mathcal{C} & & & & \\ & & I_2 & & & \\ \dot{\mathcal{C}} & & & \mathcal{C} & & \\ & \dot{\mathcal{C}} & & & \mathcal{C} & \\ & & & & & I_2 \\ & & & & & & I_2 \end{bmatrix}. \quad (4.69)$$

The last step for the Coleman's transformation of the system is to take the average over the period of the transformed state transition matrix, so as to remove any residual periodicity.

$$\bar{\mathbf{A}}_{\mathcal{C}} = \frac{1}{T} \int_0^T \mathbf{T}^{-1} \mathbf{A}_C \mathbf{T} - \mathbf{T}^{-1} \dot{\mathbf{T}} dt. \quad (4.70)$$

³The definition (4.67) is not coherent with that of $\mathcal{C}^{-1}$, due to the first column. We inserted that column to recover the average values of the degrees of freedom, which is important here, while it was unnecessary (and conceptually wrong) for the IPC.

4.7 The Modal Assurance Criterion

The design of the Campbell diagram requires to solve a large number of eigenvalue problems. It is well known that the eigenvalue solvers doesn't return the eigenvalues always in the same order, therefore there is the need of a tool that is able to reorder them. A widespread tool used to accomplish this task is the Modal Assurance Criterion (MAC). The MAC has been developed in the early eighties by R. J. Allemang, and it is based on the concept of following the evolution of the eigenvectors, instead of the eigenvalues or the frequencies; for an updated review see [1].

Here we will employ the standard MAC algorithm to follow the evolution of the eigenvalues of the $\bar{\mathbf{A}}_C$ matrix, with only a little modification aimed to increase its robustness. Subsequently we will extend the MAC to the periodic systems.

4.7.1 The linear time invariant case

In the LTI case the eigenvalues $\bar{\lambda}$ and eigenvectors $\bar{\mathbf{V}}$ of the $\bar{\mathbf{A}}_C$ dynamic matrix are computed for each value of the parameter. The modal assurance criterion is used to compare two set a and b of eigenvalues and eigenvectors.

The first step is to reduce the list of possible matches by comparing the frequencies of two eigenvalues i and j belonging to the two set. Since two complex conjugate modes share the same frequency we chose to compute the frequency of the difference of eigenvalues:

$$\Delta\bar{\lambda}_{i,j} = \bar{\lambda}_{b_j} - \bar{\lambda}_{a_i}, \quad (4.71a)$$

$$\Delta f_{i,j} = \sqrt{\text{Re}(\Delta\bar{\lambda}_{i,j})^2 + \text{Im}(\Delta\bar{\lambda}_{i,j})^2}. \quad (4.71b)$$

The MAC value is evaluated only over the eigenvectors i and j associated to a $\Delta f_{i,j}$ below a specified threshold. The MAC value can be computed in many equivalent ways, and we prefer the following

$$MAC_{i,j} \equiv \frac{\mathbf{v}_{a_i}^H \mathbf{v}_{b_j} \cdot \mathbf{v}_{b_j}^H \mathbf{v}_{a_i}}{\mathbf{v}_{a_i}^H \mathbf{v}_{a_i} \cdot \mathbf{v}_{b_j}^H \mathbf{v}_{b_j}}, \quad (4.72)$$

where $(*)^H$ denotes the complex conjugate transpose (i.e. Hermitian) operator, and $\mathbf{v}_{a_i}$ and $\mathbf{v}_{b_j}$ are columns of the respective eigenvectors matrices $\bar{\mathbf{V}}_a$ and $\bar{\mathbf{V}}_b$. When the MAC value is near 1 the two eigenvectors are consistent, and so it is likely that the eigenvalues $\bar{\lambda}_{a_i}$ and $\bar{\lambda}_{b_j}$ are associated to the same mode.

4.7.2 The linear time periodic case

In the LTP case the algorithm explained before is still valid as long as the matrix $\bar{A}_C$ is exchanged with the monodromy matrix Ψ . Similarly the eigenvalues $\bar{\lambda}$ became the characteristic multipliers θ , and the eigenvectors matrix $\bar{V}$ is that associated to the monodromy matrix S .

There is only a little aspect to be taken into account. The MAC is employed to follow the characteristic exponents, which are usually sorted from the highest super-harmonic to the lowest, in such a way that the 0-harmonic (associated to the highest participation) lies in the middle. Sometimes a participation exchange happens between the participation of what was the 0-harmonic and one of the associated super-harmonics. To avoid jumps from the 0-harmonic to one of the super-harmonics it is sufficient to do a frequency check inside the mode.

4.8 Comparison between the Floquet theory and the Coleman approximation

The comparison between the Floquet theory and the Coleman approximation has been done through the periodic Campbell diagram. We have drawn the Campbell diagrams in various operating conditions, listed hereafter.

1. Rotor operating in vacuo.
2. Rotor subject to axial wind, without the control system.
3. Rotor subject to axial wind, with the control system.
4. Rotor subject to cross flow, without the control system.
5. Rotor subject to cross flow, with the control system.
6. Rotor subject to different wind shears, without the control system.
7. Rotor subject to different wind shears, with the control system.

Our model gives some flexibility in deciding the conditions in which the machine will work. We chose to draw the Campbell diagram from the minimum angular speed planned for the machine till 1.3 of the rated value. The wind has a direct proportionality relationship with Ω , starts from 3 m/s and rises at a constant slope of 0.7836 m/s/rpm. The collective pitch has been fixed to -0.5268° .

In the next paragraphs we will report standard Campbell diagrams obtained with the Floquet theory, displaying for each mode only the principal harmonic. Every mode will then be studied separately by means of a periodic Campbell

diagram, which will show the 0Ω harmonic and two super-harmonics on each side. In the periodic Campbell diagrams the results obtained with Floquet theory will be plotted in continuous line, while those obtained with the Coleman approximation in dashed line.

4.8.1 Rotor operating in vacuo

When the wind turbine is operating in vacuo only the inertial effects are present, and the individual pitch controller hasn't any effect.

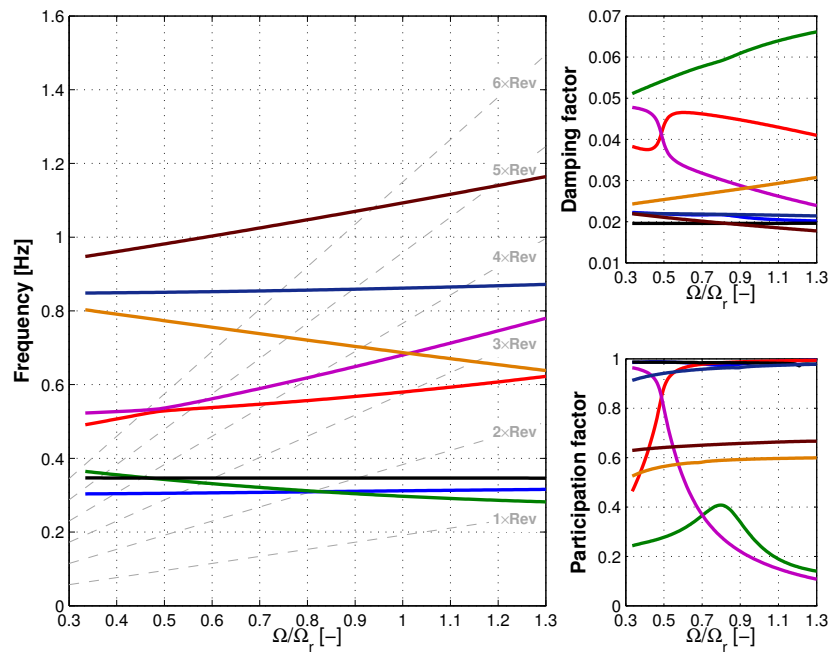


Figure 4.7: Standard Campbell diagram of the rotor operating on vacuo.

In figure 4.7 we can notice a frequency exchange between the forward flapwise whirling mode and the symmetric flapwise, moreover the symmetric flapwise is not centered between the whirling modes. This second behavior finds confirmation in [27], where the author examined a multi-body model of a 600 kW wind turbine, observing also that the natural frequencies and the mode shapes of a 5MW wind turbine are quite close to those of the wind turbine under study. Likewise the symmetric edgewise mode is not centered between the whirling modes.

From the damping factor box of figure 4.7 we can see that the absence of air has caused little dampings for all modes.

A variety of observation can be inferred from the periodic Campbell diagrams of figures 4.8 and 4.9.

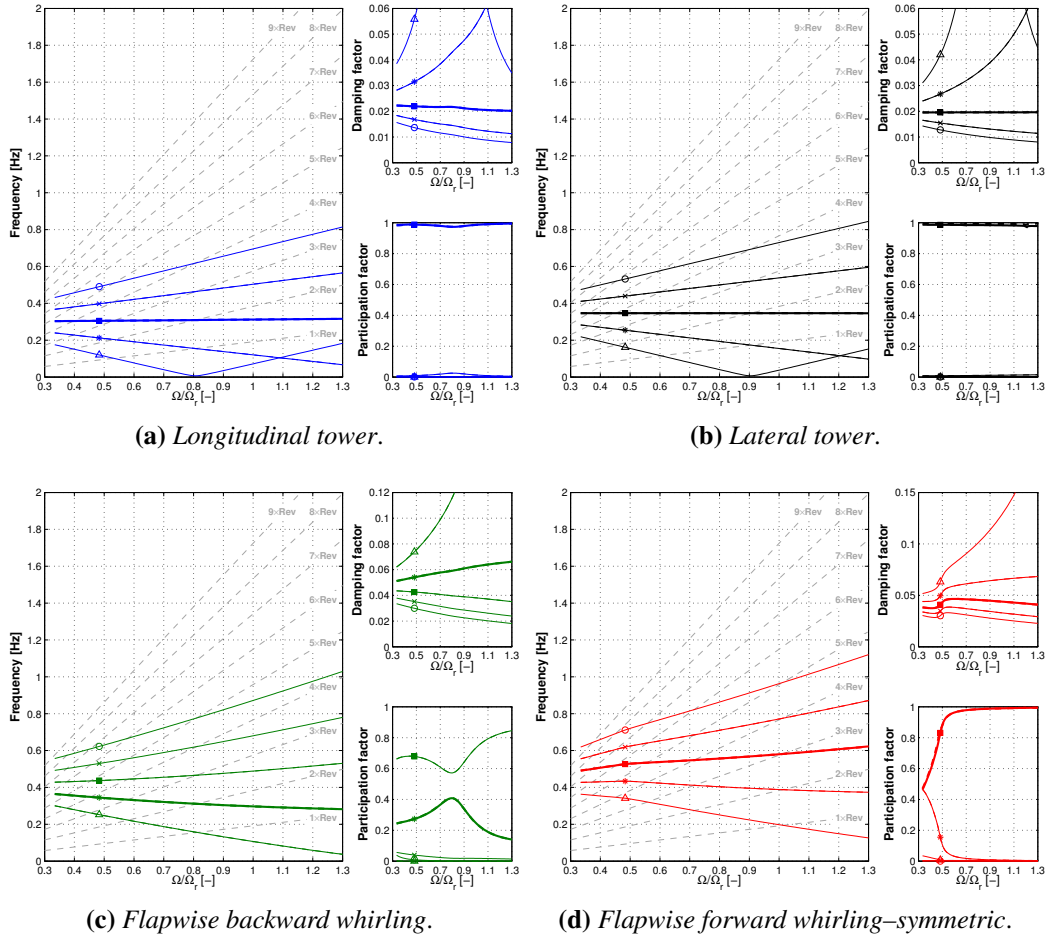


Figure 4.8: Periodic Campbell diagrams of the rotor operating on vacuo.

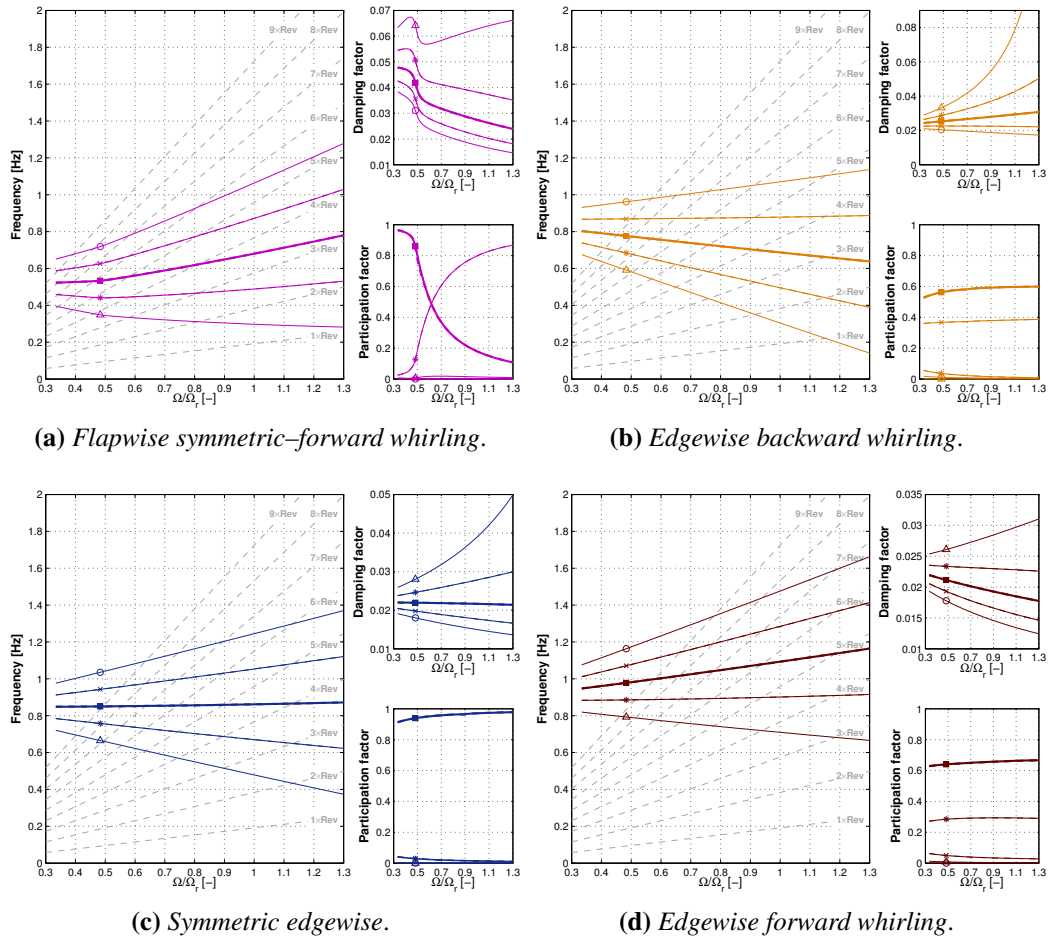


Figure 4.9: Periodic Campbell diagrams of the rotor operating on vacuo (continued).

- In figures 4.8a and 4.8b the participation associated to the principal harmonic is close to 1, therefore the two tower modes behave as time invariant.
- In the flapwise backward whirling mode, figure 4.8c, the principal harmonic isn't associated to the highest participation. Despite this the Coleman approximation is still able to see the $+1\Omega$ super-harmonic, that is associated to the highest peak in the FFT response of the system.
- The flapwise backward whirling mode and the longitudinal tower mode intersects at $\Omega/\Omega_r \simeq 0.8$ (see figure 4.7), and this reflects in their participations.
- In the symmetric to forward whirling flapwise mode, figure 4.9a, the participation of the principal harmonic drops at $\Omega/\Omega_r \simeq 0.5$, and the -1Ω super-harmonic becomes the principal one.
- The symmetric edgewise mode, figure 4.9c, is almost non-periodic.

For the backward whirling modes the $+1\Omega$ super-harmonic is the most prominent after the principal one, while for the forward whirling modes the -1Ω super-harmonic is the most prominent. Moreover we can see that all the whirling modes have a high periodic content, hence their behavior cannot be adequately described by a single frequency [16].

In all cases the Coleman approximation has shown to give extremely good results.

4.8.2 Rotor subject to axial wind, without the IPC

Here the wind turbine is subject to an axial wind having a velocity that varies with the angular speed (as explained in the beginning of this section). The wind simulates the atmospheric boundary layer through a wind shear having a power law exponent set to 0.2. Since the individual pitch controller isn't active the pitch is fixed to the collective value.

By looking at figure 4.10 we can see that the frequency exchange between the symmetric flapwise mode and the forward whirling has disappeared.

From the damping factor box in figure 4.10 we can see that the lateral tower mode, and the edgewise modes, are characterized by a little damping, while an elevated damping level characterize the flapwise modes and the longitudinal tower. This fact is due to the aerodynamic model that we chose to use. The aerodynamic model developed in [23] neglects the aerodynamic drag, hence the only damping associated to the edgewise and lateral tower modes is the small structural one.

From figures 4.11a and 4.11b we can see that the two tower modes are still time invariant.

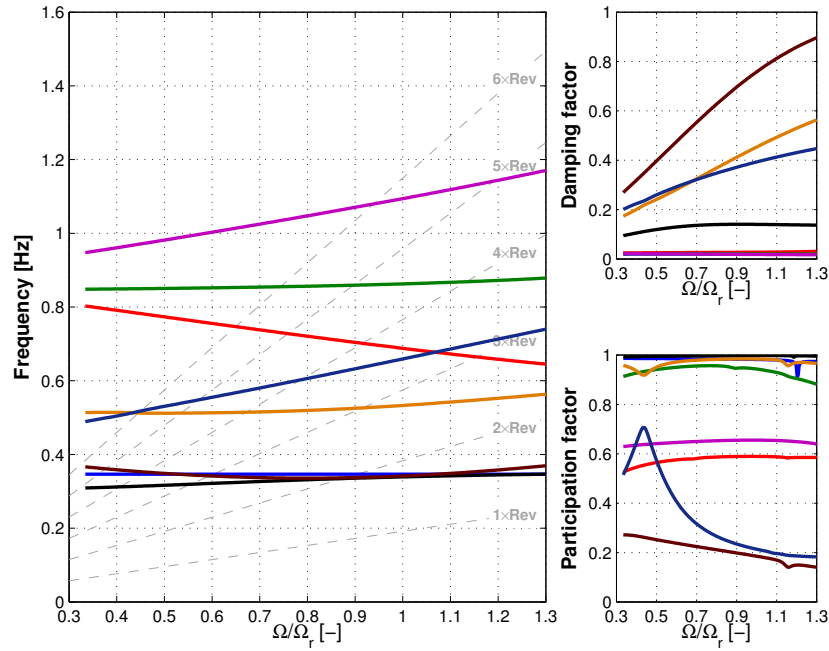


Figure 4.10: Standard Campbell diagram of the rotor subject to axial wind, without the IPC.

Figure 4.11c allow us to see a shortcoming of using the Coleman approximation. This is a case in which the Coleman approximation has seen only the 0Ω , -1Ω and -2Ω harmonics, hence ignoring the $+1\Omega$ and $+2\Omega$. In this case, in the over-speed regime, the two unseen harmonics arrive at a 25% of participation.

In figure 4.12d we can recognize that the Coleman approximation have made a little error in estimating the damping in the overspeed condition, however since it predicted a lower damping level the result is conservative.

Sometimes the participations obtained with the Coleman approximation exceeds the value 1. This is because they have been scaled with respect to the sum of the participations obtained with Floquet theory.

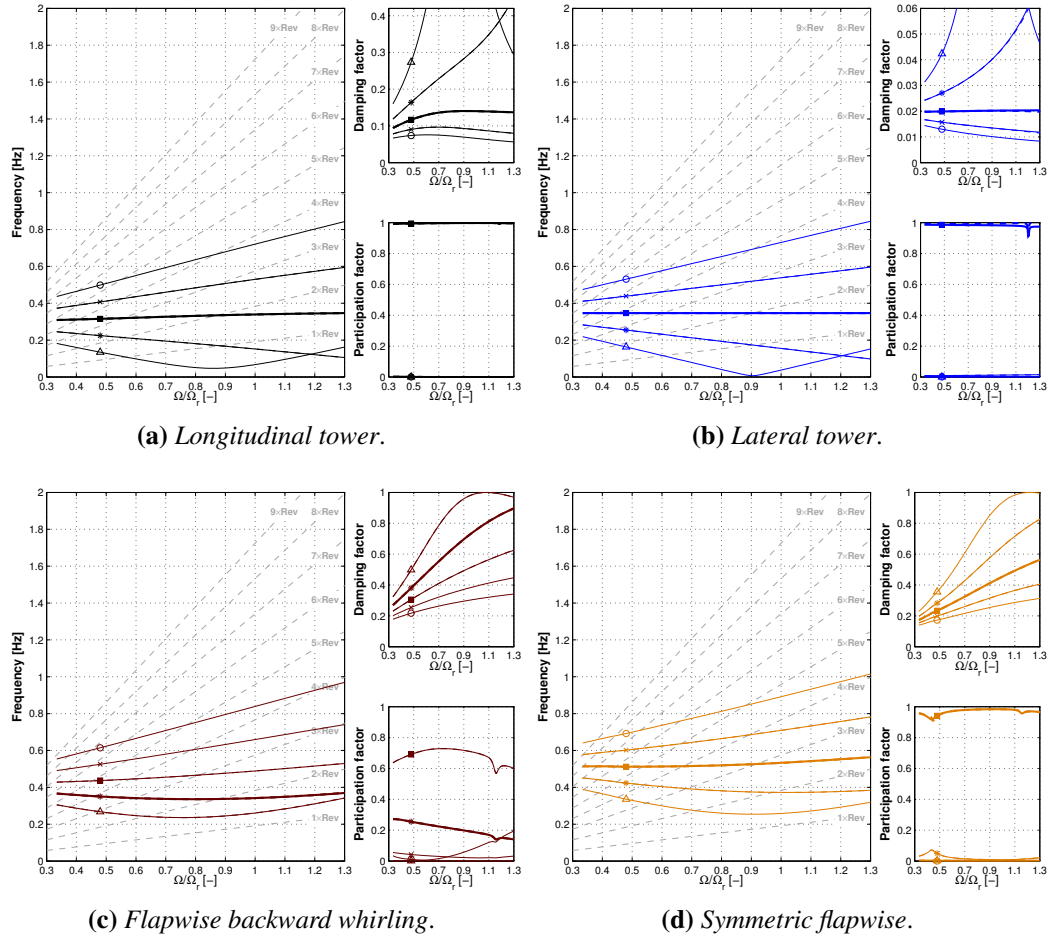


Figure 4.11: Periodic Campbell diagrams of the rotor subject to axial wind, without the IPC.

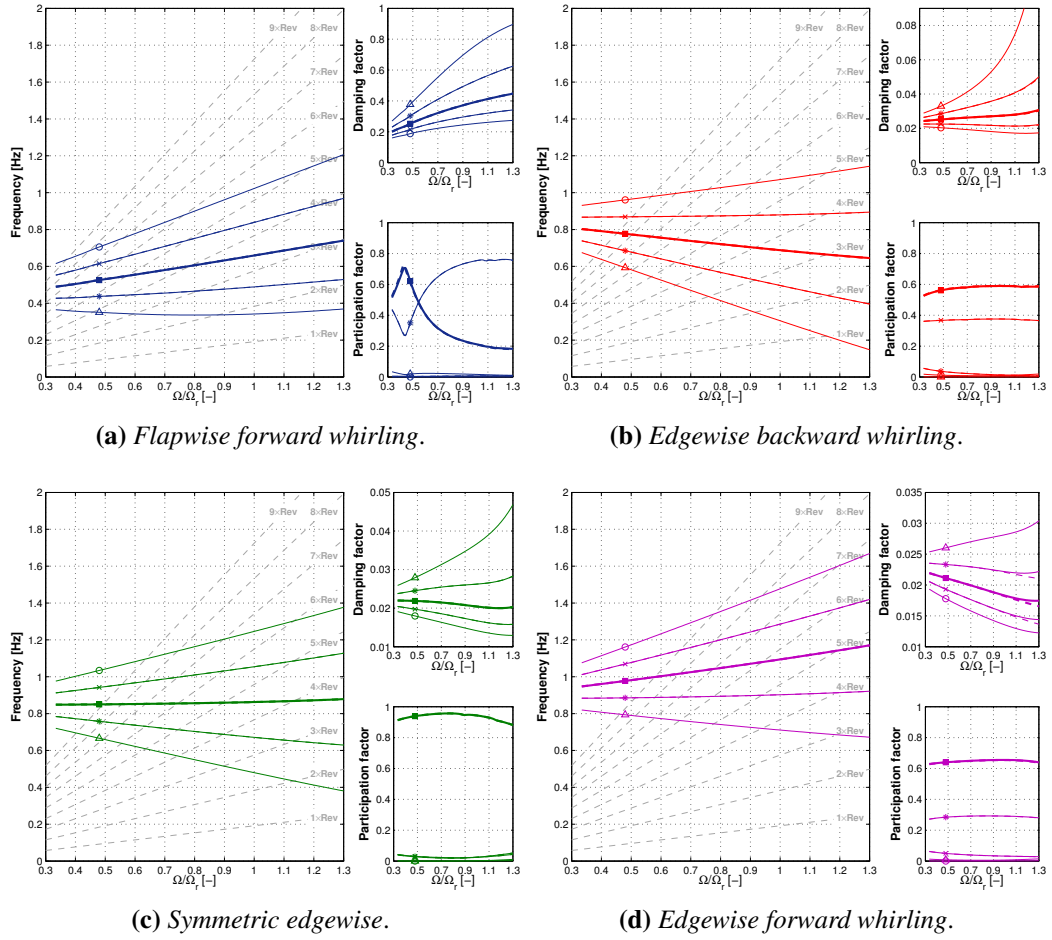


Figure 4.12: Periodic Campbell diagrams of the rotor subject to axial wind, without the IPC (continued).

4.8.3 Rotor subject to axial wind, with the IPC

We analyzed the behavior of the wind turbine in the same scenario of the previous paragraph, but this time the IPC gains were set to $K_P = -2 \times 10^{-8}$ and $K_I = -5 \times 10^{-9}$, i.e. the same values used to obtain figure 4.5.

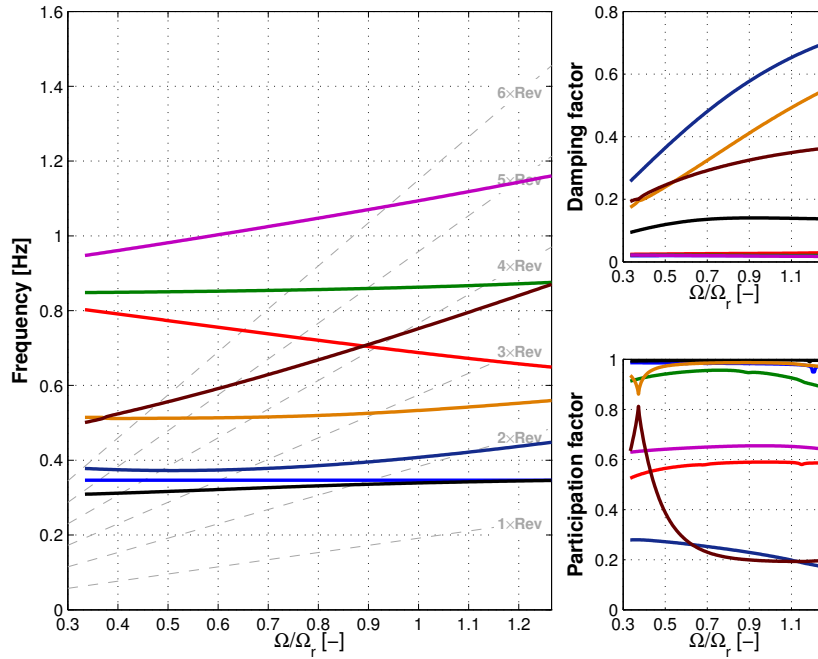


Figure 4.13: Standard Campbell diagram of the rotor subject to axial wind, with the IPC.

The effect of the individual pitch controller can be better understood by displaying side by side the frequency box obtained without and with the IPC. From figure 4.14 we can see that the IPC has caused the frequencies branch characterizing the flapwise backward and forward whirling modes to rise, while leaving all the others almost untouched. In particular the symmetric flapwise mode has not been affected by the IPC.

Next the periodic Campbell diagrams of all modes are shown.

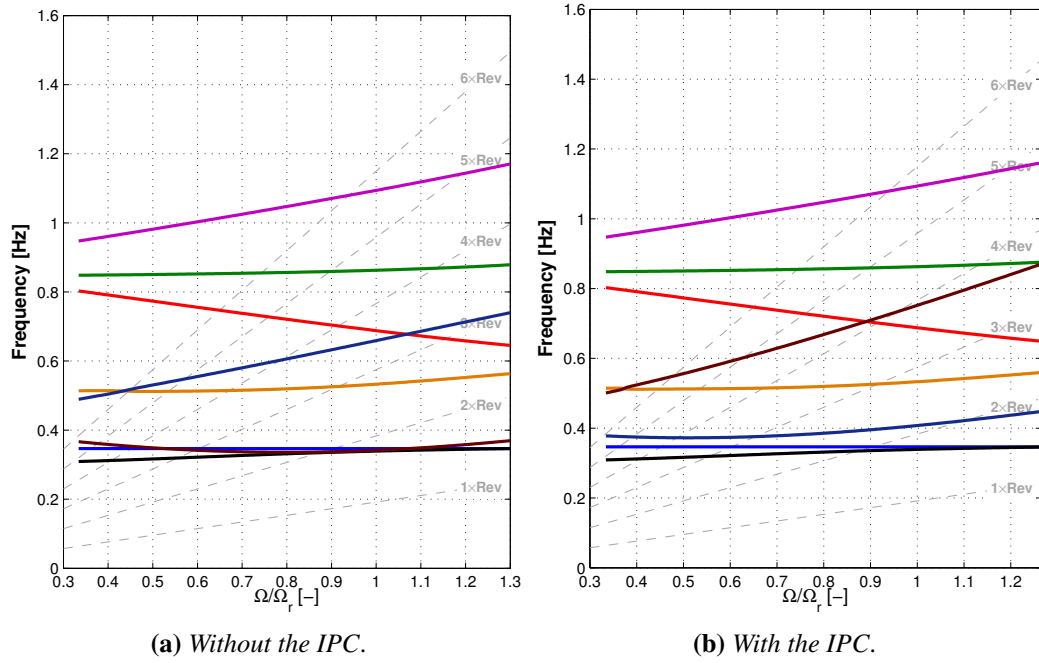


Figure 4.14: Effect of the individual pitch controller over the frequencies.

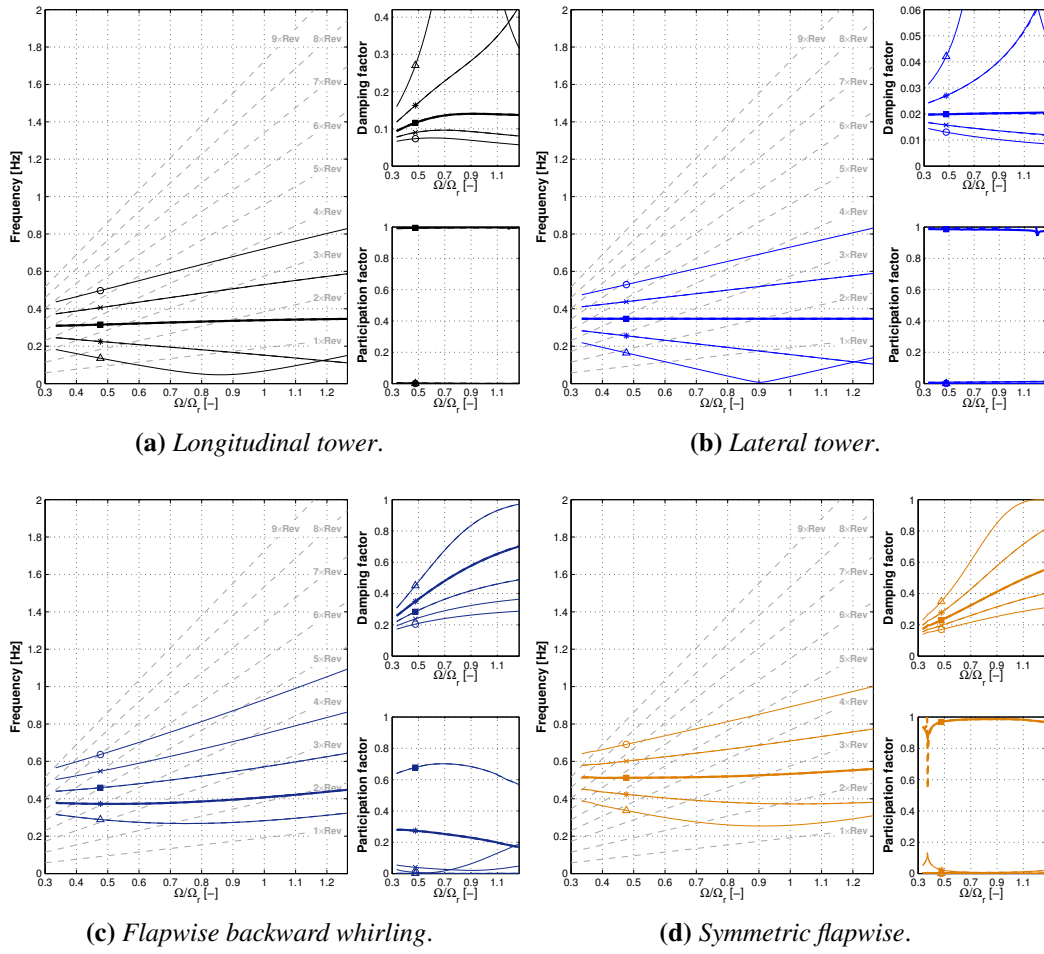


Figure 4.15: Periodic Campbell diagrams of the rotor subject to axial wind, with the IPC.

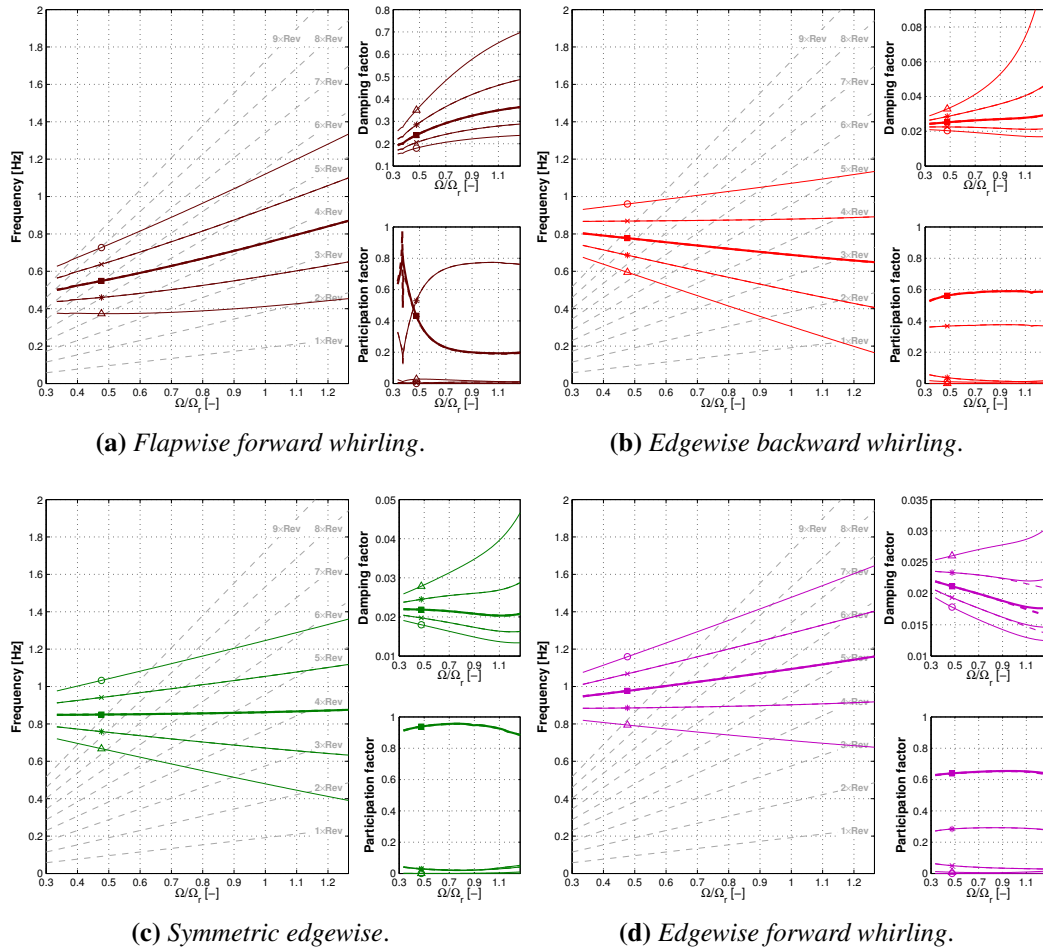


Figure 4.16: Periodic Campbell diagrams of the rotor subject to axial wind, with the IPC (continued).

4.8.4 Rotor in cross-flow conditions

To analyze the stability of the rotor in horizontal cross-flow conditions we selected a wind having a magnitude of 10 m/s, and a power law exponent of 0.2. Then we rotated it from -60° to $+60^\circ$, with respect to the shaft. The rotor speed has been set to the rated value.

The complete Campbell diagram without and with the IPC is reported in figures 4.17 and 4.18. These figures show that the frequencies and dampings are not affected by the cross-flow angle.

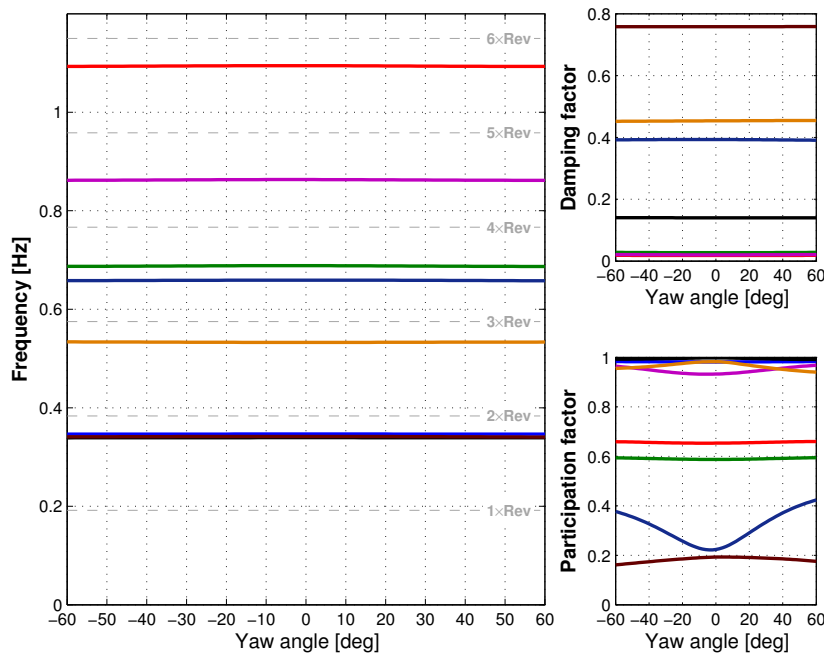


Figure 4.17: Standard Campbell diagram of the rotor subject to horizontal cross-flow, without the IPC.

We have drawn the periodic Campbell diagrams mode by mode also in this case, but for the sake of brevity we don't show them here. In all cases the Coleman approximation provided results perfectly similar to those of the Floquet theory, only sometimes we have observed errors lower than 2% on the dampings, and ignored participations till a 15%.

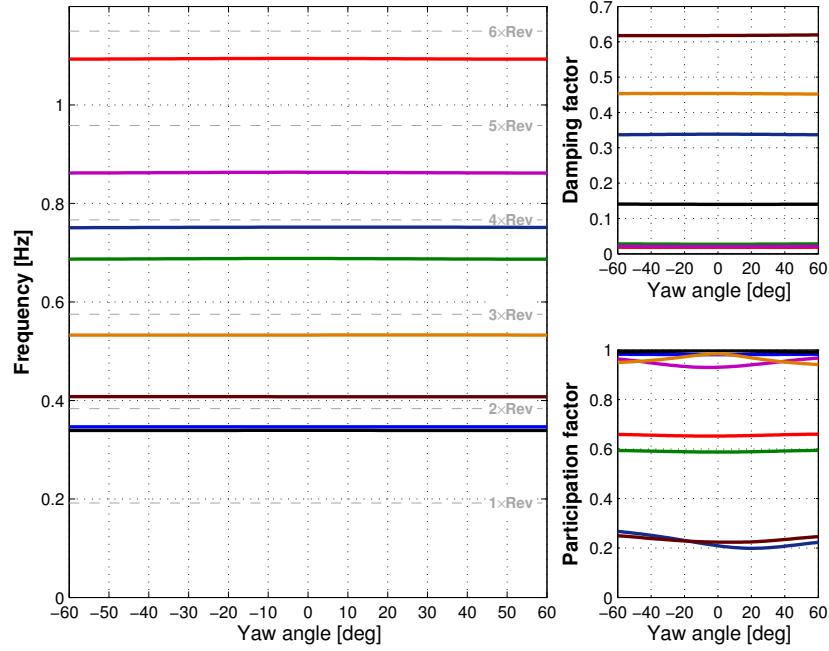


Figure 4.18: Standard Campbell diagram of the rotor subject to horizontal cross-flow, with the IPC.

4.8.5 Rotor in different wind shear conditions

A standard way of simulating the atmospheric boundary layer is through the so called power law, which has the following expression

$$V(h; \alpha) = V_{\text{ref}} \left(\frac{h}{h_{\text{ref}}} \right)^{\alpha}. \quad (4.73)$$

In the aerodynamic model of [23] the wind shear is modeled by using a linear relation, having a slope K_1 . By deriving equation (4.73) with respect to h and evaluating the result at the hub height we obtain the following expression for K_1 :

$$K_1 = \alpha \frac{V_{\text{hub}}}{h_{\text{hub}}}. \quad (4.74)$$

We have evaluated the stability of the wind turbine with the power law exponents α between 0 (no wind shear) and 0.6 (extremely high wind shear), the rotor being put in rotation at the rated rotor speed. The complete Campbell diagram without and with the IPC is reported in figures 4.20 and 4.21. From these figures we can see that, like from the cross-flow, the frequencies and dampings are almost non-dependent from the wind shear.

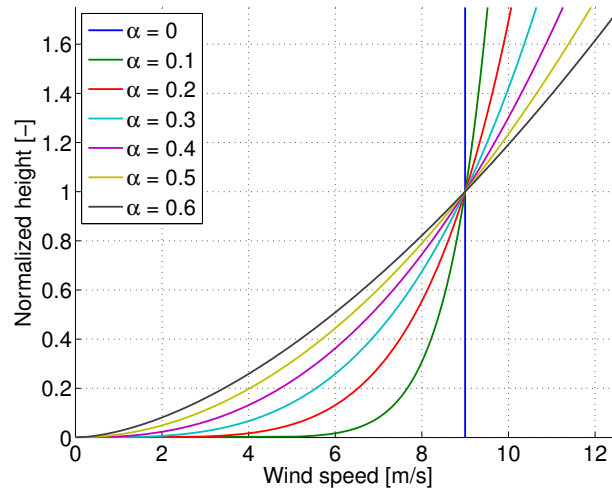


Figure 4.19: Wind velocity acting on the wind turbine obtained with the power law. The height is adimensionalized with respect to the hub height, and the wind speed at the hub height is 9 m/s.

For brevity we don't show the periodic Campbell diagram of each mode, however two of them are useful to confirm the observations made before. In figure 4.22 the Coleman approximation underestimated the damping at high wind shear by a 6%, while in figure 4.23 it has ignored participations till 30%.

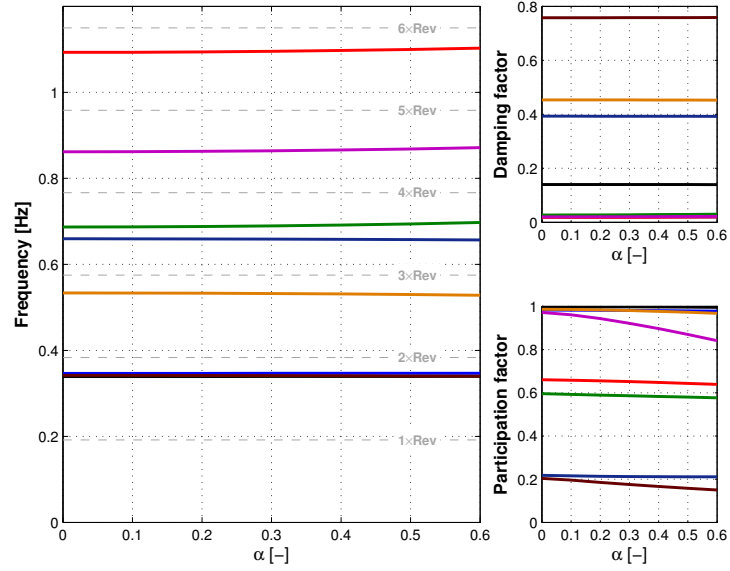


Figure 4.20: Standard Campbell diagram of the rotor subject to different vertical wind shears, without the IPC.

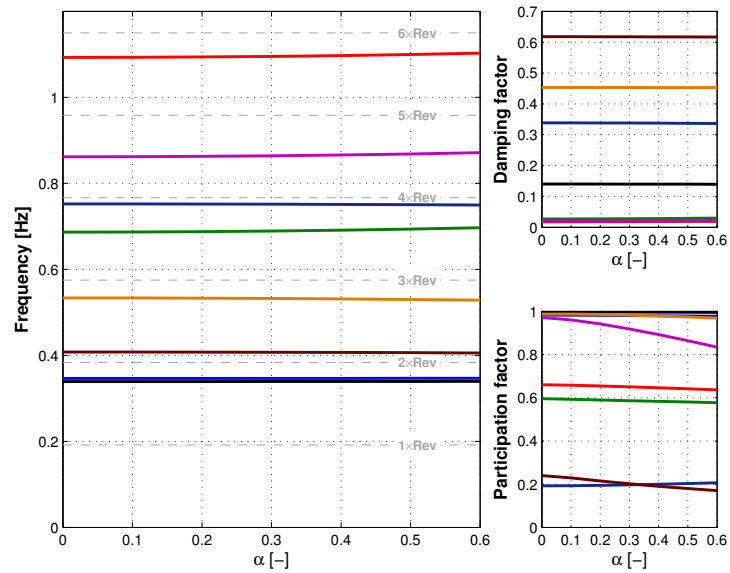


Figure 4.21: Standard Campbell diagram of the rotor subject to different vertical wind shears, with the IPC.

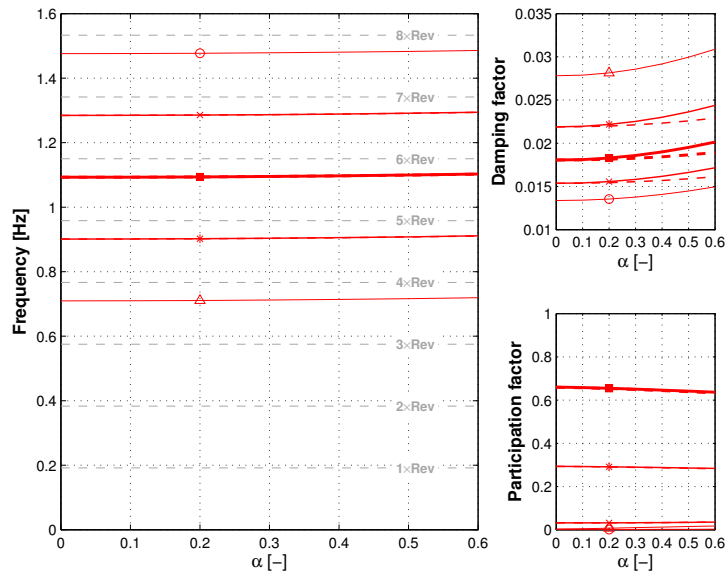


Figure 4.22: Periodic Campbell diagram of the edgewise forward whirling mode, with the IPC

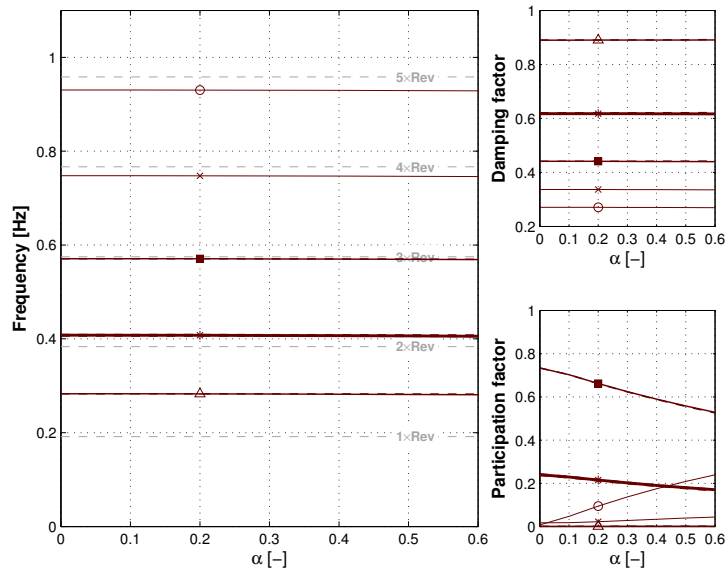


Figure 4.23: Periodic Campbell diagram of the flapwise backward whirling mode, with the IPC

4.9 Conclusions

The object of this chapter was to compare the stability analyses conducted with Floquet theory and the Coleman approximation. To accomplish this task we have constructed an analytical model of wind turbine, and we applied to it an individual pitch controller. It is well known that the Coleman transformation works perfectly, i.e. renders the system time invariant, only for isotropic rotors on isotropic supports, hence we decided to test its validity in anisotropic conditions. In our stability analyses we changed the pitch of the blades, the wind direction, and the vertical wind shear. From the results of the previous section we can draw the following conclusions.

- In all cases the Coleman approximation has given results that almost match the ones obtained with the Floquet theory. The errors caused by the use of the Coleman approximation are higher for the damping than for the frequency, and for the firsts don't exceed the 1% on the non-conservative side, and the 9% on the conservative side. These conclusions adds to the ones of [45], where the authors examined a multi-body models in highly anisotropic conditions, with the purpose of comparing the control strategies based on the Coleman transformation with those based on the Floquet theory.
- The only drawback in the use of the Coleman approximation is its inability to see more than three harmonics. Usually this is not a problem because the participations of the unseen super-harmonics are negligible, but in some cases they can become significant, and arrive at a 30% of the total.
- An important characteristic of the Coleman approximation is related to the last observation. We have seen that the whirling modes have an high periodic content, characterized by the 0Ω harmonic having the maximum participation, and a predominant $+1\Omega$ or -1Ω super-harmonic. As said in chapter 3 the Coleman approximation (without the super-harmonics reconstruction) is able to see only one harmonic, and in all cases it has seen the 0Ω harmonic or the predominant $\pm 1\Omega$ super-harmonic. After the super-harmonics reconstruction the Coleman approximation is able to see nearly all the periodic content of the mode.

Chapter 5

System identification from input-output data

In chapter 3 we explained how to perform the stability analysis of a linear time periodic system. It was shown that in order to compute the frequencies, dampings and participation factors the knowledge of the state transition matrix is necessary. Since this matrix is governed by a first order ordinary differential equation this amounts to solve N_s^2 coupled ode. The multi-body models commonly used in the industry have hundreds of degrees of freedom, and hence this is an unfeasible task.

To make periodic stability analyses of complex system we decided to identify reduced order models. These models have less then ten states, and are build on the basis of input and output time histories.

Since only input and output time histories are used for the identification our procedure is independent from the model. Hence our procedure is applicable to model of arbitrary complexity, and even to experimental data.

This chapter has been written on the basis of [20] and is divided in two sections. The first section exposes the identification of PARX and PARMAX models by means of the prediction error method, while the second section demonstrate the equivalence between the PARMAX and the state-space representation.

5.1 The prediction error method

Consider a time series $z(t)$ describing the time evolution of a measure affected by noise, and sample it at a constant sampling rate $1/\Delta t$ so that it becomes $z(k)$ with $k = 1, 2, \dots, N$. Then choose a model that is appropriate for the system being studied; an ample although not complete list of ARMAX models is given in [30]. At the chosen model is associated a predictor, that from the knowledge of the measures at the time steps before k is able to predict the measure at that time step.

The *prediction error* is defined as

$$\epsilon(k) \equiv z(k) - \hat{z}(k), \quad (5.1)$$

being $\hat{z}(k)$ the predicted measure.

The Prediction Error Method (PEM) problem is the one of finding the parameters of the predictor that minimize the following cost function

$$J = \frac{1}{N} \sum_{k=1}^N (\epsilon(k))^2. \quad (5.2)$$

In the next paragraphs we will show how to apply the prediction error method to identify PARX and PARMAX models.

A PARX model is an ARX model having time varying periodic coefficients. The PARX models are suitable to simulate signals in which the process noise is not present.

A PARX model is composed of two blocks:

- the Auto-Regressive, that gives to the model the ability to reproduce the decay of the response;
- and the eXogenous, that allow the input to enter the model.

We will consider two ways of identify PARX models: the equation-error and the output-error. The output-error is preferable because of its superior statistical properties, but its solution require a sequence of nonlinear iterations. The equation-error is thus used to furnish a suitable initial condition.

5.1.1 The equation-error method

The equation-error assumes that the measures obey the following difference equation

$$z(k) = \sum_{i=1}^{n_a} a_i(k) z(k-i) + \sum_{j=0}^{n_b} b_j(k) u(k-j) + \eta(k). \quad (5.3)$$

The n_a parameters $a_i(k)$ form the autoregressive part, while the n_b parameters b_j form the exogenous one. Both the parts are subject to the periodicity constraint:

$$a_i(k) = a_i(k+K), \quad (5.4a)$$

$$b_j(k) = b_j(k+K), \quad (5.4b)$$

in which K is the system's period.

The process noise $\eta(k)$ by hypothesis has a Gaussian distribution, with zero mean and nonsingular periodic variance $\sigma(k)^2$.¹

From [20] the optimal one-step ahead predictor associated to the process (5.3) is

$$\hat{z}(k) = \sum_{i=1}^{n_a} a_i(k)z(k-i) + \sum_{i=1}^{n_b} b_i(k)u(k-i). \quad (5.5)$$

In general the autoregressive and exogenous parts can have any trend that satisfies the periodicity constraint, however as observed in [8] the number of unknowns can greatly reduced by expanding $a_i(k)$ and $b_j(k)$ in truncated Fourier series:

$$a_i(k) = a_{i_0} + \sum_{l=1}^{N_{Fa}} (a_{i_l}^c \cos(l\psi(k)) + a_{i_l}^s \sin(l\psi(k))), \quad (5.6a)$$

$$b_j(k) = b_{j_0} + \sum_{m=1}^{N_{Fb}} (b_{j_m}^c \cos(m\psi(k)) + b_{j_m}^s \sin(m\psi(k))). \quad (5.6b)$$

The coefficients a_{i_0} , $a_{i_l}^c$, $a_{i_l}^s$, b_{j_0} , $b_{j_m}^c$, and $b_{j_m}^s$ are then collected in an array of parameters to be identified.

$$\boldsymbol{\theta} \equiv (\dots, a_{i_0}, a_{i_l}^c, a_{i_l}^s, \dots, b_{j_0}, b_{j_m}^c, b_{j_m}^s, \dots)^T, \quad (5.7)$$

Our purpose is to write the predicted measure at each time step in a compact form. To achieve this result we define the following row arrays

$$\boldsymbol{\varphi}_a(\psi(k)) \equiv [1 \quad \cos(\psi(k)) \quad \sin(\psi(k)) \quad \dots \quad \cos(N_{Fa}\psi(k)) \quad \sin(N_{Fa}\psi(k))], \quad (5.8a)$$

$$\boldsymbol{\varphi}_b(\psi(k)) \equiv [1 \quad \cos(\psi(k)) \quad \sin(\psi(k)) \quad \dots \quad \cos(N_{Fb}\psi(k)) \quad \sin(N_{Fb}\psi(k))], \quad (5.8b)$$

by multiplying these arrays by the measures and the input we obtain

$$\mathbf{a}_{r,s} = z(r)\boldsymbol{\varphi}_a(\psi(s)), \quad (5.9)$$

$$\mathbf{b}_{r,s} = u(r)\boldsymbol{\varphi}_b(\psi(s)). \quad (5.10)$$

We now use equation (5.5) to predict the measure at each time step (after $n+1$), and we obtain the following overdetermined system

$$\hat{\mathbf{z}} = \mathbf{T}\boldsymbol{\theta}, \quad (5.11)$$

¹The variance is supposed to have the same period of the system, in formula $\sigma(k)^2 = \sigma(k+K)^2$.

with $\mathbf{T}$ given by

$$\mathbf{T} = \begin{bmatrix} \mathbf{a}_{r-1,r} & \cdots & \mathbf{a}_{r-n_a,r} & \mathbf{b}_{r,r} & \cdots & \mathbf{b}_{r-n_b,r} \\ \mathbf{a}_{r,r+1} & \cdots & \mathbf{a}_{r-n_a+1,r+1} & \mathbf{b}_{r+1,r+1} & \cdots & \mathbf{b}_{r-n_b+1,r+1} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \mathbf{a}_{N-1,N} & \cdots & \mathbf{a}_{N-n_a,N} & \mathbf{b}_{N,N} & \cdots & \mathbf{b}_{N-n_b,N} \end{bmatrix}, \quad (5.12)$$

$\hat{\mathbf{z}} = (\hat{z}(r), \hat{z}(r+1), \dots, \hat{z}(N))^T$ and $r \equiv \max(n_a, n_b) + 1$.

The prediction error is calculated by applying equation (5.1) at each time step between r and N

$$\boldsymbol{\epsilon} = \mathbf{z} - \hat{\mathbf{z}} = \mathbf{z} - \mathbf{T}\boldsymbol{\theta}. \quad (5.13)$$

This vector is used to compute the cost function, which apart from an unnecessary normalization is given by

$$J = \boldsymbol{\epsilon}^T \boldsymbol{\epsilon}. \quad (5.14)$$

The array of parameters which yield minimum the cost function is obtained by computing the pseudo-inverse of $\mathbf{T}$ as

$$\boldsymbol{\theta} = (\mathbf{T}^T \mathbf{T})^{-1} \mathbf{T}^T \mathbf{z}. \quad (5.15)$$

5.1.2 The output-error method

The output-error method assume that a signal $y(k)$ unaffected by noise obeys the following PARX process

$$y(k) = \sum_{i=1}^{n_a} a_i(k)y(k-i) + \sum_{j=0}^{n_b} b_j(k)u(k-j). \quad (5.16)$$

A noise $\eta(k)$, supposed to be white with zero mean and periodic variance, affects the measures $z(k)$

$$z(k) = y(k) + \eta(k). \quad (5.17)$$

From [20] the optimal predictor for system (5.16–5.17) is

$$\hat{z}(k) = \sum_{i=1}^{n_a} a_i(k)y(k-i) + \sum_{i=1}^{n_b} b_i(k)u(k-i). \quad (5.18)$$

Differently from (5.5) this is a nonlinear predictor because the sensitivity derivatives $\partial \hat{z} / \partial \boldsymbol{\theta}$ depends on $y(k-i)$, which in turn depends on the parameters.

Given the predictor (5.18) and the measures $z(k)$ the prediction error can be computed for all time steps between r and N by using equation (5.1). From the prediction error $\boldsymbol{\epsilon}$ we can evaluate the cost function J through equation (5.2). The cost function can then be minimized by using standard algorithms, and the initial condition given by (5.15).

5.1.3 Identification of PARMAX models

A Periodic AutoRegressive Moving Average with eXogenous input model assumes that the time history of the measures has the following structure

$$z(k) = \sum_{i=1}^{n_a} a_i(k)z(k-i) + \sum_{j=0}^{n_b} b_j(k)u(k-j) + \sum_{w=1}^{n_c} c_w(k)\eta(k-w) + \eta(k). \quad (5.19)$$

The coefficients $c_w(k)$ compose the moving average part and are periodic of period K , i.e. $c_w(k) = c_w(k+K)$. From [20, 10, 8] the optimal one step ahead predictor associated to the PARMAX process (5.19) is

$$\hat{z}(k) = \sum_{i=1}^{n_a} a_i(k)z(k-i) + \sum_{j=1}^{n_b} b_j(k)u(k-j) + \sum_{w=1}^{n_c} c_w(k)(z(k-w) - \hat{z}(k-w)). \quad (5.20)$$

As can be seen this predictor is nonlinear in the parameters since the predicted measure at time k depends from its previous values, which in turn depend on the parameters. Moreover we can notice that, while in equation (5.19) the moving average part model the way in which the process noise enters the model, in the predictor it models the way in which the prediction error enters the model.

As done in the PARX case for the AR and X parts we expand the MA part in truncated Fourier series

$$c_w(k) = c_{w_0} + \sum_{n=1}^{N_{Fc}} (c_{w_n}^c \cos(n\psi(k)) + c_{w_n}^s \sin(n\psi(k))), \quad (5.21)$$

so that the array of the parameters becomes

$$\theta \equiv (\dots, a_{i_0}, a_{i_l}^c, a_{i_l}^s, \dots, b_{j_0}, b_{j_m}^c, b_{j_m}^s, \dots, c_{w_0}, c_{w_n}^c, c_{w_n}^s, \dots)^T. \quad (5.22)$$

Before computing the prediction error and minimizing the cost function it is convenient to pose the attention to the stability characteristics of the problem, and to the initial conditions for the minimization algorithm.

Stability of a PARMAX process

The discussion of the stability of a linear time periodic system would require to use the concept of *harmonic transfer function* introduced by Wereley in [49]. Here we will give only an heuristic demonstration, which will lead us to write the correct expression for the stability constraint.

We begin by defining the back-shift operator q such that

$$z(k)q^i \equiv z(k-i). \quad (5.23)$$

The PARMAX process (5.19) can be rewritten in terms of q and we obtain

$$z(k) = \sum_{i=1}^{n_a} a_i(k) q^i z(k) + \sum_{j=0}^{n_b} b_j(k) q^j u(k) + \sum_{w=1}^{n_c} c_w(k) q^w \eta(k) + \eta(k), \quad (5.24)$$

that is no more a difference equation, but a polynomial equation in q . We now expand the sums, and upon collecting $z(k)$, $u(k)$ and $\eta(k)$ we get

$$[1 - a_1(k)q - a_2(k)q^2 + \dots]z(k) = [b_1(k)q + b_2(k)q^2 + \dots]u(k) + [1 + c_1(k)q + c_2(k)q^2 + \dots]\eta(k). \quad (5.25)$$

The expression can be simplified by defining the autoregressive, exogenous and moving average polynomials

$$A(q, k) \equiv 1 - a_1(k)q - a_2(k)q^2 + \dots, \quad (5.26a)$$

$$B(q, k) \equiv b_1(k)q + b_2(k)q^2 + \dots, \quad (5.26b)$$

$$C(q, k) \equiv 1 + c_1(k)q + c_2(k)q^2 + \dots, \quad (5.26c)$$

so that the PARMAX process becomes the following (see figure 5.1 for a graphical representation)

$$z(k) = \frac{B(q, k)}{A(q, k)} u(k) + \frac{C(q, k)}{A(q, k)} \eta(k). \quad (5.27)$$

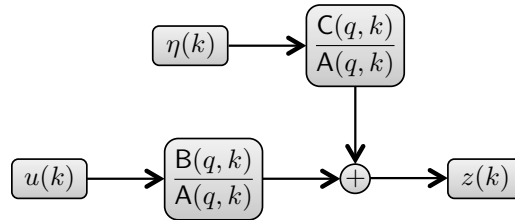


Figure 5.1: Block diagram of a PARMAX process.

From (5.27) we can easily see that the stability of the PARMAX process (5.19) is connected to the $A(q, k)$ polynomial, and hence to its autoregressive part.

We proceed by studying the stability of the PARMAX predictor. The predictor (5.20) can be rewritten as

$$\hat{z}(k) = - \sum_{i=1}^{n_c} c_i(k) \hat{z}(k-i) + \sum_{i=1}^{\max(n_a, n_b)} (a_i(k) + c_i(k)) \hat{z}(k-i) + \sum_{i=1}^{n_b} (b_i(k)) u(k-i). \quad (5.28)$$

By repeating the same passages did before we obtain

$$\begin{aligned} [1 + c_1(k)q + c_2(k)q^2]\hat{z}(k) = \\ [a_1(k)q + a_2(k)q^2 + \dots + c_1(k)q + c_2(k)q^2 + \dots]z(k) \\ + [b_1(k)q + b_2(k)q^2 + \dots]u(k), \end{aligned} \quad (5.29)$$

and by substituting the definitions (5.26) we get the following system, depicted in figure 5.2,

$$\hat{z}(k) = \frac{C(q, k) - A(q, k)}{C(q, k)}z(k) + \frac{B(q, k)}{C(q, k)}u(k). \quad (5.30)$$

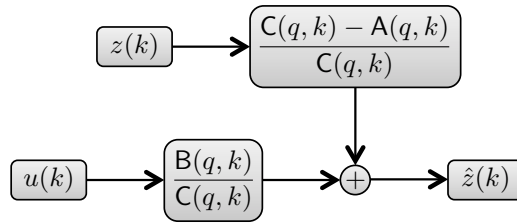


Figure 5.2: Block diagram of a PARMAX predictor.

From (5.30) we can easily see that the stability of the PARMAX predictor (5.20-5.28) is connected to the $C(q, k)$ polynomial, and hence to its moving average part. This should not be surprising since as said before the moving average part in the PARMAX predictor models the way in which the prediction error enters the predicted output, hence requiring an asymptotically stable dynamics for the MA part imply to require an asymptotically stable dynamics of the prediction error.

The risk of instability of the predictor is the reason for which some care must be put during the minimization of the cost function.

There are essentially three ways to perform the minimization of J in the correct manner, here we will give only a brief summary; the interested reader can find more details in [20, 8, 39]. If during the minimization of J an unstable predictor is found, do one of the following.

1. Halve repeatedly the coefficients $c_w(k)$ till a stable predictor is achieved. This method has the shortcoming of practically re-initializing the minimization of the cost function.
2. Solve a periodic Riccati equation to compute a minimum phase representation of the given (non-minimum phase) PARMAX model, so that the impulsive responses of both models have the same autocorrelation function.

3. Compute an equivalent minimum phase representation by using the Rissanen's projection algorithm. This technique is similar to one used for ARMAX systems, where the Bauer's projection algorithm is employed.

In this work we developed an innovative approach to do the identification of PARMAX models. We considered the identification a constrained optimization problem, defined as follows

$$\min_{\boldsymbol{\theta}} J(z(k) - \hat{z}(k); \boldsymbol{\theta}), \quad (5.31a)$$

$$\text{s.t.: } |\mathcal{P}(\boldsymbol{\theta})| < 1, \quad (5.31b)$$

where $|\mathcal{P}(\boldsymbol{\theta})|$ are the modulus of the poles of the PARMAX predictor. The problem (5.31) has been solved with MATLAB's function `fmincon`, utilized with the *interior-point* algorithm [19].

The previous passages have only an illustrative function to let the reader comprehend the stability problem. Some more implementative details will be given in the following.

As said before the stability of the predictor is associated to that of its moving average part, then we consider a new system where the moving average part plays the role of the autoregressive one,

$$y(k) = \sum_{i=1}^{N_c} -c_i(k)y(k-i). \quad (5.32)$$

In section 5.2 we will see that the process (5.32) can be written in state-space canonical observational form as

$$\mathbf{x}(k+1) = \mathbf{A}(k)\mathbf{x}(k), \quad (5.33a)$$

$$y(k) = \mathbf{C}(k)\mathbf{x}(k). \quad (5.33b)$$

The state $\mathbf{x}(k)$ in (5.33) has its elements sorted from n_c to 1, i.e. $\mathbf{x}(k) = (x_{n_c}, x_{n_c-1}, \dots, x_1)^T$, the matrix $\mathbf{C}(k)$ is constructed so that $y(k) = x_1(k)$, and the dynamic matrix is given by

$$\mathbf{A}(k) = \begin{bmatrix} 0 & 0 & \cdots & 0 & \alpha_n(k) \\ 1 & 0 & \cdots & 0 & \alpha_{n-1}(k) \\ 0 & 1 & \cdots & 0 & \alpha_{n-2}(k) \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & \alpha_1(k) \end{bmatrix}, \quad (5.34)$$

where $\alpha_j(k) = -c_j(k+j)$, $\forall j = 1, \dots, n_c$.

The monodromy matrix associated to the discrete time system (5.33) is (see [8])

$$\Psi = A(K)A(K-1) \cdots A(1). \quad (5.35)$$

The eigenvalues of the monodromy matrix are by definition the characteristic multipliers, which are the poles $\mathcal{P}(\theta)$ of the PARMAX predictor.

Exploration of the local minima and initial conditions

The minimization of the cost function is performed, as said above, with an algorithm called interior-point. This algorithm mixes in a sophisticated way the techniques belonging to the Sequential Quadratic Programming (SQP) and the trust region, however it has not been conceived to explore the local minima of the objective function; minima that the tests that we have conducted have shown to exist.

Another problem associated to all Newton based methods is the one of the initial conditions. In [39] are indicated three strategies to set the initial conditions.

1. Use null initial conditions.
2. Initialize the parameters belonging to the moving average part with null values, and the autoregressive and axogenous ones with the values obtained by the identification of the data with a PARX model.
3. Identify the data with an ARMAX model, and set to zero the parameters of the periodic parts.

All these strategies presents shortcomings and peculiarities, that are discussed in [39]. In this works we opted for another approach, that work in conjunction with the problem of avoiding the local minima.

We chose to use MATLAB's `MultiStart` class. The function `run` of this class starts a series of local solvers from a set of initial points $\{\theta_1, \theta_2, \dots, \theta_n\}$, with the purpose of exploring the basin of the cost function. The parameters inside each θ_j are picked randomly inside the unit disk.

During a run some of the initial set of parameters will correspond to unstable predictors, and during the local minimization process it is possible that the solver won't be able reach a local minimum, however by selecting an high enough number of initial conditions the global minimum is likely to be found.

5.2 Realization of input-output models in state-space form

Time series are a way of expressing dynamical system which leave aside the state.

The usual way of representing dynamical systems is through the so called state-space representation. Although useful for many application this form is not particularly suited for the purpose of doing identifications, since the knowledge of the state is usually limited.

During this thesis we will use time series in polynomial form to perform the identifications and the state-space one to compute the eigenvalues, to find a relation between the two is thus mandatory.

The demonstration hereafter will follow the one presented in [43], and will extend it to the more general case of time varying systems with weighted process noise in output and measurement noise. The following result has been demonstrated in different ways also in [16] and [47].

For simplicity we will restrict our demonstration to the order 3, but the final result won't suffer of this limitation.

Let us consider the following dynamical system

$$\mathbf{x}(k+1) = \mathbf{A}(k)\mathbf{x}(k) + \mathbf{B}(k)u(k) + \mathbf{E}(k)\eta(k), \quad (5.36a)$$

$$y(k) = \mathbf{C}(k)\mathbf{x}(k) + \mathbf{D}(k)u(k) + \mathbf{F}(k)\eta(k) + \mathbf{G}(k)r(k). \quad (5.36b)$$

The state vector is defined as $\mathbf{x}(k) \equiv (x_3(k), x_2(k), x_1(k))^T$, and the matrices in the state transition equation are

$$\mathbf{A}(k) \equiv \begin{bmatrix} 0 & 0 & \alpha_3(k) \\ 1 & 0 & \alpha_2(k) \\ 0 & 1 & \alpha_1(k) \end{bmatrix}, \quad \mathbf{B}(k) \equiv \begin{bmatrix} \beta_3(k) \\ \beta_2(k) \\ \beta_1(k) \end{bmatrix}, \quad \mathbf{E}(k) \equiv \begin{bmatrix} \epsilon_3(k) \\ \epsilon_2(k) \\ \epsilon_1(k) \end{bmatrix}. \quad (5.37)$$

The matrices in the measurement equation are defined as

$$\begin{aligned} \mathbf{C}(k) &\equiv [0 \quad 0 \quad 1], \quad \mathbf{D}(k) \equiv [\delta_0(k)], \\ \mathbf{F}(k) &\equiv [\varphi_0(k)], \quad \mathbf{G}(k) \equiv [\gamma_0(k)]. \end{aligned} \quad (5.38)$$

We now substitute the previous definitions in (5.36)

$$\begin{pmatrix} x_3(k+1) \\ x_2(k+1) \\ x_1(k+1) \end{pmatrix} = \begin{bmatrix} 0 & 0 & \alpha_3(k) \\ 1 & 0 & \alpha_2(k) \\ 0 & 1 & \alpha_1(k) \end{bmatrix} \begin{pmatrix} x_3(k) \\ x_2(k) \\ x_1(k) \end{pmatrix} + \begin{bmatrix} \beta_3(k) \\ \beta_2(k) \\ \beta_1(k) \end{bmatrix} u(k) + \begin{bmatrix} \epsilon_3(k) \\ \epsilon_2(k) \\ \epsilon_1(k) \end{bmatrix} \eta(k), \quad (5.39a)$$

$$y(k) = [0 \quad 0 \quad 1] \begin{pmatrix} x_3(k) \\ x_2(k) \\ x_1(k) \end{pmatrix} + \delta_0 u(k) + \varphi_0 \eta(k) + \gamma_0 r(k), \quad (5.39b)$$

and we evaluate the product of the $\mathbf{A}(k)$ and $\mathbf{C}(k)$ matrices by the state vector

$$\begin{pmatrix} x_3(k+1) \\ x_2(k+1) \\ x_1(k+1) \end{pmatrix} = \begin{bmatrix} \alpha_3(k)x_1(k) \\ x_3(k) + \alpha_2(k)x_1(k) \\ x_2(k) + \alpha_1(k)x_1(k) \end{bmatrix} + \begin{bmatrix} \beta_3(k) \\ \beta_2(k) \\ \beta_1(k) \end{bmatrix} u(k) + \begin{bmatrix} \epsilon_3(k) \\ \epsilon_2(k) \\ \epsilon_1(k) \end{bmatrix} \eta(k), \quad (5.40a)$$

$$y(k) = x_1(k) + \delta_0 u(k) + \varphi_0 \eta(k) + \gamma_0 r(k). \quad (5.40b)$$

The output equation can be used to get $x_1(k)$

$$x_1(k) = y(k) - \delta_0(k)u(k) - \varphi_0(k)\eta(k) - \gamma_0(k)r(k), \quad (5.41)$$

which allow us to define a shorthand for the next formulae, namely

$$\sigma(k) \equiv y(k) - \delta_0(k)u(k) - \varphi_0(k)\eta(k) - \gamma_0(k)r(k). \quad (5.42)$$

We can now express each component of the state vector independently from the others

$$x_3(k+1) = \alpha_3(k)\sigma(k) + \beta_3(k)u(k) + \epsilon_3(k)\eta(k), \quad (5.43a)$$

$$\begin{aligned} x_2(k+1) &= \alpha_3(k-1)\sigma(k-1) + \beta_3(k-1)u(k-1) + \epsilon_3(k-1)\eta(k-1) \\ &\quad + \alpha_2(k)\sigma(k) + \beta_2(k)u(k) + \epsilon_2(k)\eta(k), \end{aligned} \quad (5.43b)$$

$$\begin{aligned} x_1(k+1) &= \alpha_3(k-2)\sigma(k-2) + \beta_3(k-2)u(k-2) + \epsilon_3(k-2)\eta(k-2) \\ &\quad + \alpha_2(k-1)\sigma(k-1) + \beta_2(k-1)u(k-1) + \epsilon_2(k-1)\eta(k-1) \\ &\quad + \alpha_1(k)\sigma(k) + \beta_1(k)u(k) + \epsilon_1(k)\eta(k). \end{aligned} \quad (5.43c)$$

The previous equations written in matrix form become

$$\begin{aligned} \begin{pmatrix} x_3(k+1) \\ x_2(k+1) \\ x_1(k+1) \end{pmatrix} &= \begin{bmatrix} \alpha_3(k) & 0 & 0 \\ \alpha_2(k) & \alpha_3(k-1) & 0 \\ \alpha_1(k) & \alpha_2(k-1) & \alpha_3(k-2) \end{bmatrix} \begin{pmatrix} \sigma(k) \\ \sigma(k-1) \\ \sigma(k-2) \end{pmatrix} \\ &\quad + \begin{bmatrix} \beta_3(k) & 0 & 0 \\ \beta_2(k) & \beta_3(k-1) & 0 \\ \beta_1(k) & \beta_2(k-1) & \beta_3(k-2) \end{bmatrix} \begin{pmatrix} u(k) \\ u(k-1) \\ u(k-2) \end{pmatrix} \\ &\quad + \begin{bmatrix} \epsilon_3(k) & 0 & 0 \\ \epsilon_2(k) & \epsilon_3(k-1) & 0 \\ \epsilon_1(k) & \epsilon_2(k-1) & \epsilon_3(k-2) \end{bmatrix} \begin{pmatrix} \eta(k) \\ \eta(k-1) \\ \eta(k-2) \end{pmatrix}. \end{aligned} \quad (5.44)$$

Taking the last line and substituting the left hand side with $\sigma(k+1)$ gives

$$\begin{aligned} \sigma(k+1) &= [\alpha_1(k) \quad \alpha_2(k-1) \quad \alpha_3(k-2)] \begin{pmatrix} \sigma(k) \\ \sigma(k-1) \\ \sigma(k-2) \end{pmatrix} \\ &\quad + [\beta_1(k) \quad \beta_2(k-1) \quad \beta_3(k-2)] \begin{pmatrix} u(k) \\ u(k-1) \\ u(k-2) \end{pmatrix} \\ &\quad + [\epsilon_1(k) \quad \epsilon_2(k-1) \quad \epsilon_3(k-2)] \begin{pmatrix} \eta(k) \\ \eta(k-1) \\ \eta(k-2) \end{pmatrix}. \end{aligned} \quad (5.45)$$

By expanding for $\sigma(k)$ and rearranging we obtain

$$\begin{aligned}
& \begin{bmatrix} 1 & -\alpha_1(k) & -\alpha_2(k-1) & -\alpha_3(k-2) \end{bmatrix} \begin{pmatrix} y(k+1) \\ y(k) \\ y(k-1) \\ y(k-2) \end{pmatrix} = \\
& \begin{bmatrix} \delta_0(k+1) & \eta_1(k) & \eta_2(k-1) & \eta_3(k-2) \end{bmatrix} \begin{pmatrix} u(k+1) \\ u(k) \\ u(k-1) \\ u(k-2) \end{pmatrix} \\
& \begin{bmatrix} \varphi_0(k+1) & \vartheta_1(k) & \vartheta_2(k-1) & \vartheta_3(k-2) \end{bmatrix} \begin{pmatrix} \eta(k+1) \\ \eta(k) \\ \eta(k-1) \\ \eta(k-2) \end{pmatrix} \\
& \begin{bmatrix} \gamma_0(k+1) & \lambda_1(k) & \lambda_2(k-1) & \lambda_3(k-2) \end{bmatrix} \begin{pmatrix} r(k+1) \\ r(k) \\ r(k-1) \\ r(k-2) \end{pmatrix}, \quad (5.46)
\end{aligned}$$

where we have used the following shorthands

$$\eta_j(k) \equiv \beta_j(k) - \alpha_j(k)\delta_0(k), \quad (5.47a)$$

$$\vartheta_j(k) \equiv \epsilon_j(k) - \alpha_j(k)\varphi_0(k), \quad (5.47b)$$

$$\lambda_j(k) \equiv -\alpha_j(k)\gamma_0(k). \quad (5.47c)$$

The last equation can be written in more concise way as

$$\begin{aligned}
y(k+1) = & \sum_{i=1}^3 \alpha(k-i+1)y(k-i+1) \\
& + \eta_i(k-i+1)u(k-i+1) + \delta_0(k+1)u(k+1) \\
& + \vartheta_i(k-i+1)\eta(k-i+1) + \varphi_0(k+1)\eta(k+1) \\
& + \lambda_i(k-i+1)r(k-i+1) + \gamma_0(k+1)r(k+1). \quad (5.48)
\end{aligned}$$

We are now in the position of being able to remove the constraint on the number of states, and upon shift one step back in time we finally get

$$\begin{aligned}
y(k) = & \sum_{i=1}^n \alpha(k-i)y(k-i) \\
& + (\beta_i(k-i) - \alpha_i(k-i)\delta_0(k-i))u(k-i) + \delta_0(k)u(k) \\
& + (\epsilon_i(k-i) - \alpha_i(k-i)\varphi_0(k-i))\eta(k-i) + \varphi_0(k)\eta(k) \\
& - \alpha_i(k-i)\gamma_0(k-i)r(k-i) + \gamma_0(k)r(k). \quad (5.49)
\end{aligned}$$

Equation (5.49) is a difference equation that express the relation between the input and the output in the time domain, exactly as the transfer function do in the frequency domain. This equation has been derived by starting from a system with n states, we are now interested in starting from a time varying ARMAX model and write it in state space form.

A time varying ARMAX model is defined by the following difference equation, but it should be noted that common ARMAX models doesn't include the measurement noise.

$$\begin{aligned}
 y(k) = & \sum_{i=1}^{n_a} a_i(k)y(k-i) \\
 & + \sum_{i=1}^{n_b} b_i(k)u(k-i) + b_0(k)u(k) \\
 & + \sum_{i=1}^{n_c} c_i(k)\eta(k-i) + c_0(k)\eta(k) \\
 & + \sum_{i=1}^{n_d} d_i(k)r(k-i) + d_0(k)r(k).
 \end{aligned} \tag{5.50}$$

Time varying ARMAX models, with different orders for two or more of its parts, can be written in state-space form by simply setting to zero the values of the non-existing coefficients.

A direct comparison of the terms in the equations (5.49) and (5.50) gives the result we were looking for

$$\alpha_i(k) = a_i(k+i), \tag{5.51a}$$

$$\delta_0(k) = b_0(k), \tag{5.51b}$$

$$\beta_i(k) = b_i(k+i) + a_i(k+i)b_0(k), \tag{5.51c}$$

$$\varphi_0(k) = c_0(k), \tag{5.51d}$$

$$\epsilon_i(k) = c_i(k+i) + a_i(k+i)c_0(k), \tag{5.51e}$$

$$\gamma_0(k) = d_0(k), \tag{5.51f}$$

$$d_i(k) = -\alpha_i(k+i)\gamma_0(k+i). \tag{5.51g}$$

The $A(k)$, $B(k)$ and $E(k)$ matrices of the state-space form associated to

equations (5.50, 5.51) are

$$\mathbf{A}(k) = \begin{bmatrix} 0 & 0 & \cdots & 0 & \alpha_n(k) \\ 1 & 0 & \cdots & 0 & \alpha_{n-1}(k) \\ 0 & 1 & \cdots & 0 & \alpha_{n-2}(k) \\ \vdots & \ddots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & 1 & \alpha_1(k) \end{bmatrix}, \quad \mathbf{B}(k) = \begin{bmatrix} \beta_n(k) \\ \beta_{n-1}(k) \\ \beta_{n-2}(k) \\ \vdots \\ \beta_1(k) \end{bmatrix}, \quad \mathbf{E}(k) = \begin{bmatrix} \epsilon_n(k) \\ \epsilon_{n-1}(k) \\ \epsilon_{n-2}(k) \\ \vdots \\ \epsilon_1(k) \end{bmatrix}, \quad (5.52)$$

the $\mathbf{C}(k)$ matrix is a row matrix filled with zeros and the last element equal to 1, and the $\mathbf{D}(k)$, $\mathbf{F}(k)$ and $\mathbf{G}(k)$ matrices are the same of equation (5.38).

Chapter 6

Stability analysis of a detailed multi-body model of a 6MW wind turbine

In chapter 5 we exposed the mathematical passages necessary to conduct the identifications. In this chapter we will explain how to translate those passages in a real, although virtual environment.

In [16] appeared for the first time the notion of *periodic Campbell diagram*. In this pioneering work a large number of identifications have been conducted in ideal wind conditions, that is to say in a constant axial wind. This thesis started with the idea of extending the method proposed in [16] to allow the drawing of the periodic Campbell diagram also when the wind turbine is operating in turbulent wind conditions. To achieve this result we developed a new procedure, which account for the statistical nature of the problem.

The procedure can be summarized in the following steps, and is schematized in figure 6.1.

1. When the rotor's speed exhibits little variations, perturb the system, so as to excite the response of the desired mode.
2. Record the response of a suitable load, where the response of the excited mode is most prominent.
3. When the high frequency oscillations caused by the excitation method dampened, estimate the rotor's angular speed.
4. When the excitation method has finished its action start the PARMAX identification.

5. Repeat steps 1 to 4 until the rotor speed (or the wind speed) range is entirely covered.
6. Fit with polynomials the principal harmonics for frequency and damping, and all the harmonics for the participations.
7. Check if the convergence for the coefficients of the polynomials has been reached, and in the negative case add more points.
8. Reconstruct the super harmonics for frequency and damping.

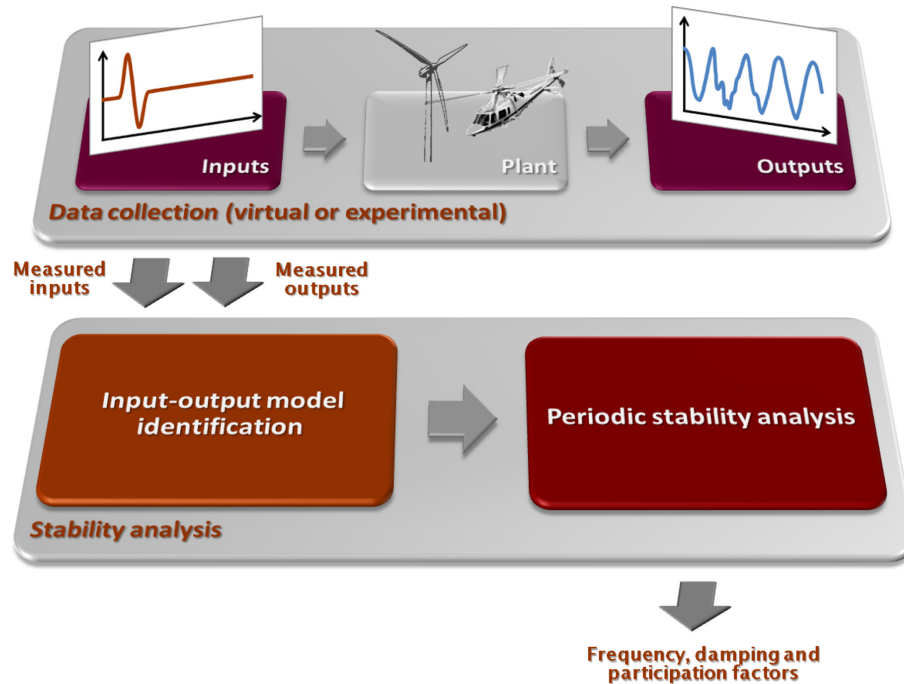


Figure 6.1: Model-independent stability analysis process. The post-processing phase is not indicated. Reproduced from [16].

The steps of the procedure just outlined will be described in detail in this chapter, the various problems that arise in turbulent wind conditions will be discussed, and at the end we will show the the periodic Campbell diagram relative to the first edgewise mode of a 6MW wind turbine.

6.1 The numerical simulation

In this work we relied on numerical simulations to estimate the stability characteristics of the wind turbine. These simulations use a high-fidelity multi-body model of wind turbine, representing most of its geometry and the relevant mechanics. The employed wind turbine is of direct-drive type, and produces up to 6 MW. The winds that we used are stationary or turbulent, and respectful of the IEC 61400 rules.

6.1.1 The Cp-Lambda and TurbSim software

Cp-Lambda [17] is a software employed in the analysis and design of wind turbine, developed at the Dipartimento di Scienze e Tecnologie Aerospaziali (DAST) of the Politecnico di Milano. Its name is the acronym of Code for Performance, Loads, Aeroelasticity by Multi-Body Dynamics Analysis. This software employs nonlinear finite elements, based on geometrically exact formulation, and impose the constraints through scaled Lagrange multipliers. For the time integration it uses an implicit non-linearly unconditionally stable energy decaying scheme [4]. The aerodynamic loads are obtained with a blade-element momentum (BEM) model, which is based on the annular stream tube theory with wake swirl, corrected with tip and hub losses, unsteady behaviors and dynamic stall. Cp-Lambda contains models of sensors and actuator, and is coupled with external codes for the implementation of the control laws. Its flexibility allow to study systems having complex topologies.

The wind fields that we give in input to Cp-Lambda are generated with TurbSim, a software developed at the National Renewable Energy Laboratory (NREL). TurbSim [33] is a stochastic, full-field turbulent-wind simulator. Three-component wind-speed vectors at points in a two-dimensional vertical rectangular grid are generated with a statistical model. TurbSim writes the spectra of the velocity components and the spatial coherence in the frequency domain, and then utilize the inverse Fourier transform to generate the time series. This software use the main characteristics of the wind turbine to generate turbulent winds of specified intensity, which also simulate the atmospheric boundary layer.

6.1.2 Wind generation

We performed the simulations with two kinds of winds: stationary and turbulent, from 3 to 10 m/s. This range of winds makes the wind turbine able to cover the entire range of angular speed.

The stationary wind is constant in time, but nonuniform over the rotor-disc, because it simulates the atmospheric boundary layer by using the power law with

an exponent of 0.2.

The turbulent wind uses a Kaimal spectral model, and has been set to have a “B” turbulence intensity. From IEC 61400, with the given velocity range this amount to considering turbulence intensities that goes from 36% to 14%. TurbSim uses a random seed to generate turbulent wind history, this seed is essentially a phase shift between the sinusoids that compose the velocity. In the course of our identifications we considered 7 different seeds, so as to remove from the results every dependence from the particular used wind field.

6.2 Excitation of modes

The identification of reduced models imply that the identified models will be able to represent only a small amount of the behaviors of the system. In particular our identifications are able to represent only one mode. This restriction obliged us to excite only one mode at a time.

In [29, 46] the authors demonstrated that the first eight wind turbine’s modes (with the exception of the lateral tower mode) can be excited with suitable movements of the pitch actuators.

To excite the lag mode we have chosen to use two force doublets, in the edgewise direction, located at mid span and near the tip of the blade. The doublets have the trend depicted in figure 6.2, in which can be seen that the peak values are reached in a equal azimuth angle $\Delta\psi/2$ of a few degrees. The maximum applied force has been set to a fraction of the resistant torque. In Cp-Lambda the doublets

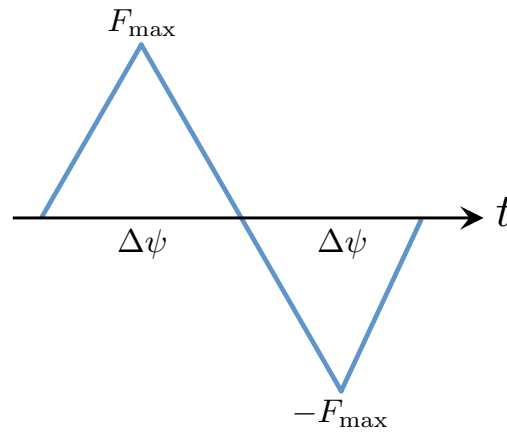


Figure 6.2: Time evolution of the force doublet used to excite the lag mode.

have been modeled by concentrated forces, while in the real world they could be obtained with pyrotechnic exciters.

6.3 Sampling process and identification time

The theory of linear time periodic systems states that a system is periodic if and only if the matrices that describe its dynamic are periodic (with the same period). In discrete time this translates in the fact that the coefficients in the matrices that describe the system repeat themselves after a fixed number of time steps. In this work all identifications are performed in discrete time, and the previous sentence implies that each period must contain the same number of time steps.

The purpose of the identifications is to perform modal analysis; since the modes and the frequencies are defined in continuous time a conversion must be performed on the characteristic exponents. This conversion is expressed by equation (3.49), here reported for convenience

$$\eta_{j\text{continuous}} = \frac{1}{\Delta t} \ln(\eta_{j\text{discrete}}).$$

We can see that, in order to perform the discrete to continuous conversion of the characteristic exponents, not only each period must contain the same number of time steps, but also that the time step must be constant.

The identifications carried on in this work have been performed on time histories coming from numerical analysis. For numerical reasons the time step used by Cp-Lambda, although constant, is really small (on the order of a hundredth of a second). It is not possible to perform identifications with so many points in each period, thus a re-sampling has been done on the time histories. The sampling has been carried on by splines, and the number of points in each period has been lowered to 50÷90.

The sampling process is necessary to perform the identifications, but it has an important side effect that must be taken into account. The maximum resolved frequency depends on the time step, therefore although reducing the number of points in each period make the identification easier, it also lower the highest frequency that can be identified. When dealing with wind turbines the frequencies that can be identified depend on the mode of interest. For a highly damped mode such as the flap one only the 1P can be find out in the response, while for a low damped one such as the lag also the 2P and 3P can be seen. With 90 points in a period the 3P frequency can still be observed, while with 50 it disappears as can be seen in figure 6.3, which show the FFT of the edgewise bending moment.

The effect of the sampling process is not only that of deleting the high frequencies. It as been observed that it also change in random way the identifications's output, namely the frequency, the damping factor and the participation factor. Example of this behaviors are given in figure 6.4 and 6.5, which show two typical PARX and PARMAX identifications. In figure 6.4 we can see that probably only one period was not enough to capture the relevant informations. Moreover the

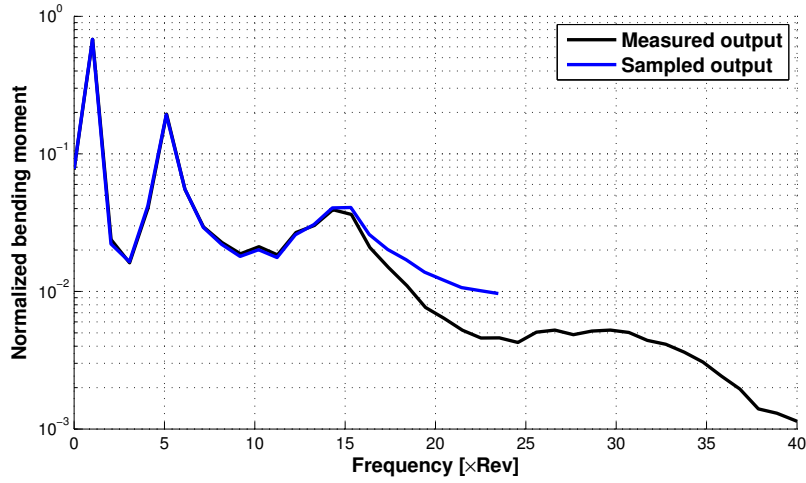


Figure 6.3: Effect of sampling with 50 points on the period. $\Omega/\Omega_r = 83.6\%$, average wind speed 8 m/s, B turbulence level.

outputs diverge with more than eighty points in each period. Figure 6.5 highlights that no deterministic trend can be observed by varying the sampling frequency and the identification time, however there are some combinations which lead to similar results.

A low sampling frequency make the identification easier, while an high one sometimes make the results more accurate. By a trial and error approach 65 to 75 points in a period has been found to be a good compromise, between accuracy of the result and easiness of identification.

The number of periods used to identify the system seems to have the same effect of the sampling frequency, but the causes are truly different. In a turbulent environment we can't expect that the wind turbine's speed will remain almost constant for more than a few seconds, and this is exactly an hypothesis that we are breaking by performing identifications over more than a few periods. We have found that one or two periods are the right choice to obtain accurate results.

From an extensive identification campaign we have seen that the confidence level on the identification's output decreases from PARX to PARMAX, and from frequency to damping factor to participation factor. For an outlook of the accuracy of the identifications's output see figure 6.6.

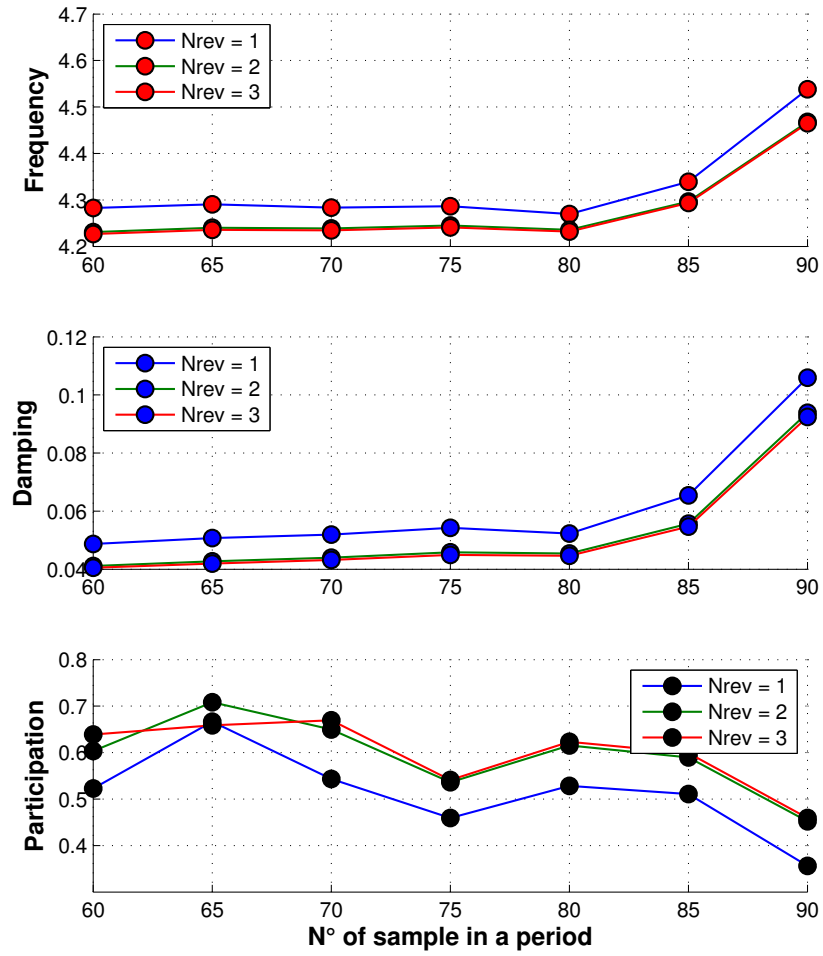


Figure 6.4: Result of a typical PARX identification, executed at the rated rotor speed. The identification time goes from 1 to 3 periods, and the sampling frequency from 60 to 90 points in a period.

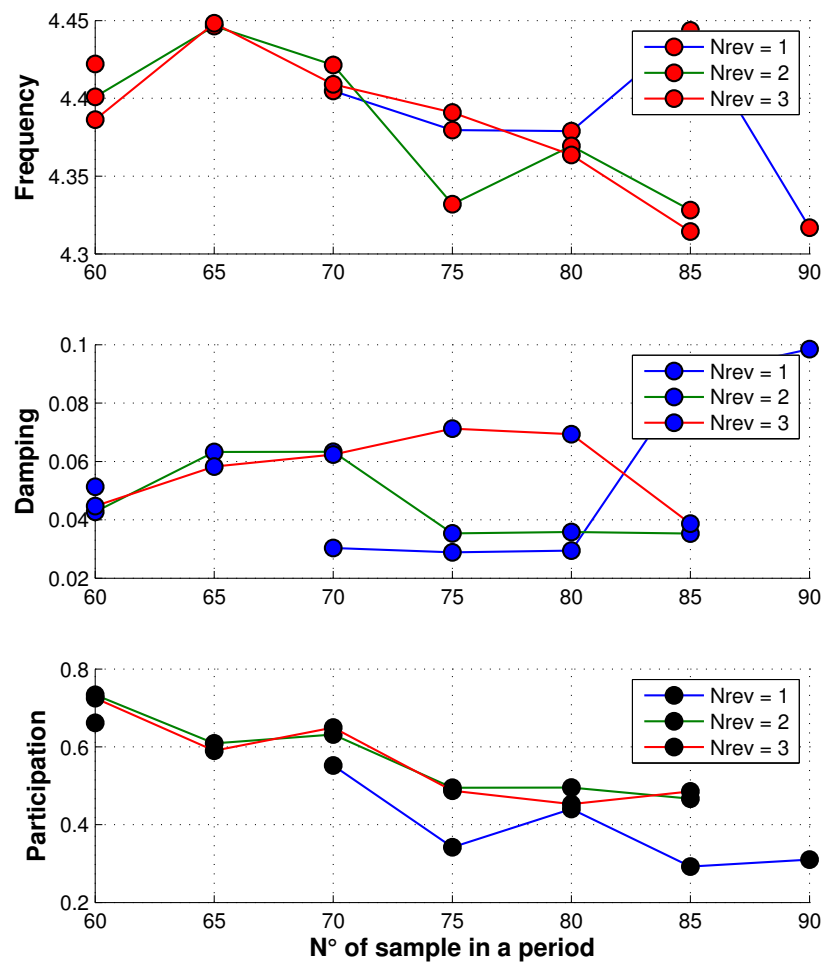


Figure 6.5: Result of a typical PARMAX identification, executed at 89% of the rated rotor speed. The identification time goes from 1 to 3 periods, and the sampling frequency from 60 to 90 points in a period.

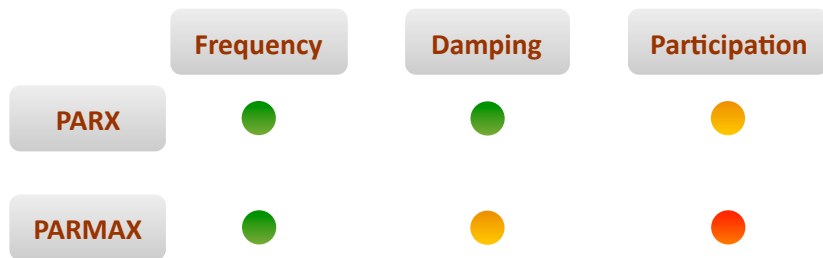


Figure 6.6: Confidence level on the identification's output.

6.4 Angular speed estimation

In the previous sections we stated that in order to work with a LTP system the period must be constant, and hence also the angular speed. With a wind turbine this is never the case, not even in the non-turbulent case.

In non-turbulent wind conditions the angular speed experience small 3P oscillations, due to the load filtering operated by the rotor (see [15]). In the turbulent wind case the variations of the angular speed increase with the standard deviation of the wind velocity, which in turn depends on the turbulence level and the asymptotic wind velocity, as figure 6.7 shows.

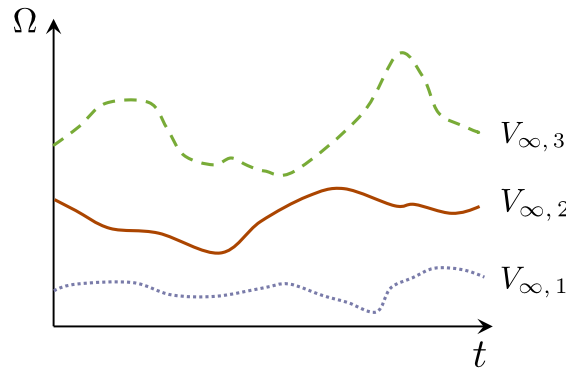


Figure 6.7: Possible angular speed histories for three turbulent winds with increasing mean velocity.

In order to perform the identifications the angular speed should be as constant as possible for the entire identification time.

Consider now the idealized situation in which the wind is constant in time and uniform over the rotor disc, and no gravity is present. Even in this case the rotor speed would not be constant because of the forcing imposed to excite the mode of interest. Figure 6.8 shows the effect that the forcing has on the angular speed. We can conclude that the *right* angular speed associated to a single identification doesn't exist. This consideration pose a serious problem on how to draw the Campbell diagram. To overcome this difficulty we have had to estimate the angular speed for each case. There are two ways to estimate Ω .

The first way is to perform a simple average:

$$\hat{\Omega} = \begin{cases} \frac{1}{t_f - t_i} \int_{t_i}^{t_f} \Omega(t) dt, & \text{in continuous time,} \\ \frac{1}{n} \sum_{k=k_i}^{k_f} \Omega(k), & \text{in discrete time.} \end{cases} \quad (6.1)$$

The initial estimation time t_i should be placed a second or less after the end of the forcing, so that the high frequency oscillations showed in figure 6.8 are not

included in the average. The final identification time t_f is located after 1 to 3 periods from t_i .

The second way of estimating Ω consists in fitting a sine function to the measured quantity. This method should be used when the angular speed's time history is not available, and stems from the fact that in ideal wind conditions the loads measured in the moving reference frame have an almost sinusoidal trend, with 1P frequency. This method should be used carefully because in non-ideal wind conditions various harmonics are present in the response, and turbulence will manifest as random variations superimposed on the sinusoidal trend. Despite these shortcomings this method showed to be quite robust and provided results in agreement with those of the first method.

We will now investigate the effect that the angular speed has on the identification. For this task we have considered an angular speed which is the 60% of the rated one. A standard eigenvalue analysis showed that the corresponding frequency of the first edgewise mode is 7.53 per rev. Starting from this point we considered an interval with Δf equal to 1% of the central frequency, and the associated $\Delta\Omega$ equal to 45.8% of the central angular speed, see figure 6.9.

By selecting the two angular speeds at the extremes of the interval we got the results of tables 6.1. In table 6.1a it is impossible to understand which is the principal harmonic, and in table 6.1b completely wrong results are shown.

Table 6.1: Two PARX identifications of the first edgewise mode, at $0.6 \Omega_r$, executed with completely wrong angular speeds.

(a) $\hat{\Omega} = 0.47\Omega_r$			(b) $\hat{\Omega} = 0.74\Omega_r$		
Freq [$\times$ Rev]	Damping	Participation	Freq [$\times$ Rev]	Damping	Participation
7.9306	0.2043	0.0728	2.9047	0.3376	0.0253
9.1833	0.1764	0.1304	3.6723	0.2670	0.0252
10.4417	0.1551	0.1304	4.4530	0.2202	0.1095
11.7039	0.1384	0.0785	5.2410	0.1871	0.0911
12.9688	0.1249	0.0360	6.0335	0.1625	0.0638

The selection of the correct angular speed gave the results of table 6.2, where as we can see the error is extremely low. Other identifications showed that errors in the angular speed below $\pm 5\%$ causes errors in the estimated frequency below 1%, nevertheless the accuracy of the participation factors is highly affected from $\hat{\Omega}$.

From the previous study we can say that, apart from rough errors in the choice of the angular speed, the estimated frequencies won't be wrong.

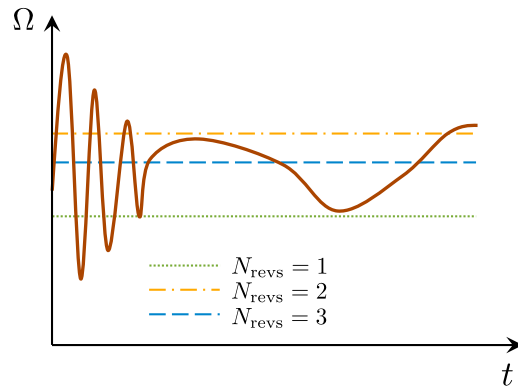


Figure 6.8: Angular speed estimation after the imposition of the doublets.

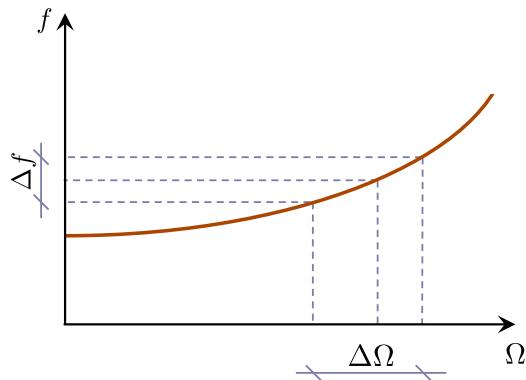


Figure 6.9: Qualitative plot of the first edgewise frequency.

Table 6.2: PARX identification of the first edgewise mode, at $0.6\Omega_r$, executed with the correct angular speed.

Freq [$\times$ Rev]	Damping	Participation
9.5378	0.0194	0.0092
8.5402	0.0217	0.1695
7.5427	0.0245	0.5549
6.5452	0.0283	0.1251
5.5478	0.0333	0.0296

6.5 Choice of model complexity

The complexity of the models used to identify the system has been set accordingly to a few “rules”, followed by a trial and error approach of the goodness of the identification.

The first step has been to set the complexity of the AutoRegressive part. This part models the decaying response of the system, and has been set accordingly to the number of peaks in the FFT of the response [36]. In particular for a low damped mode such as the lag one 3 peaks are present in the response, remembering that each of these is due to a couple of complex conjugate poles we set N_a equal to 6.

In the deterministic wind case only one Exogenous coefficient can be identified, therefore we set N_b equal to 1. We kept this choice also for the turbulent wind case, trusting that the wind variations would have been absorbed in the Moving Average part.

The Moving Average part models the variations in the response caused by atmospheric’s turbulence. There are no physical reasons to consider these variations periodic (furthermore with the same period of the wind turbine’s rotor), therefore we set N_{F_c} equal to 0.

All the remaining parameters, namely N_{F_a} , N_{F_b} and N_c have been set accordingly to the rule that a model must be as simple as possible. In particular we imposed $N_{F_a} = 1$, $N_{F_b} = 1$ and $N_c = 2$.

6.6 Fitting process

As exposed in section 6.3 the sampling frequency and the identification’s time could have a non-negligible effect on the quality of the results. With care in choosing the parameters these effects can be limited, but even in the ideal case the

engineer can't go beyond the white noise limit. To overcome this inaccuracy we resolved to fit the results with polynomials.

We have seen that low degree polynomials, usually lower than fourth order, works extremely well. The approximating polynomials have been obtained by a robust least-squares algorithm, which seeks to reduce the influence of the out-of-trend (named hereafter outlier) values.

In the following we will report the details of the employed algorithms, the reader can find more detailed explanations in the documentation of the MATLAB's curve fitting toolbox.

6.6.1 Robust least-squares algorithm

Previously we have discussed the influence that: the turbulence seed, the sampling rate, the identification time and the estimated Ω have on the results. In an effort to minimize those influences we resolved to do many identifications, and fit the results by using a robust least-squares algorithm.

A standard least-squares algorithm find the coefficients of the approximating function by minimizing the summed square of residuals S , calculated as

$$r_i \equiv y_i - \hat{y}_i, \quad (6.2)$$

$$S = \sum_{i=1}^n r_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2, \quad (6.3)$$

where y_i are the response data, $\hat{y}_i$ are the predicted values and r_i is by definition the residual. For linear least squares the measures are given by

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (6.4)$$

where $\mathbf{X}$ is the design matrix, $\boldsymbol{\beta}$ is the vector of the coefficients and $\boldsymbol{\epsilon}$ is the vector of the errors. After some passages the *estimated* vector of coefficients is found to be

$$\mathbf{b} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \quad (6.5)$$

and the predicted measures are $\hat{\mathbf{y}} = \mathbf{X}\mathbf{b}$.

During a least-squares fitting it is usually assumed that the data y_i follows a normal (Gaussian) distribution with zero mean and constant variance. This assumption is not strictly needed by the algorithm, however when it is verified it causes better fits. This behavior is due to the out of trend measures, named hereafter as outliers, which have a significant impact on the fit and are seldom find in a Gaussian distribution. If the variance of the measurement error is not constant

then, to improve the fit, it is better to give different weights to the measures, so that the function to be minimized becomes

$$S = \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2, \quad (6.6)$$

in which w_i are the weights. In this way the estimated vector of coefficients becomes

$$\mathbf{b} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}, \quad (6.7)$$

where $\mathbf{W} = \text{diag}(w_i)$.

An automatic way of choosing the weights is by adopting the robust least-squares algorithm called bisquare. This algorithm assign automatically a weight to each measure, basing on how far it is from the fitting line. The algorithm has an iterative nature, and it is composed by the following steps.

1. Fit the model by weighted least-squares, or standard least-squares if no weights are supplied.
2. Calculate the adjusted residuals. The adjusted residuals are given by

$$r_{\text{adj}} = \frac{r_i}{\sqrt{1 - h_i}}, \quad (6.8)$$

where h_i are the leverages. A high leverage indicate that the given measure will have a high influence on the fit.

3. Standardize the adjusted residuals through

$$u = \frac{r_{\text{adj}}}{Ks}, \quad (6.9)$$

where K is a tuning constant set to 4.685 and s is the robust variance, calculated by dividing the median absolute deviation of the residuals by 0.6745.

4. Compute the robust weights in the following way

$$w_i = \begin{cases} (1 - u_i^2)^2, & |u_i| < 1, \\ 0, & |u_i| \geq 1. \end{cases} \quad (6.10)$$

5. If the fit converge the iterations concludes, otherwise go back to the first step.

6.6.2 Residual analysis

The order of the polynomials is chosen basing on residual analysis, and on the well accepted rule that a fitting should be as simple as possible. In a perfect fit the residual r_i should be a white noise, while in a wrong fit r_i will exhibits a trend.

There are various statistical quantity to test the whiteness of the residual, examples are the sum of squares due to errors (SSE) and the R-square. In the following we will report their definitions, as explained in the MATLAB's manual.

The SSE is also called summed square of residual, and is a measure of the global deviation of the y_i from the fit. A well done fit has SSE near 0.

$$SSE \equiv \sum_{i=1}^n w_i (y_i - \hat{y}_i)^2. \quad (6.11)$$

The R-square measure how well the fit can explain the variation of the data, and is defined by the ratio between the sum of squares of the regression (SSR) and the total sum of squares (SST)

$$\text{R-square} \equiv \frac{SSR}{SST} = 1 - \frac{SSE}{SST}. \quad (6.12)$$

From its definition the R-square is the square of the correlation of the response values y_i and the predictions $\hat{y}_i$, and is comprised between 0 and 1, the latter value indicating a better fit. The sum of squares of the regression is given by

$$SSR \equiv \sum_{i=1}^n w_i (\hat{y}_i - \bar{y})^2, \quad (6.13)$$

where $\bar{y}$ is the average of the observations, i.e. $\bar{y} \equiv 1/n \sum_{i=1}^n y_i$. The total sum of squares is defined similarly as

$$SST \equiv \sum_{i=1}^n w_i (y_i - \bar{y})^2. \quad (6.14)$$

Other statistical quantity used to judge the goodness of a fit are the degrees of freedom adjusted R-square and the root mean squared error, which often give results in accord with the previous indicators.

6.6.3 Coefficients confidence bounds

The coefficients confidence bounds gives an indication of how high it is the uncertainty over the coefficients. An ample bound could indicate that too few points y_i

were used to do the fit. The coefficients confidence bounds are symmetric with respect to the coefficients b and are given by

$$C_{\text{upper, lower}} = b \pm t\sqrt{S}. \quad (6.15)$$

In this formula t is calculated through the inverse of the Student's cumulative distribution function, and indicates the confidence level; S is the diagonal of the estimated covariance matrix of the coefficients estimates, $(X^T X)^{-1}s^2$, where X is the design matrix and s^2 is the mean squared error.

6.6.4 Response prediction bounds

The response prediction bounds express the probability that new measures will lie close to the fitting line. In this work we have chosen to draw nonsimultaneous functional bounds, which means that we are looking for the confidence that a new measure will lie within the interval given a single predicted value. In formula we have

$$P_{n,f} = y \pm t\sqrt{xSx^T}. \quad (6.16)$$

6.7 Super-harmonics reconstruction

From Floquet's theory results that: given the characteristic exponent relative to the principal harmonics all the super-harmonics are uniquely determined. If the principal harmonic is known in terms of frequency and damping, then the real and imaginary parts of the complex characteristic exponent can be reconstructed by the following relations

$$\text{Re}(\eta_0) = -\omega_0\xi_0, \quad (6.17a)$$

$$\text{Im}(\eta_0) = \omega_0\sqrt{1 - \xi_0^2}. \quad (6.17b)$$

The super-harmonics are characterized by having their imaginary part at a distance $n\Omega$ from the imaginary part of the principal one, hence the frequency and damping of the super-harmonics are given by

$$\omega_n = \sqrt{(\text{Re}(\eta_0))^2 + (\text{Im}(\eta_0) + n\Omega)^2}, \quad n = \dots, -1, 0, 1, \dots, \quad (6.18a)$$

$$\xi_n = -\frac{\text{Re}(\eta_0)}{\omega_n}. \quad (6.18b)$$

6.8 Results of the identifications

We will now show the results of the identifications conducted to draw the periodic Campbell diagram relative to the first edgewise mode.

6.8.1 Non-turbulent wind conditions

The firsts identifications were conducted in non-turbulent wind conditions, by using a PARX structure. Figure 6.10 shows the comparison between measured (in solid line) and predicted (in dashed line) normalized blade root edgewise bending moment, in the time (left) and frequency (right) domains. From the excellent superposition of the curves the reader can notice the goodness of the identification.

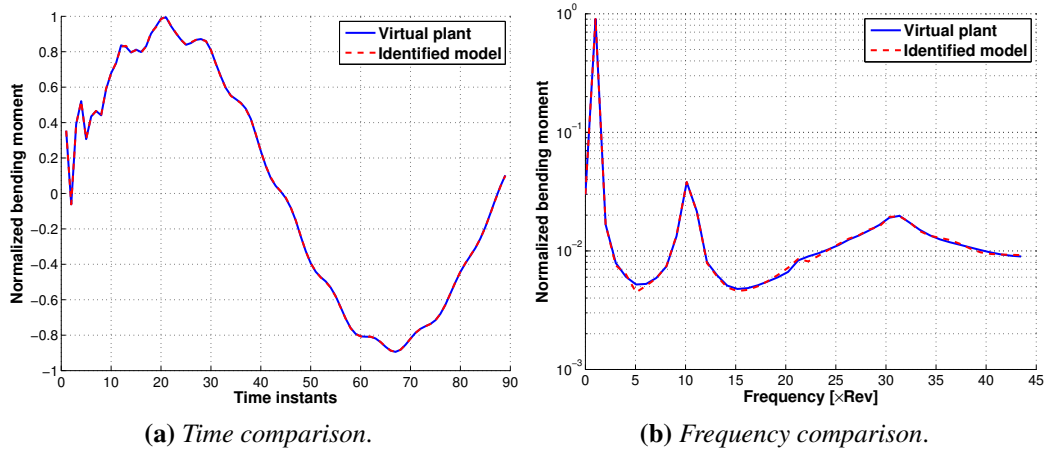


Figure 6.10: Example of PARX identification, executed at about $0.43\Omega_r$.

These identifications lead us to draw the periodic Campbell diagram in figure 6.11, from the minimum rotor speed till the rated one. This diagram differs from the one reported in [16] because in that article the authors decided to adopt an axial wind, while here a horizontal wind with atmospheric boundary layer is used.

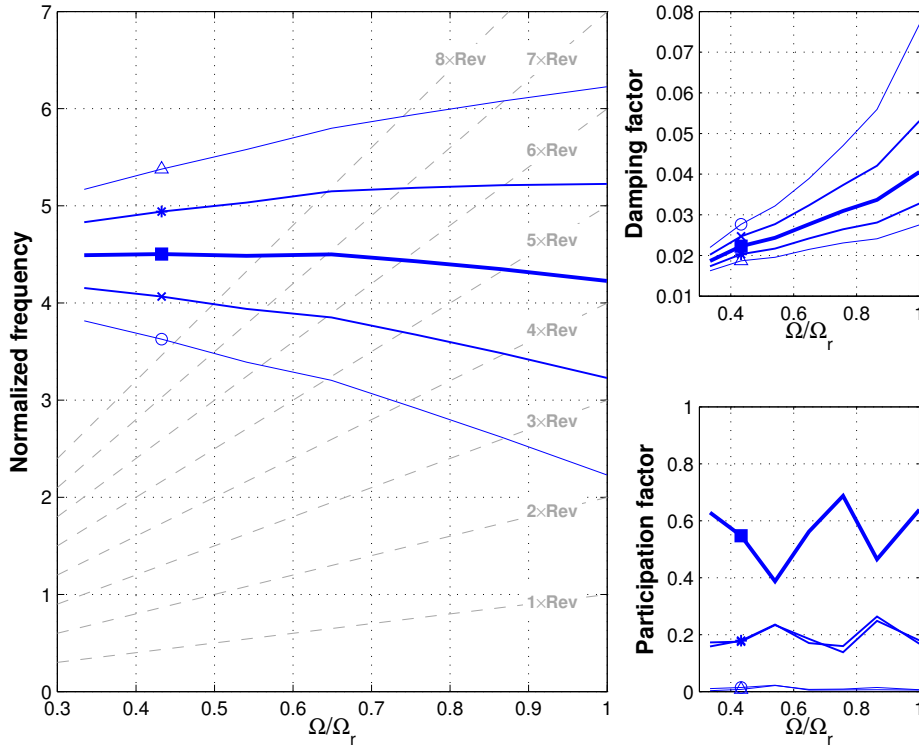


Figure 6.11: Periodic Campbell diagram of the first edgewise mode, drew in non-turbulent wind conditions.

6.8.2 Turbulent wind conditions

The work continued with the PARMAX identifications, examples of which can be seen in figures 6.12 and 6.13. In the first figure the perturbation was light, and dampened in less than one period, while in the second it was stronger, nevertheless in both cases the PARMAX predictor has been able to follow the time histories.

We executed the identification procedure 47 times, so as to have available an ample population of data, on which make statistical analysis. The residual analysis on these data showed that the best way of fitting the frequencies and the participations is by using 2nd degree polynomials, while the dampings are better fitted by a straight line. The resulting periodic Campbell diagram is shown in figure 6.14, together with the points used for the fitting and the computed prediction bounds. As we can see the frequencies are characterized by a high accuracy, while a noticeable spread in the data is present in the dampings, and an even higher one is present in the participations.

In a real environment it is not possible to perform the identifications at a given rotor speed, instead it is possible to do the identifications in a specified range of

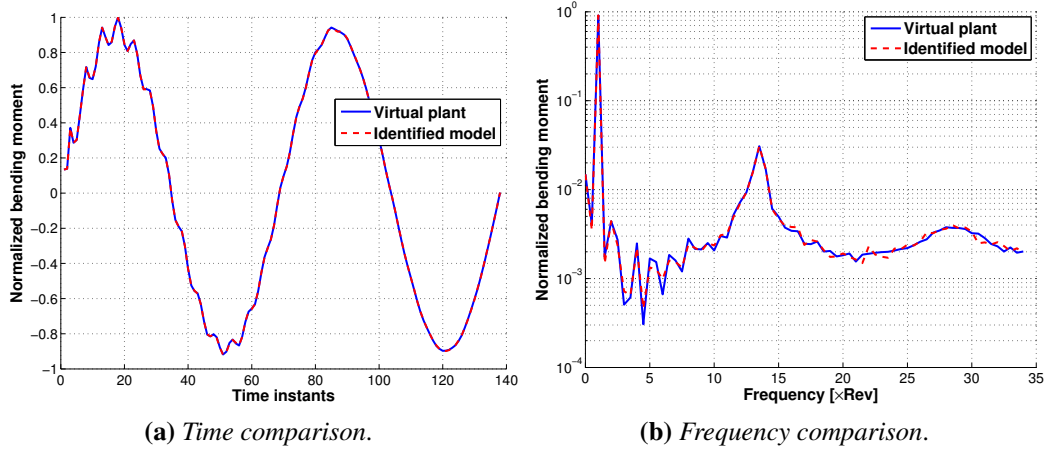


Figure 6.12: Example of PARMAX identification with a light perturbation, executed at about $0.33\Omega_r$.

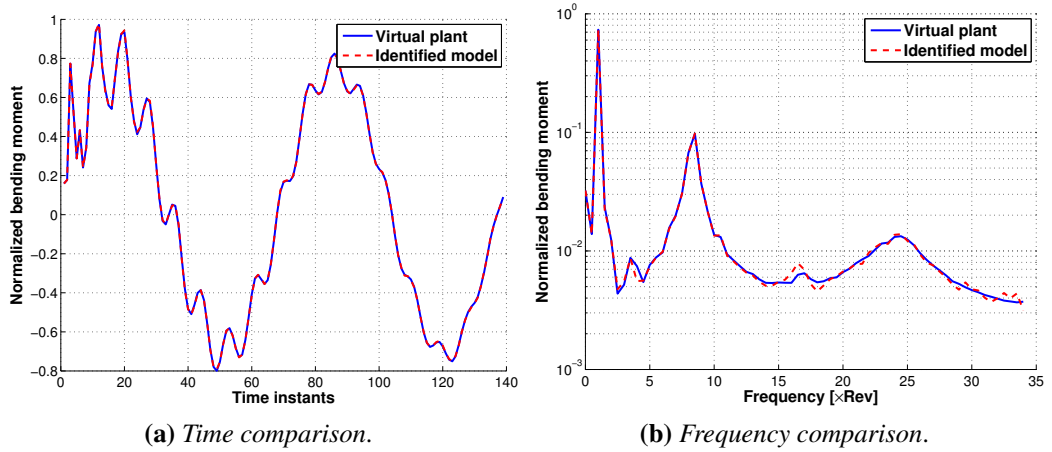


Figure 6.13: Example of PARMAX identification with a strong perturbation, executed at about $0.55\Omega_r$.

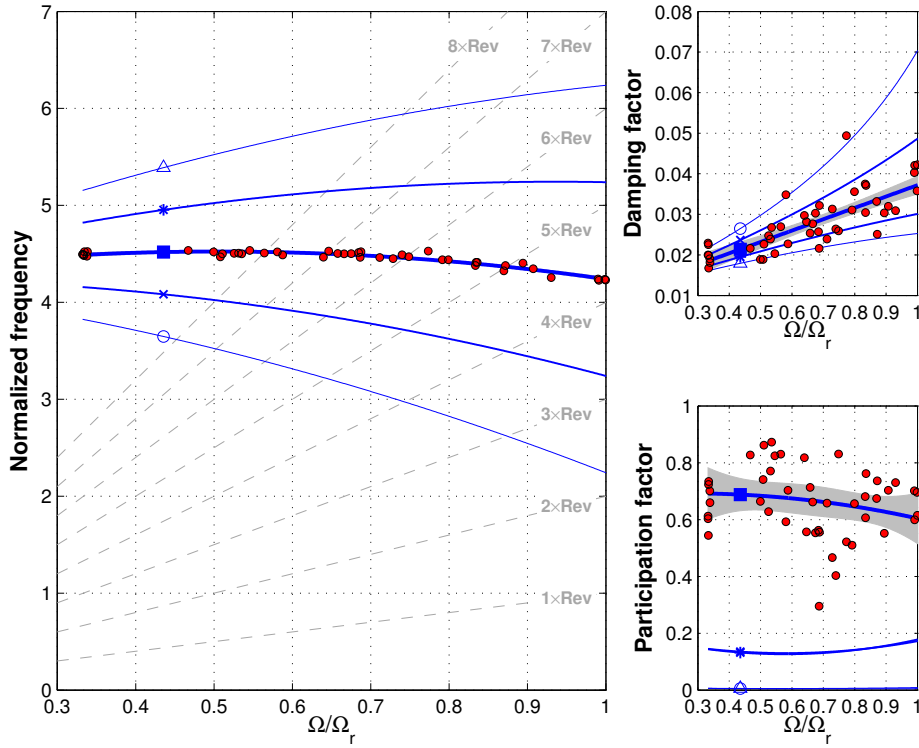


Figure 6.14: Periodic Campbell diagram of the first edgewise mode, drew in turbulent wind conditions. The grey bands represent the 2σ non-simultaneous functional prediction bounds.

rotor speeds. This consideration brought us to conclude that, the right way of conducting the convergence analysis on our population of data is the following

1. Randomly peak three identification points.
2. Perform the fitting of the principal harmonics.
3. Check the convergence of the coefficients. If the convergence has not been reached add one point and repeat steps 2 and 3.

In figure 6.15 we have reported the medians of the coefficients of the fitting polynomials and the medians of the lower and upper 2σ confidence bounds. What we can see is that the medians of the coefficients converge really quickly, but a lot of identification points are necessary to reduce the confidence bounds till a satisfactory amplitude.

To discover what happens when the identification points are taken completely random over the rotor speed range we have repeated the fitting process a large number of times. The result can be seen in figures 6.16, 6.17 and 6.18, where the

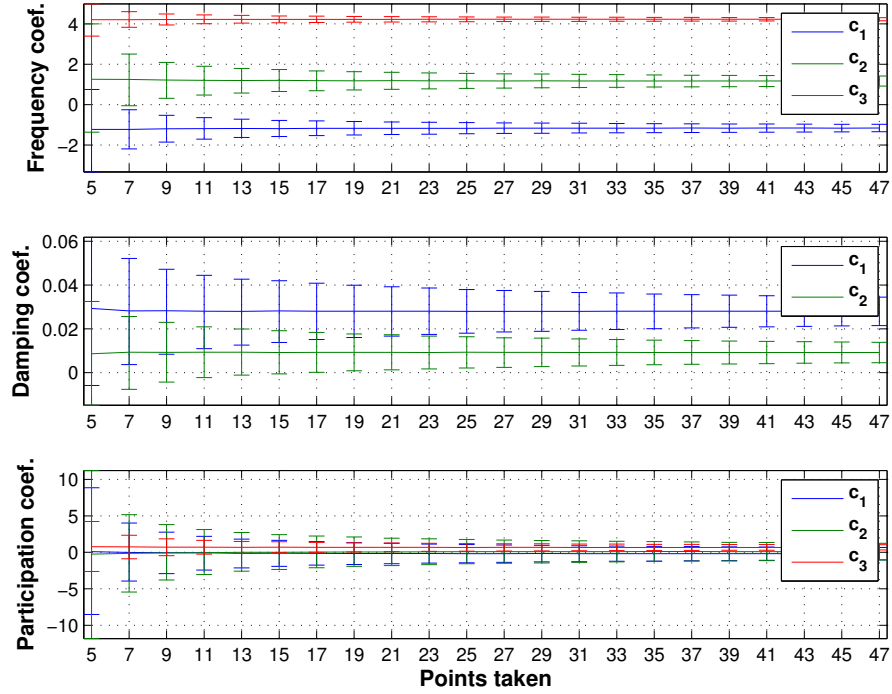


Figure 6.15: Convergence study of the coefficients of the fitting polynomials. The polynomials have the form: $y = c_1x^n + c_2x^{n-1} + \dots + c_{n+1}$.

statistics of the populations are shown by means of box plots. From these figures we can notice that if only a few identification points are available, and if these points were choose with little care, then completely misleading fitted curves are obtained.

From the convergence study we can conclude that with 20 identification points three results will follow:

- the angular speed range will be well covered;
- the convergence of the coefficients will be obtained;
- the confidence bounds of the coefficients will be quite narrow.

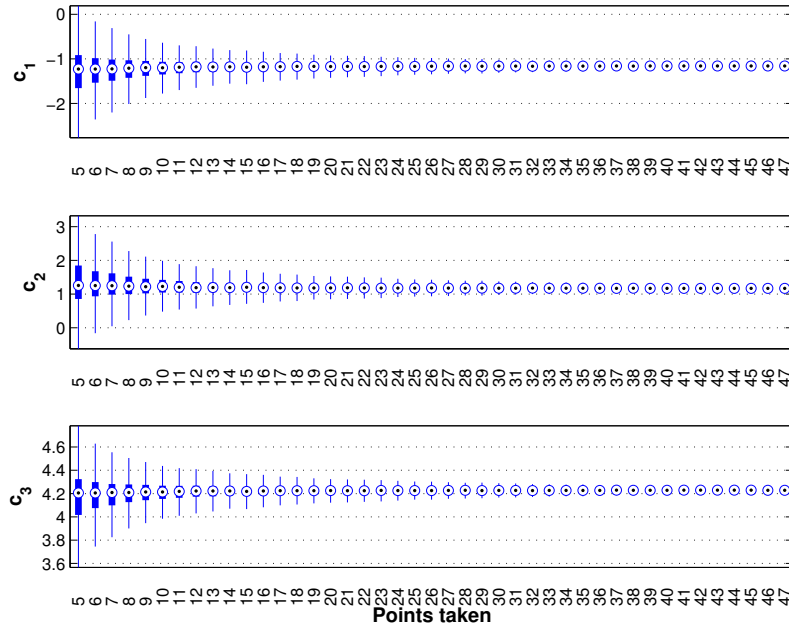


Figure 6.16: Population of the values of the coefficients that fit the frequencies.

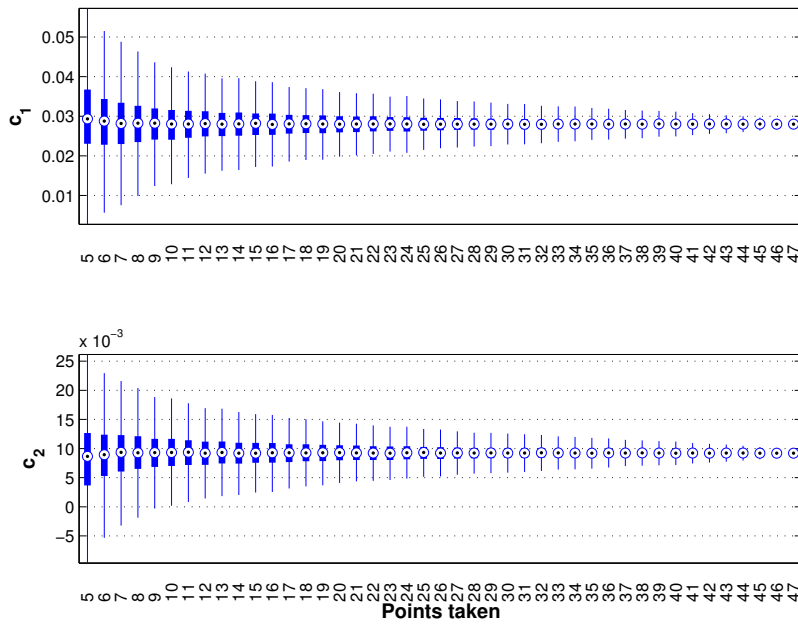


Figure 6.17: Population of the values of the coefficients that fit the dampings.

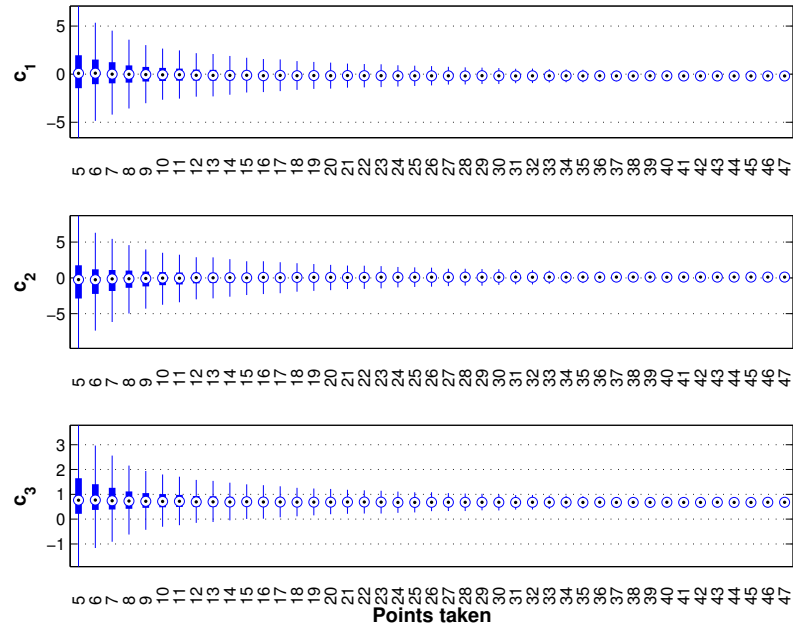


Figure 6.18: Population of the values of the coefficients that fit the participations.

6.8.3 Comparison between turbulent and non-turbulent wind conditions

The two Campbell diagrams of figures 6.11 and 6.14 have been superimposed, and the result is displayed in figure 6.19. By examining this figure we can see that the turbulence level doesn't influence the frequencies nor the dampings of the first edgewise mode.

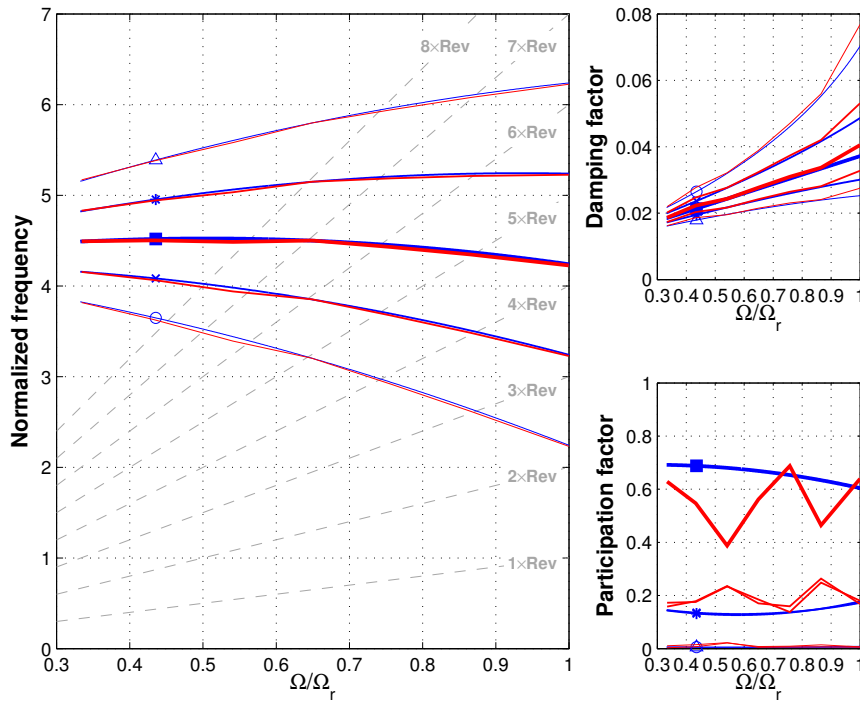


Figure 6.19: Comparison between the periodic Campbell diagrams drawn in non-turbulent (red) and turbulent (blue) wind conditions.

6.9 Conclusions

In this chapter we have demonstrated how it is possible to perform periodic stability analyses of real wind turbines, by means of techniques belonging to the field of system identification. We have exposed the various steps of the developed procedure, and analyzed the sources of errors, and how to tackle with them. The procedure has been applied to study the first edgewise mode of the blade, and we

have discussed the Campbell diagrams obtained in turbulent and non-turbulent wind conditions.

We can draw the following conclusions.

- From the comparison between figures 6.11 and 6.14 it is evident that the turbulence level doesn't influence the frequencies, nor the dampings, of the first blade edgewise mode.
- The identified frequencies are characterized by a high level of precision, while a greater uncertainty is present in the estimated dampings, and an even higher one in the participations.
- The sampling rate, the number of periods used to do identify, and the estimated angular speed have a considerable influence on the quality of the identification. Nevertheless the angular speed appears to show its influence much more on the participation, than on the frequencies and dampings.
- The inaccuracy of the results can be lowered by performing a robust curve fitting of the identification points.
- By performing the convergence study of the curve fittings we have seen that an elevated number of identifications is necessary to bring the coefficients confidence bounds below an acceptable threshold, while a less restrictive constraint is posed on the value of the coefficients.
- The Moving Average part is responsible of the excellent superposition of the measured and predicted time histories, and it should thus be preferred with respect to a precise tuning of the eXogenous part.

Chapter 7

Conclusions and future improvements

In this thesis we have addressed to solve the currents needs, enlighten in the introduction, for the modal analysis of wind turbines.

Our first task has been to evaluate the error committed in performing the modal analysis with the Coleman approximation instead of the Floquet theory. To accomplish this we have built an analytical model of wind turbine, that admits periodic steady states, and we applied to it an individual pitch controller [13]. After having analyzed the model in a variety of anisotropic conditions we draw the following conclusions.

- Even in high anisotropic conditions the Coleman approximation furnishes results that almost match the ones obtained by using the Floquet theory, both in terms of frequencies, than of dampings and participations.
- The only drawback of the Coleman approach is its inability to see more than three harmonics, and occasionally this consists in ignoring a significant part of the periodic content of the modes.
- The alteration that the IPC [13] causes at the frequencies is now understood, in fact we have observed that the IPC causes a steepen of the frequency branches characterizing the flapwise whirling modes, while leaving the others almost unaffected.
- Crossflows and wind shears causes little changes in the frequency and damping content of the modes of our analytical model.

The other need cited in the introduction is to perform periodic stability analysis on real wind turbines. In [16] the authors treated the problem by identifying reduced order models on the sole basis of input-output time histories, so as to

develop a model-independent tool, but supposed ideal wind conditions. With this work we extended [16], and opened the way to perform periodic stability analysis of wind turbines operating in the field. After an extended campaign of identifications we made the following conclusions.

- PARMAX models are able to closely follow the time histories of loads recorded in turbulent wind condition.
- The quality of the identification results is highly dependent from: the identification time, the number of samples per period and the estimated angular speed; and this dependence increases from PARX to PARMAX, and from frequency to damping factor to participation factor. These effects can be limited by performing a robust least-squares fitting of the data.
- PARMAX identifications are highly reliable to estimate the frequencies, but a noticeable spread is present in the dampings, and an even higher one in the participations.
- The turbulence level doesn't influence the frequencies nor the dampings of the first blade edgewise mode.

This work can be enhanced and improved in a few manners.

During this thesis we stressed that our identification procedure is applicable to experimental data, nevertheless we have only studied a numerical (multi-body) model. That model has been created with the goal of accurately describe the physics of the machine and wind, and evaluating the loads, thus it is representative of a real wind turbine. In [3] the authors developed a method with the same objective of our, i.e. perform periodic stability analyses of real wind turbines. Their method nevertheless is based on the Operational Modal Analysis, a theory that has different hypothesis with respect to the one employed here. At present [3] represents the state of the art, and an important improvement of this work would be to compare the results obtainable with our procedure with those obtainable with the OMA, so as to understand the pros and cons of each method.

A straightforward improvement is to apply the identification procedure to modes other than the first edgewise one. This would increase the level of knowledge over the machine, and would make it possible other comparisons with [16].

The way in which the identifications are made can be improved by allowing for multiple output. In fact in [16] and in chapter 3 it was shown that each mode can appear more or less periodic if it is observed from one signal instead of another. This suggest that the identification of multiple modes at the same time would lower the effort needed to do a complete stability analysis.

It could also be useful to implement other identification techniques, such as: Kalman filtering, subspace model identification and wavelets. This would permit

to test the robustness of the various identification techniques, and discover the ones that performs better for the periodic stability analysis of wind turbines operating in the field.

In terms of enhancements the dependence of the results from the angular speed variations caused by turbulence has been studied, but not yet deeply considered. A possible solution could be to identify PARMAX models that allow for slow variation of the system period.

In [42] the authors allowed their model to have blades with different flapping stiffness. This modification, together with differences in structural dampings, masses and inertia could be easily implemented in our model. This would allow to perform a better comparison between the Floquet theory and the Coleman approximation. Another improvement of the model could be to remove the flap-lag hinge, and consider a flexible blade like in [28], for example through an Ritz series expansion. Also the aerodynamics can be improved, by considering a more physical inflow.

Our analytical model in the future could also be used for control law testing. To make it adapt to this task the constraint of constant angular speed should be removed, and the generator torque added.

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